BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science

BANARAS HINDU UNIVERSITY

Department of Botany

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Botany

DEPARTMENT OF BOTANY
BANARAS HINDU UNIVERSITY, VARANASI-221005
Syllabus
NEP Course Structure for Four Year Undergraduate BotanyProgram

	Paper/Course	Credit	Name of paper
Semester I (20 Credits)	**Introductory/Foundation Level Course**		
	BOTMJ11	3	Cryptogams
	BOTMJ12	1	Lab Work Based on Paper BOTMJ11
	BOTMN11	3	Cryptogams
	BOTMN12	1	Lab Work Based on Paper BOTMN11
	BOTMD11	2	Elements of Ecology
	BOTMD12	1	Lab Work Based on Paper BOTMD11
	AE1	2	Offered by Institute/University
	BOTSE11	2	Waste Management
	BOTSE12	1	Lab Work Based on Paper BOTSE11
	VA1	2	Offered by Institute/University
	VA2	2	Offered by Institute/University
	Total	**20 Credits**	

	Paper/Course	Credit	Name of paper
Semester II (20 Credits)	**Introductory/Foundation Level Course**		
	BOTMJ21	3	General Microbiology
	BOTMJ22	1	Lab Work Based on Paper BOTMJ21
	BOTMN21	3	General Microbiology
	BOTMN22	1	Lab Work Based on PaperBOTMN21
	BOTMD21	2	Herbal Medicine
	BOTMD22	1	Lab Work Based on Paper BOTMD21
	AE2	2	Offered by Institute/University
	BOTSE21	2	Preservation and Management of Microbial Resource
	BOTSE22	1	Lab Work Based on Paper BOTSE21
	VA3	2	Offered by Institute/University
	VA4	2	Offered by Institute/University
	Total	**20 Credits**	

	Paper/Course	Credit	Name of paper
Semester III (20 Credits)	**Intermediate Level Course**		
	BOTMJ31	3	Cytology and Genetics
	BOTMJ32	1	Lab Work Based on Paper BOTMJ31
	BOTMJ33	3	Plant Anatomy and Embryology
	BOTMJ34	1	Lab Work Based on Paper BOTMJ33
	BOTMV31	3	Mushroom Cultivation
	BOTMV32	1	Lab Work Based on Paper BOTMV31
	BOTMD31	2	Industrial Microbiology
	BOTMD32	1	Lab Work Based on Paper BOTMD31
	AE3	2	Offered by Institute/University
	BOTSE31	2	Ethnobotany
	BOTSE32	1	Lab Work Based on Paper BOTSE31
		Total-	**20 Credits**

	Paper/Course	Credit	Name of paper
	Intermediate Level Course		
Semester IV (20 Credits)	BOTMJ41 BOTMJ42	3 1	Angiosperm Taxonomy Lab Work Based on PaperBOTMJ41
	BOTMJ43 BOTMJ44	3 1	Gymnosperm and Paleobotany Lab Work Based on PaperBOTMJ43
	BOTMJ45 BOTMJ46	3 1	Plant Ecology Lab Work Based on PaperBOTMJ45
	BOTMJ47 BOTMJ48	1 1	Microbial Resource Lab Work Based on PaperBOTMJ47
	BOTMN41/BOTMN43/ BOTMN45 BOTMN42/ BOTMN44/ BOTMN46	3 1	*Students may opt for any of the following: BOTMN41/BOTMN43/BOTMN45 Lab Work Based on PaperBOTMN41/BOTMN43/BOTMN45
	AE4	2	Offered by Institute/University
	Total	**20 Credits**	

	Paper/Course	Credit	Name of paper
	High Level Course		
Semester V (20 Credits)	BOTMJ51 BOTMJ52	3 1	Plant Physiology Lab Work Based on Paper BOTMJ51
	BOTMJ53 BOTMJ54	3 1	Plant Pathology Lab Work Based on Paper BOTMJ53
	BOTMJ55 BOTMJ56	3 1	Recombinant DNA Technology Lab Work Based on Paper BOTMJ55
	BOTMJ57 BOTMJ58	1 1	Plant and Microbial Techniques Lab Work Based on PaperBOTMJ57
	BOTMV51 BOTMV52	3 1	Herbarium Preparation and Preservation Lab Work Based on Paper BOTMV51
	BOTIN51	2	Internship
	Total	**Credits**	

	Paper/Course	Credit	Name of paper
	High Level Course		
Semester VI (20 Credits)	BOTMJ61 BOTMJ62	3 1	Comparative Cryptogams Lab Work Based on Paper BOTMJ61
	BOTMJ63 BOTMJ64	3 1	Comparative Phanerogams Lab Work Based on Paper BOTMJ63
	BOTMJ65 BOTMJ66	3 1	Biodiversity and Biostatistics Lab Work Based on Paper BOTMJ65
	BOTMJ67 BOTMJ68	3 1	Cytogenetics and Evolutionary Processes Lab Work Based on Paper BOTMJ67
	BOTMV61 BOTMV62	3 1	Plant Micropropagation Lab Work Based on Paper BOTMV61
	Total	**20 Credits**	

Honours without Research

	Paper/Course	Credit	Name of paper
Semester VII (20 Credits) B.Sc. (Hons.) without Research	**Advance Level Course**		
	BOTMJ71 BOTMJ72	3 1	Plant Biochemistry and Molecular Biology Lab Work Based on Paper BOTMJ71
	BOTMJ73 BOTMJ74	3 1	Plant Biotechnology and Genetic Engineering Lab Work Based on Paper BOTMJ73
	BOTMJ75	2	Field Excursion
	BOTMJ76 BOTMJ77	3 1	Photoecophysiology Lab Work Based on Paper BOTMJ76
	BOTMJ78	2	Research Methodology
	BOTMN71/BOTMN73/BOTMN76 BOTMN72/BOTMN74/ BOTMN77	3 1	*Students may opt for any of the following: BOTMN71/BOTMN73/ BOTMN76 Lab Work Based on Paper BOTMN71/ BOTMN73/BOTMN76
	Total	**20 Credits**	

	Paper/Course	Credit	Name of paper
Semester VIII (20 Credits) B.Sc. (Hons.) without Research	**Advance Level Course**		
	BOTMJ81 BOTMJ82	3 1	Microbiology Lab Work Based on Paper BOTMJ81
	BOTMJ83 BOTMJ84	3 1	Stress Biology Lab Work Based on Paper BOTMJ83
	BOTMJ85 BOTMJ86	3 1	Ecology,Pollution and Toxicology Lab Work Based on Paper BOTMJ85
	BOTMJ87 BOTMJ88	3 1	Plant Diseases and Protection Lab Work Based on Paper BOTMJ87
	BOTMN81/BOTMN83/BOTMN85/ BOTMN87 BOTMN82/BOTMN84/BOTMN86/ BOTMN88	3 1	*Students may opt for any of the following: BOTMN81/BOTMN83/BOTMN85/BOTMN 87 Lab work based on paper BOTMN81/BOTMN83/BOTMN85/BOTMN 87
	Total	**20 Credits**	

Honours with Research

	Paper/Course	Credit	Name of paper
	Advance Level Course		
Semester VII (20 Credits) B.Sc. (Hons.) with Research	BOTMJ71R BOTMJ72R	3 1	Plant Biochemistry and Molecular Biology Lab Work Based on Paper BOTMJ71R
	BOTMJ73R BOTMJ74R	3 1	Plant Biotechnology and Genetic Engineering Lab Work Based on Paper BOTMJ73R
	BOTMJ75R BOTMJ76R	3 1	Ecosystem Dynamics Lab Work Based on Paper BOTMJ75R
	BOTMJ77R	2	Field Excursion
	BOTMJ78R	2	Research Methodology
	BOTMN71R/BOTMN73R/BOTMN75R BOTMN72R/BOTMN74R/BOTMN76R	3 1	*Students may opt for any of the following: BOTMN71R/BOTMN73R/BOTMN75R Lab work based on paper BOTMN71R/BOTMN73R/BOTMN75R
	Total	**20 Credits**	

	Paper/Course	Credit	Name of paper
Semester VIII (20 Credits) B.Sc. (Hons.) with Research	**Advance Level Course**		
	BOTDS81R	12	Dissertation/Project
	BOTMJ82R BOTMJ83R	3 1	Aquatic Ecology Lab Work Based on PaperBOTMJ82R
	BOTMN84R BOTMN85R	3 1	Aquatic Ecology Lab Work Based on Paper BOTMN84R
	Total:	**20 Credits**	

BANARAS HINDU UNIVERSITY
(Detailed Syllabus of different Courses under NEP 2020)

Department of Botany, Institute of Science

Faculty of Science, BHU

SEMESTER: I

Course Title	**BOTMJ11 : Cryptogams**
Category of Course[1]	**Major**/ Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation
Credits[2]**& Hour of Teaching**	

	Theory	Practical	Cumulative
Credits	3	1	4
Hours of Teaching	**45**	**30**	**75**

Course Objectives	The course on cryptogams aims to provide students with a comprehensive understanding of the general characteristics, classification, and life cycles of various cryptogamic groups. It covers detailed studies of representative genera from algae, fungi, bryophytes, and pteridophytes, exploring their structure, reproduction, and ecological significance. Through this course, students will gain foundational knowledge essential for further studies in plant sciences and biodiversity.

Units	**Course Content**	**Hours of Teaching:45 h**
I	General characteristics and classification of cryptogams	5.0 h
II	Structure, reproduction and life cycles of representative genera of algae: • Chlorophyceae: *Volvox, Oedogonium* • Xanthophyceae: *Vaucheria* • Phaeophyceae: *Ectocarpus* • Rhodophyceae: *Polysiphonia* • Cyanophyceae: *Nostoc, Scytonema*	12.0 h
III	Structure, reproduction and life cycles of representative genera of fungi: • Mastigomycotina:*Albugo* • Zygomycotina: *Rhizopus* • Ascomycotina: *Peziza* • Basidiomycotina: *Agaricus* • Deuteromycotina: *Cercospora, Alternaria*	12.0 h
IV	Structure, reproduction and life cycles of representative genera of Bryophytes: • Hepaticopsida: *Marchantia* • Anthocerotopsida: *Anthoceros* • Bryopsida: *Funaria*	8.0 h
V	Structure, reproduction and life cycles of representative genera of Pteridophytes: • Lycophyta: *Selaginella* • Sphenophyta: *Equisetum* • Filicophyta: *Pteris*	8.0 h

Texts/ References	1. F.E. Fritsch (1935, reprinted in 1979). Structure and Reproduction of the Algae. Vol I.Vikash Publishing House Pvt. Ltd, New Delhi 2. F.E. Fritsch (1945, reprinted in 1979). Structure and Reproduction of the Algae. Vol I.Vikash Publishing House Pvt. Ltd, New Delhi

[1] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[2] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

3. Morris I (1967) Introduction to Algae, Hutchinson University Library, London.
4. Lee RE (2018) Phycology, Fifth Edition. Cambridge University Press, New Delhi.
5. Smith GM (1972) Cryptogamic Botany Vol I and II, Tata McGraw Hill Publishing Co Ltd, New Delhi.
6. Alexopolous CJ, Mims CW, Blackwell M (2007 Reprinted in 2013) Introductory Mycology 4th edition,Wiley, India, New Delhi..
7. Parihar NS (1973) An Introduction to Embryophyta, Vol I (Bryophyta) and Vol II (Pteridophyta), Central Book Department, Allahabad.
8. Rashid A (2011, reprinted in 2019) An Introduction to Pteridophyta 2nd edition, Vikas Publishing House Pvt Ltd, Noida.
9. Gangulee HC, Kar AK (2015) College Botany VolII(Algae, Fungi, Brophyta, Pteridophyta), New Central Book Agency, Kolkata.
10. Kumar HD (2007) Introductory Phycology, 2nd edition, Affiliated East-West Press Pvt. Ltd., New Delhi
11. Dubey HC (2017) An Introduction to Fungi, 3rd edition, Vikash Publication House Pvt Ltd, Noida.
12. Bold HC, Wynne MJ (1985) Introduction to the Algae, 2nd edition, Prentice-Hall Inc, New Jersey.
13. L.E. Graham, J.M. Graham (1009) Algae, 2nd Edition, Benjamin Cummings.
14. J. Weber and RW Weber (2007, reprinted in 1016). Introduction to Fungi. Cambridge, New Delhi.

Learning Outcomes	By the end of this course on cryptogams, students will understand and classify various groups of cryptogams, describe the structure, reproduction, and life cycles of key genera in algae, fungi, bryophytes, and pteridophytes, and identify their ecological roles and evolutionary significance. This foundational knowledge will enable students to apply their understanding of cryptogams to broader studies in plant sciences and biodiversity.
Course Title	**BOTMJ12 : Lab Work Based on Paper BOTMJ11**

Units	**Practicals**	**Hours of Teaching:30 h**
I	General instructions for staining, slide preparation, and microscopic observation	6.0 h
II	Observation of morphological and reproductive structure of following Algae: *Volvox* *Oedogonium* *Vaucheria* *Ectocarpus* *Polysiphonia* *Nostoc* *Scytonema*	6.0 h
III	Observation of morphological and reproductive structure of following Fungi: *Albugo* *Rhizopus* *Peziza* *Agaricus* *Cercospora* *Alternaria*	6.0 h
IV	Observation of morphological and reproductive structure of following Bryophytes: *Marchantia* *Anthoceros* *Funaria*	6.0 h
V	Observation of morphological and reproductive structure of following Pteridophytes: *Selaginella* *Equisetum* *Pteris*	6.0 h

Course Title	BOTMN11: Cryptogams
Category of Course[3]	Major / **Minor** / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation
Credits[4]& Hour of Teaching	<table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr><tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr></table>
Course Objectives	This course aims to provide students with a comprehensive understanding of cryptogams by exploring their general characteristics, classification, and life cycles. Covering key genera in algae, fungi, bryophytes, and pteridophytes, students will study the structure and reproduction of these organisms. By the end of the course, students will gain a foundational knowledge essential for further studies in plant sciences, ecology, and biodiversity, enabling them to appreciate the ecological and evolutionary significance of these diverse groups.

Units	Course Content	Hours of Teaching:45 h
I	General characteristics and classification of cryptogams	5.0 h
II	Structure, reproduction and life cycles of representative genera of algae: • Chlorophyceae: *Volvox, Oedogonium* • Xanthophyceae: *Vaucheria* • Phaeophyceae: *Ectocarpus* • Rhodophyceae: *Polysiphonia* • Cyanophyceae: *Nostoc, Scytonema*	12.0 h
III	Structure, reproduction and life cycles of representative genera of fungi: • Mastigomycotina:*Albugo* • Zygomycotina: *Rhizopus* • Ascomycotina: *Peziza* • Basidiomycotina: *Agaricus* • Deuteromycotina: *Cercospora* , *Alternaria*	12.0 h
IV	Structure, reproduction and life cycles of representative genera of Bryophytes: • Hepaticopsida: *Marchantia* • Anthocerotopsida: *Anthoceros* • Bryopsida: *Funaria*	8.0 h
V	Structure, reproduction and life cycles of representative genera of Pteridophytes: • Lycophyta: *Selaginella* • Sphenophyta: *Equisetum* • Filicophyta: *Pteris*	8.0 h

Texts / References	1 F.E. Fritsch (1935, reprinted in 1979). Structure and Reproduction of the Algae. Vol I.Vikash Publishing House Pvt. Ltd, New Delhi 2 F.E. Fritsch (1945, reprinted in 1979). Structure and Reproduction of the Algae. Vol I.Vikash Publishing House Pvt. Ltd, New Delhi 3 Morris I (1967) Introduction to Algae, Hutchinson University Library, London. 4 Lee RE (2018) Phycology, Fifth Edition. Cambridge University Press, New Delhi. 5 Smith GM (1972) Cryptogamic Botany Vol I and II, Tata McGraw Hill Publishing Co Ltd, New Delhi. 6 Alexopolous CJ, Mims CW, Blackwell M (2007 Reprinted in 2013) Introductory Mycology 4th edition,Wiley, India, New Delhi..

[3] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[4] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	7 Parihar NS (1973) An Introduction to Embryophyta, Vol I (Bryophyta) and Vol II (Pteridophyta), Central Book Department, Allahabad. 8 Rashid A (2011, reprinted in 2019) An Introduction to Pteridophyta 2nd edition, Vikas Publishing House Pvt Ltd, Noida. 9 Gangulee HC, Kar AK (2015) College Botany VolII(Algae, Fungi, Brophyta, Pteridophyta), New Central Book Agency, Kolkata. 10 Kumar HD (2007) Introductory Phycology, 2nd edition, Affiliated East-West Press Pvt. Ltd., New Delhi 11 Dubey HC (2017) An Introduction to Fungi, 3rd edition, Vikash Publication House Pvt Ltd, Noida. 12 Bold HC, Wynne MJ (1985) Introduction to the Algae, 2nd edition, Prentice-Hall Inc, New Jersey. 13 L.E. Graham, J.M. Graham (1009) Algae, 2nd Edition, Benjamin Cummings. 14 J. Weber and RW Weber (2007, reprinted in 1016). Introduction to Fungi. Cambridge, New Delhi.
Learning Outcomes	By the end of this course on cryptogams, students will be able to classify and describe the general characteristics, structure, reproduction, and life cycles of key genera in algae, fungi, bryophytes, and pteridophytes. They will gain an understanding of the ecological roles and evolutionary significance of these groups, providing a strong foundation for further studies in plant sciences and biodiversity. This knowledge will enable students to appreciate the diversity and complexity of cryptogamic organisms and their importance in various ecosystems.

Course Title	**BOTMN12: Lab Work Based on Paper BOTMN11**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	General instructions for staining, slide preparation, and microscopic observation	6.0 h
II	Observation of morphological and reproductive structure of following Algae: *Volvox* *Oedogonium* *Vaucheria* *Ectocarpus* *Polysiphonia* *Nostoc* *Scytonema*	6.0 h
III	Observation of morphological and reproductive structure of following Fungi: *Albugo* *Rhizopus* *Peziza* *Agaricus* *Cercospora* *Alternaria*	6.0 h
IV	Observation of morphological and reproductive structure of following Bryophytes: *Marchantia* *Anthoceros* *Funaria*	6.0 h
V	Observation of morphological and reproductive structure of following Pteridophytes: *Selaginella* *Equisetum* *Pteris*	6.0 h

Course Title	BOTMD11 : Elements of Ecology
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation
Credits& Hour of Teaching	

	Theory	Practical	Cumulative
Credits	2	1	3
Hour of Teaching	30	30	60

Course Objectives	This course aims to provide students with a foundational understanding of ecology by exploring key concepts such as types of population interactions, ecosystem components, types and functions, and the significance of biodiversity. Additionally, the course will examine ecological adaptations in hydrophytes and xerophytes. By the end of the course, students will gain essential knowledge for further ecological studies and an appreciation of the complexity and interdependence of life within various ecosystems.

Units	Course Content	Hours of Teaching:30 h
I	Introduction to ecology	4.0 h
II	Types of population interactions	4.0 h
III	Ecosystem: Components, types and function	8.0 h
IV	Significance ofbiodiversity	7.0 h
V	Ecological adaptation: Hydrophytes and xerophytes	7.0 h

Texts/ References	1. Odum EP, Barrett GW (2005) Fundamentals of Ecology, Thomson Ed. Brooks/Cole, Cengage Learning India Pvt. Ltd, New Delhi. 2. Odum EP (1983) Basic Ecology Ed. Saunders College Pub, New York. 3. Smith TM, Smith RL (2008) Elements of Ecology, 7th edition, Benjamin- Cummings New York. 4. Miller GT (2004) Essentials of Ecology, 3rd edition, Brooks, Cole, New York. 5. Smith RL (2001) Ecology and Field Biology, Benjamin Cummings, New York. 6. Vallero D (2014) Fundamentals of Air Pollution, 3rd edition, Academic Press, Elsevier, Burlington. 7. Klassen C, Watkins JB (2010) Essentials of Toxicolgy, 2nd edition, McGraw- Hill Companies, New York. 8. Singh JS, Singh SP, Gupta SR (2014) Ecology Environmental Science and Conservation, S Chand & Co, New Delhi. 9. Mueller-Dombois D and Ellenberg H (1974) Aims and Methods of Vegetation Ecology. Wiley & Sons., NY, USA
Learning Outcomes	By the end of this course, students will be able to understand and explain the fundamentals of ecology, including types of population interactions, ecosystem components and functions, and the significance of biodiversity. They will also be able to identify and describe ecological adaptations in hydrophytes and xerophytes. This knowledge will provide a solid foundation for further ecological studies and foster an appreciation for the complexity and interdependence of organisms within various ecosystems.

Course Title	BOTMD12 : Lab Work Based on Paper BOTMD11

Units	Practicals	Hours of Teaching:30 h
I	Soil properties analysis: • soil texture • pH • Carbonate and nitrate content	10.0 h
II	Demonstration of instruments involved in measurement of abiotic factors	10.0 h
III	Ecological study of plant adaptations in response to: • Hydrophytes • Xerophytes	10.0 h

Course Title	**AE1: Offered by Institute/University (2 Credits)**
Category of Course[5]	Major / Minor / Minor (Vocational) / SEC / **AEC** / VAC / MD/Internship/Dissertation

Course Title	**BOTSE11: Waste Management**
Category of Course[6]	Major / Minor / Minor (Vocational) / **SEC** / AEC / VAC / MD/Internship/Dissertation

Credits[7] & Hour of Teaching

	Theory	Practical	Cumulative
Credits	2	1	3
Hour of Teaching	30	30	60

Course Objectives

The course on waste management aims to provide students with a comprehensive understanding of the principles and practices for handling solid and liquid wastes. It covers the characteristics, segregation, and management of different types of wastes, along with effective treatment and disposal methods. Students will also learn about the environmental and health impacts of wastes and the relevant rules and regulations governing waste management. By the end of the course, students will be equipped with the knowledge and skills necessary to implement sustainable waste management practices and ensure compliance with legal standards.

Units	**Course Content**	**Hours of Teaching: 30 h**
I	Introduction to solid and liquid wastes Electronic and Medical Wastes	5.0 h
II	Solid wastes: Electronic and Medical Wastes, Characteristics, segregation and management	9.0 h
III	Liquid wastes: Characteristics and management	9.0 h
IV	Waste management: Rules and regulations	7.0 h

Texts/ References

1. Singh JS, Singh SP and Gupta SR. 2014. Ecology, Environmental Science and Conservation. S. Chand & Company Pvt. Ltd., New Delhi
2. Hassan A. 2022. Introduction to waste management: a text book. Wiley publishers
3. Linda G. 2021. Waste management practices in developing countries (Eds). MDPI

Learning Outcomes

Upon completing this course, students will be able to identify and classify different types of solid and liquid wastes, implement effective segregation and management techniques, and understand their environmental and health impacts. They will be skilled in applying appropriate treatment and disposal methods while ensuring regulatory compliance. Additionally, students will develop sustainable waste management strategies, navigate relevant rules and regulations, and address waste management challenges with innovative solutions that prioritize environmental protection and public health.

Course Title	**BOTSE12: Lab Work Based on Paper BOTSE11**

Units	**Practicals**	**Hours of Teaching:30 h**
I	Collection of solid wastes	6.0 h
II	Segregation of solid wastes	6.0 h
III	Visit to solid waste treatment plants	6.0 h
IV	Visit to wastewater treatment plants	6.0 h
V	Water quality analysis	6.0 h

[5] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[6] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[7] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Title	**VA1: Offered by Institute/University(2 Credits)**
Category of Course[8]	Major / Minor / Minor (Vocational) / SEC / AEC / **VAC -1**/ MD/Internship/Dissertation

Course Title	**VA2: Offered by Institute/University(2 Credits)**
Category of Course[9]	Major / Minor / Minor (Vocational) / SEC / AEC / **VAC -2**/ MD/Internship/Dissertation

SEMESTER: II

Course Title	**BOTMJ21: General Microbiology**
Category of Course[10]	**Major** / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation
Credits[11] **& Hour of Teaching**	<table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr><tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr></table>
Course Objectives	This course aims to provide students with a comprehensive understanding of microbiology, covering its history, scope, and the role of microbes in extreme environments. It explores the ecological, morphological, physiological, biochemical, and molecular criteria for bacterial classification, and details the structure of bacterial cells and bacteriophages. Additionally, the course examines genetic recombination in bacteria, the role of microorganisms in carbon and nitrogen cycling, antibiotic production, alcohol and beverage production, xenobiotic degradation, and their impact on human health. Through this course, students will gain essential knowledge for advanced studies and applications in microbiology and related fields.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• History and scope of microbiology • Microbes in extreme environment	8.0 h
II	• Position of microorganisms in the living world; Ecological, morphological, physiological, biochemical and molecular criteria for the classification of bacteria (scheme not required)	7.0 h
III	• Structure of a bacterial cell: flagella, cell envelope, cell membrane, chromosome, plasmid and endospore • Structure of bacteriophages T4 and Lambda; Lysogenic and lytic cycle	8.0 h
IV	• A brief account of genetic recombination in bacteria (transformation, conjugation and transduction) • Role of microorganisms in carbon and nitrogen cycling	10.0 h
V	• Microorganisms and their role in the production of antibiotics • Role of microbes in alcohol and beverages production • Microbes in xenobiotic degradation • Microorganisms and human health	12.0 h

[8] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[9] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[10] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[11] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Texts/ References	1. Willey J, Sandman K, Wood D (2020) Prescott's Microbiology, 11^{th} edition, McGraw-Hill, New York. 2. Madigan MT, Bender KS, Buckley DH, Sattley WM, Stahl DA (2020) Brock Biology of Microorganism, 16^{th} edition, Pearson Education, London, United Kingdom 3. Stanier RY, Ingraham JL, Wheelis ML, Painter PR (1987) General Microbiology, 5^{th} edition, MacMillan Press Ltd, New Jersey. 4. Dubey RC, Maheshwari DK (2014) A Textbook of Microbiology, Revised edition, S Chand and Company Ltd, New Delhi
Learning Outcomes	By the end of this course, students will be able to understand the history and scope of microbiology, describe the role of microbes in extreme environments, and classify bacteria based on ecological, morphological, physiological, biochemical, and molecular criteria. They will be able to explain the structure of bacterial cells and bacteriophages, understand genetic recombination in bacteria, and describe the roles of microorganisms in carbon and nitrogen cycling, antibiotic production, alcohol and beverage production, xenobiotic degradation, and human health. This knowledge will prepare students for advanced studies and practical applications in microbiology.

Course Title	**BOTMJ22: Lab Work Based on Paper BOTMJ21**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Study of free-living nitrogen fixing cyanobacteria (*Nostoc*, *Anabaena* and *Aulosira*)	6.0 h
II	• Study of symbiotic nitrogen fixers in *Cycas* and *Azolla* plants	6.0 h
III	• Calibration of microscope using ocular and stage micrometers, and the measurement of dimensions of microbial cells	6.0 h
IV	• Study of cyanobacterial blooms • Gram's staining of bacteria	6.0 h
V	• Demonstration of equipment's (autoclave, laminar flow, thermal cycler, spectrophotometer and bacteriological incubators)	6.0 h

Course Title	**BOTMN21: General Microbiology**			
Category of Course[12]	Major / **Minor**/ Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation			
Credits[13]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide students with a thorough understanding of microbiology by covering its history, scope, and the role of microbes in extreme environments. It includes an exploration of microbial classification, the structure of bacterial cells and bacteriophages, and genetic recombination processes. The course also examines the impact of microorganisms on carbon and nitrogen cycling, their roles in antibiotic production, alcohol and beverage production, xenobiotic degradation, and human health. By the end, students will be well-equipped for advanced studies and practical applications in microbiology.			

Units	**Course Content**	**Hours of Teaching:45 h**
I	• History and scope of microbiology • Microbes in extreme environment	8.0 h
II	• Position of microorganisms in the living world; Ecological, morphological, physiological, biochemical and molecular criteria for	7.0 h

[12] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[13] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	the classification of bacteria (scheme not required)	
III	• Structure of a bacterial cell: flagella, cell envelope, cell membrane, chromosome, plasmid and endospore • Structure of bacteriophages T4 and Lambda; Lysogenic and lytic cycle	8.0 h
IV	• A brief account of genetic recombination in bacteria (transformation, conjugation and transduction) • Role of microorganisms in carbon and nitrogen cycling	10.0 h
V	• Microorganisms and their role in the production of antibiotics • Role of microbes in alcohol and beverages production • Microbes in xenobiotic degradation • Microorganisms and human health	12.0 h

Texts/ References	1. Willey J, Sandman K, Wood D (2020) Prescott's Microbiology, 11th edition, McGraw-Hill, New York. 2. Madigan MT, Bender KS, Buckley DH, Sattley WM, Stahl DA (2020) Brock Biology of Microorganism, 16th edition, Pearson Education, London, United Kingdom 3. Stanier RY, Ingraham JL, Wheelis ML, Painter PR (1987) General Microbiology, 5th edition, MacMillan Press Ltd, New Jersey. 4. Dubey RC, Maheshwari DK (2014) A Textbook of Microbiology, Revised edition, S Chand and Company Ltd, New Delhi
Learning Outcomes	Upon completion of this course, students will be able to describe the history and scope of microbiology, classify microorganisms based on various criteria, and explain the structure and function of bacterial cells and bacteriophages. They will understand genetic recombination in bacteria, and the roles of microorganisms in carbon and nitrogen cycling, antibiotic production, alcohol and beverage production, xenobiotic degradation, and human health. This knowledge will enable students to apply microbiological concepts to practical and research contexts effectively.

Course Title	**BOTMN22: Lab Work Based on Paper BOTMN21**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Study of free-living nitrogen fixing cyanobacteria (*Nostoc*, *Anabaena* and *Aulosira*)	6.0 h
II	• Study of symbiotic nitrogen fixers in *Cycas* and *Azolla* plants	6.0 h
III	• Calibration of microscope using ocular and stage micrometers, and the measurement of dimensions of microbial cells	6.0 h
IV	• Study of cyanobacterial blooms • Gram's staining of bacteria	6.0 h
V	• Demonstration of equipments (autoclave, laminar flow, thermal cycler, spectrophotometer and bacteriological incubators)	6.0 h

Course Title	BOTMD21: Herbal Medicine
Category of Course[14]	Major / Minor/ Minor (Vocational) / SEC / AEC / VAC / **MD**/Internship/Dissertation
Credits[15]& Hour of Teaching	<table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>2</td><td>1</td><td>3</td></tr><tr><td>Hour of Teaching</td><td>30</td><td>30</td><td>60</td></tr></table>
Course Objectives	This course aims to provide students with a comprehensive understanding of herbal medicine, including its fundamentals and integration into complementary, alternative, and integrative medicine. Students will learn about the standardization of herbal drugs through pharmacognostical and organoleptic studies, and techniques for analyzing herbal drugs using chromatographic methods. The course will cover the commercial cultivation, diagnostic features, bioactive molecules, and therapeutic values of various medicinal plants. Additionally, students will explore herbs' therapeutic actions, conservation efforts, bioprospection, and protection of traditional medicinal knowledge. By the end, students will be equipped to critically evaluate and apply herbal medicine in various contexts.

Units	Course Content	Hour of Teaching:30 h
I	• Fundamentals of herbal medicine:MateriaMedica- an introduction to herbal medicine; herbal medicine an integral part of complementary, alternative and integrative medicine	4.0 h
II	• Standardization of herbal drugs:Pharmacognostical and organoleptic study of the herbal medicines; Factors affecting the quality of herbs ingredients, adulterations of herbal drugs, deterioration of herbal drugs, substitution of herbs, counterfeiting of herbal medicine	6.0 h
III	• Analysis of herbal drugs: Chromatographic analysis of herbal drugs using TLC, High-Performance Thin-Layer Chromatography (HPTLC), HPLC	5.0 h
IV	• Commercial cultivation, diagnostic features, bioactive molecules and therapeutic values of some common medicinal plants: Giloy, Brahmi, Safed Musli, Aola, Kalmegh, Satavari, Bel, Sarpgandha, Ashwagandha, Aloe, Tulsi, Neem	7.0 h
V	• Herbs and drug action: Neuro-promontory drugs, Cardiovascular drugs, Nutraceuticals and medicinal foods; anti-inflammatory drugs and anticancer drugs • Conservation of medicinal plants • Bioprospection, biopiracy and protection of traditional medicinal knowledge	8.0 h

Texts/ References	1. SC Mandal (2011). Herbal Drugs, New Central Book Agency, New Delhi. 2. ND Prajapati, S.S. Purohit, Arun K. Sharma andTarun Kumar (2006). A Hand Book of Medicinal Plants: A Complete Source Book, Agrobios (India), Jodhpur. 3. SG Joshi (2018). Medicinal Plants,Oxford& IBH Publishing. 4. JB (2019). Herbal Medicine: Simple and Effective Natural Remedies to Heal Common Ailments,Novelty Publishing LLC 5. AK (2014). Fundamentals of Herbal Medicine: Science of Nutrition, Human Anatomy and Body Systems, Chaukhambha Orientalia. 6. PC Trivedi (2009). Indian Medicinal Plants. Aavishkar Publishers & Distributors. 7. SK Bhattacharjee (2019). Hand Book of Aromatic Plants. 3rd Edition; Pointer Publishers. 8. LD Kapoor (2005). Handbook of Ayurvedic Medicinal Plants. 9. MP Singh et al. (2003). Indigenous Medicinal Plants Social Forestry &Tribals. Daya Publishing House. 10. VV Sivarajan & I Balachandran (1994). Ayurvedic Drugs and their Plant Sources, Oxford&

[14] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[15] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	IBH. 11. HW Youngken (2018). A Text Book of Pharmacognosy, Palala Press.
Learning Outcomes	Upon completing this course, students will be able to understand the fundamentals of herbal medicine and its role in integrative health practices. They will be proficient in the standardization and analysis of herbal drugs, using techniques like TLC, HPTLC, and HPLC. Students will gain knowledge about the commercial cultivation, diagnostic features, and therapeutic values of key medicinal plants. Additionally, they will be able to assess herbs' therapeutic actions, understand issues related to conservation, bioprospection, and protection of traditional medicinal knowledge, and apply this understanding to practical and research contexts in herbal medicine.

Course Title	**BOTMD22: Lab Work Based on Paper BOTMD21**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Identification and medicinal value of locally available medicinal plants	10.0 h
II	• Locally available crude herbal raw materials and its histo- anatomical analysis	10.0 h
III	• Methods of extraction of drugs from crud herbal raw materials: Maceration, Decoction, and Hot continuous extraction (Soxhlet)	10.0 h

Course Title	**AE2: Offered by Institute/University (2 Credits)**
Category of Course[16]	Major / Minor / Minor (Vocational) / SEC / **AEC-2**/ VAC / MD/Internship/Dissertation

Course Title	**BOTSE21: Preservation and Management of Microbial Resource**			
Category of Course[17]	Major / Minor / Minor (Vocational) / **SEC** / AEC / VAC / MD/Internship/Dissertation			
Credits[18] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	2	1	3
	Hour of Teaching	30	30	60
Course Objectives	This course provides an overview of the principles and practices involved in the preservation and management of microbial resources. It covers the importance of microbial diversity, various preservation techniques, microbial culture collections, and the legal and ethical aspects of microbial resource management. Students will gain practical knowledge and skills necessary for handling, preserving, and managing microbial resources in research and industrial settings.			

Units	**Course Content**	**Hours of Teaching:30 h**
I	Introduction to Microbial Resources Types of microorganisms, their roles in the environment, and importance in various industries, Ecological, industrial, and biotechnological importance of preserving microbial diversity, Definition, examples, and the concept of microbial genetic resources.	6.0 h
II	Techniques for Preserving Microbial Cultures Refrigeration, storage on agar slants, and sub-culturing, Cryopreservation,	6.0 h

[16] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[17] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[18] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	lyophilization (freeze-drying), and storage in liquid nitrogen, Factors affecting the stability and viability of preserved cultures.	
III	Microbial Culture Collections Standards and protocols for maintaining microbial collections, Examples of major culture collections (e.g., ATCC, DSMZ) and their significance in research and industry, Importance of accurate documentation and information systems in managing culture collections.	6.0 h
IV	Legal and Ethical Considerations Patents, copyrights, and trademarks related to microbial resources, Guidelines for the safe handling and transport of microbial cultures, Access and benefit-sharing, biopiracy, and compliance with the Nagoya Protocol.	6.0 h
V	Microbial Resource Management Real-world examples of microbial resource management in research and industry, Management and preservation of microbial cultures, including creating a mini-culture collection.	6.0 h

Texts/ References	1. Atlas, R.M. (2010). Handbook of Microbiological Media. CRC Press. 2. Smith, D., & Onions, A.H.S. (1994). The Preservation and Maintenance of Living Fungi. CAB International. 3. Sneath, P.H.A., & Holt, J.G. (2001). Bergey's Manual of Systematic Bacteriology. Springer.
Learning Outcomes	This course content provides a structured approach to understanding and managing microbial resources, equipping students with both theoretical knowledge and practical skills essential for careers in microbiology and related fields.

Course Title	**BOTSE22: Lab Work Based on PaperBOTSE21**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	Isolation and Culturing of Microorganisms, Techniques for isolating microorganisms from soil, water, and air samples	10.0 h
II	Preservation of Microorganisms by freezing with glycerol. Preparation of cryoprotectants, Freezing and storage techniques for bacteria, fungi and algae	10.0 h
III	Lyophilization (Freeze-Drying) methods for preservation	5.0 h
IV	Examination of successful microbial resource management project	5.0 h

Course Title	**VA3: Offered by Institute/University(2 Credits)**
Category of Course[19]	Major / Minor / Minor (Vocational) / SEC / AEC / **VAC-3**/ MD/Internship/Dissertation

Course Title	**VA4: Offered by Institute/University(2 Credits)**
Category of Course[20]	Major / Minor / Minor (Vocational) / SEC / AEC / **VAC-4**/ MD/Internship/Dissertation

[19] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[20] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

SEMESTER: III

Course Title	**BOTMJ31: Cytology and Genetics**
Category of Course[21]	**Major** / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation
Credits[22] **& Hour of Teaching**	<table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr><tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr></table>
Course Objectives	This course aims to provide a comprehensive understanding of cytology and genetics. It covers the fundamentals of cell biology, including cell structure, organelles, and the ultrastructure of the nucleus. Students will study cell division processes, including mitosis and meiosis, and explore basic and advanced genetic principles, such as Mendelian genetics, gene interactions, and linkage. The course also addresses genetic mapping, extra-nuclear inheritance, and sex determination mechanisms. By the end, students will be equipped with a solid foundation in cytological and genetic concepts applicable to both research and practical applications.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Introduction to cell: Cell as the basic unit of life, cell theory, prokaryotic and eukaryotic cell, differences between plant and animal cell • Structural organization and function of intracellular organelles • Cell wall: primary and secondary structure, middle lamella, plasmodesmata, apoplast and symplast and their functions • Cell membrane: chemical composition, membrane models, function • Cytoskeleton: microtubules, microfilaments and intermediate filaments, structure and their functions. • Endomembrane system: endoplasmic reticulum, Golgi apparatus, lysosomes, peroxisomes, vacuoles	10.0 h
II	• Other organelles: mitochondria, chloroplast and ribosomes • Ultrastructure of nucleus: nuclear envelope, NPC, nucleolus, chromatin and chromosomes, DNA packaging, euchromatin and heterochromatin, special types of chromosomes- polytene chromosomes, lamp brush chromosomes, B chromosomes	10.0 h
III	• Cell division: Cell cycle, mitosis and meiosis, control and regulation of cell cycle, significance of meiosis	5.0 h
IV	• Basic Principles of Genetics: Pre-Mendalian concepts of Genetics, Mendel's experiments, law of segregation and law of independent assortment • Neo Mendelism: Intergenic and intragenic gene interactions, incomplete dominance and Co-dominance, lethal alleles, epistasis, complementary gene, supplementary gene, inhibitory gene, duplicate gene, multiple allelism	6.0 h
V	• Linkage, Crossing over and Genetic mapping: Chromosomal theory of inheritance, Linkage and crossing over, Cytological basis of crossing over, crossing over and linkage map • Intragenic mapping and modern concept of gene • Extra-Nuclear Inheritance: Chloroplast and mitochondrial genomes, chloroplast inheritance and variegation in Four O'clock plant, mitochondrial inheritance in *Neurospora* and petites in yeast, maternal effect; shell coiling in Snails and Kappa particles in	14.0 h

[21] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[22] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	Paramecium • Sex determination: Basis of sex determination, sex determination in plants, chromosome theory of sex determination, sex linked characters	

Texts/ References	1. Gardner, E.J., Simmons, M.J., Snustad, D.P. (1991). Principles of Genetics, 8th edition. New Delhi, Delhi: John Wiley & sons. 2. Griffiths, A.J.F., Doebley,J., Peichel, C, Wassarman D (2020). Introduction to Genetic Analysis, 12th edition. New York, NY: W.H. Freeman and Co. 3. Clark MS, Wall WJ (1996) Chromosomes: The Complex Code, Chapman & Hall, London. 4. Alberts B, Bray D, Lewis J, Ralf M, Roberts K, Watson JD (1999) Molecular Biology of the Cell, Garland Publishing Inc, New York.. 5. Campbell, N.A., Reece J.B., Urry L.A., Cain M.L., Wasserman S.A., Minorsky P.V., Jackson, R.B. (2020). Biology. San Francisco, SF: Pearson Benjamin Cummings. 6. Russell, P. J. (2010). Genetics- A Molecular Approach. 3rd Edition. Benjamin Cummings. 7. Snustad, D.P., Simmons, M.J. (2016). Principles of Genetics, 7th Edition. New Delhi, Delhi: John Wiley & sons. 8. Hartl, D.L., Ruvolo, M. (2019). Genetics: Analysis of Genes and Genomes, 9th edition, Jones and Bartlett Learning. 9. Singh, B. D. (2023). Fundamentals of Genetics, 6th edition. MedTech. 10. De Robertis EDP, De Robertis EMF (2006) Cell and Molecular Biology, 8th edition, 11. Cooper GM (2019) The Cell: A Molecular Approach, 8th edition, Sinauer Oxford University Press, New York. 12. Alberts, Heald, Johnson, Morgan, Raff, Roberts, Walter Molecular Biology of the Cell, 7th edition, W.W. Norton & Company. 13. Gerald Karp (2019). Cell and Molecular Biology, 9th edition, Wiley 14. Harvey Lodish, Arnold Berk, Chris A. Kaiser, et al., (2016) Molecular Cell Biology, 8th edition, WH Freeman.
Learning Outcomes	Upon completing this course, students will be able to understand cell structure and function, including organelles and the nucleus, and explain cell division processes such as mitosis and meiosis. They will grasp fundamental and advanced genetic principles, including Mendelian genetics, gene interactions, linkage, and genetic mapping. Students will also be familiar with extra-nuclear inheritance, sex determination mechanisms, and their applications in various organisms. This knowledge will provide a strong foundation for further studies and research in cytology and genetics.

Course Title	**BOTMJ32: Lab Work Based on Paper BOTMJ31**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Acquaintance with cytochemical techniques: fixative and stains • Study of prokaryotic (cyanobacteria) and eukaryotic cells (onion peel)	6.0 h
II	• Mitosis in root tip of onion	6.0 h
III	• Meiosis in flower bud of onion	6.0 h
IV	• Demonstration of Mendel's law using Chi-square test (Monohybrid and Dihybrid Crosses)	6.0 h
V	• Study of epistasis using Chi-square test • Demonstration and construction of linkage map using suitable data	6.0 h

Course Title	**BOTMJ33: Plant Anatomy and Embryology**			
Category of Course[23]	Major/ Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation			
Credits[24] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4

[23] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

	Hour of Teaching	45	30	75

Course Objectives	This course aims to provide students with an in-depth understanding of plant anatomy and embryology. It covers tissue classification, shoot apex organization, and the structural differences between dicot and monocot leaves and stems. Students will study cambium function, secondary growth, and anatomical anomalies in stems and roots. The course also includes detailed exploration of flower structure, reproductive processes, and gametogenesis. Additionally, students will learn about pollination, fertilization, endosperm development, and embryogeny in dicots and monocots. By the end of the course, students will gain comprehensive knowledge of plant structure and development crucial for advanced botanical studies.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Classification of tissues: simple and complex Organization of shoot apex: Apical cell theory, Histogen theory, Tunica Corpus theory	9.0 h
II	• Structure of dicot and monocot leaf, Kranz anatomy • Structure of dicot and monocot stem • Structure, function and seasonal activity of cambium; secondary growth in root and stem	10.0 h
III	• Anatomy of stem and root with special reference to plants showing anomalies: *Nyctanthes, Bignonia, Strychnos, Boerhaavia, Dracaena*; Root-*Tinospora*	6.0 h
IV	• Flower structure and reproduction • Microsporangium and microsporogenesis • Megasporangium and megasporogenesis • Male gametophyte	10.0 h
V	• Female gametophyte • Pollination and fertilization • Endosperm (different modes of development) • Embryogeny (development of typical dicot and monocot embryo)	10.0 h

Texts/ References	1. Dickson, W.C. (2000) Integrative Plant Anatomy. Harcourt Academic Press, USA. 2. Fahn, A. (1974) Plant Anatomy. Pergmon Press, USA. 3. Mauseth, J. D. (1988) Plant Anatomy. The Benjammin/ Cummings Publisher, USA. 4. Esau, K. (1977). Anatomy of Seed Plants. John Wiley and Sons, Inc., Delhi. 5. Pandey BP (2007) Angiosperms- Taxonomy, Embryology and Anatomy, S Chand and Co, New Delhi. 6. Tayal MS (1994) Plant Anatomy, Rastogi Publications, Meerut. 7. Bhojwani SS, Bhatnagar SP and Dantu PK (2015) The Embryology of Angiosperms, 6th Edition, Vikas Publishing House Pvt Ltd, Noida. 8. Johri, BM (2011) Embryology of Angiosperms, Springer Berlin Heidelberg.
Learning Outcomes	Upon completing this course, students will be able to identify and classify plant tissues, understand shoot apex organization theories, and describe the structural differences between dicot and monocot leaves and stems. They will also be able to explain cambium function, secondary growth, and anatomical anomalies in stems and roots. Students will gain knowledge of flower structure, reproductive processes, gametogenesis, and the development of endosperm and embryos in dicots and monocots. This comprehensive understanding will enable students to apply anatomical and embryological concepts to advanced botanical research and practical applications.

[24] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Title	BOTMJ34: Lab Work Based on Paper BOTMJ33	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• T.S. of stems:*Nyctanthes, Bignonia, Strychnos, Boerhaavia* and *Dracaena*	15.0 h
II	• T.S of *Tinospora* root	5.0 h
III	• T.S of dicot leaf and monocot leaf	5.0 h
IV	• Adaptive Anatomy of xerophytes and hydrophytes	5.0 h

Course Title	**BOTMV31: Mushroom Cultivation**
Category of Course[25]	Major/ Minor / **Minor (Vocational-1)**/ SEC / AEC /VAC / MD/Internship/Dissertation
Credits[26]& Hour of Teaching	<table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr><tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr></table>
Course Objectives	This course aims to provide students with a comprehensive understanding of mushroom cultivation, including identification of edible and medicinal species, and various cultivation methods such as substrate preparation, spawning, casing, and outdoor techniques. It covers environmental factors affecting cultivation, common diseases and pests along with their control strategies, and best practices for harvesting, handling, packaging, and storage. Additionally, students will explore the economic and nutritional importance of mushrooms. By the end of the course, students will be equipped with practical knowledge and skills for successful mushroom cultivation and management.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Introduction to Mushroom Cultivation	7.0 h
II	• Mushroom Identification: Edible and medicinal mushroom species	8.0 h
III	• Cultivation Methods: Substrate preparation, spawning, casing and outdoor cultivation methods	8.0 h
IV	• Environmental Factors affecting mushroom cultivation • Common diseases and pests affecting mushroom cultivation and its control strategies	12.0 h
V	• Harvesting, handling, packaging, and storage of mushrooms • Economic and nutritional importance of mushrooms	10.0 h

Texts/ References	1. Mushroom Cultivation and Its Use in Rural Development by J. H. Wasson. 2. The Mushroom Cultivator: A Practical Guide to Growing Mushrooms at Home by PaulStamets and J. S. Chilton. 3. Mushroom Biology: Concise Basics and Current Developments by Philip G. Miles and Shu-Ting Chang. 4. Scientific Paper: "Cultivation of Edible Mushrooms by Authors: M. M. R. Khan, M. A. Kabir, and M. A. Hannan. 5. Website: Mushroom Expert - Mushroom Cultivation (http://www.mushroomexpert.com/) 6. Description: This website offers a wealth of information on mushroom cultivation, including tutorials, tips, and troubleshooting guides
Learning Outcomes	Upon completing this course, students will be able to identify edible and medicinal mushrooms and understand various cultivation methods, including substrate preparation, spawning, and casing. They will gain knowledge of environmental factors affecting mushroom growth, and strategies for managing diseases and pests. Students will also learn effective techniques for harvesting, handling, packaging, and storing mushrooms. Additionally, they will understand the economic and nutritional significance of mushrooms, preparing them to apply their knowledge in both practical cultivation and the broader context of mushroom production and management.

[25] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[26] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Title	BOTMV32: Lab Work Based on Paper BOTMV31	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Hands-on lab sessions on substrate preparation	10.0 h
II	• Spawning and casing layer application	10.0 h
III	• Field visits and case Studies	10.0 h

Course Title	**BOTMD31: Industrial Microbiology**
Category of Course[27]	Major / Minor / Minor (Vocational) / SEC / AEC /VAC / **MD**/Internship/Dissertation
Credits[28] **& Hour of Teaching**	

	Theory	Practical	Cumulative
Credits	2	1	3
Hour of Teaching	30	30	30

Course Objectives	This course aims to provide students with a comprehensive understanding of industrial microbiology, including its definition, scope, and historical development. Students will explore the role of microorganisms in industrial processes, microbial growth and physiology, and the applications of microbial biotechnology in agriculture, environment, pharmaceuticals, and energy. The course covers food and beverage microbiology, including fermentation processes and biotechnological advancements, as well as the role of microbes in food spoilage, enzyme production, and biofuels. Additionally, students will study pharmaceutical microbiology, focusing on biopharmaceutical production, microbial contamination, sterilization, and regulatory compliance.

Units	**Course Content**	**Hours of Teaching:30 h**
I	• Introduction to Industrial Microbiology:Definition and scope of industrial microbiology, historical development and milestones, microorganisms and their role in industrial processes, importance of industrial microbiology in the modern world • Microbial Growth and Physiology:Microbial cell structure and function, growth phases and factors influencing microbial growth, measurement of microbial growth, microbial metabolism and energy production	6.0 h
II	• Microbial Biotechnology:Microbial biotechnology in the context of agriculture, environment, pharmaceuticals and energy - An Overview. Industrial applications of genetic engineering for production of biopharmaceuticals and industrial enzymes; synthetic biology and its industrial implications	6.0 h
III	• Food and Beverage Microbiology:Microorganisms as source of food such as single cell protein, curd, yogurt, bakery products and mushrooms; biotechnological advancements in the food industry, fermentation processes, basic design of a fermenter and production of alcohol, vinegar, organic acids and amino acids	6.0 h
IV	• Microbes in spoilage of foods and food products: Cereals, vegetables, pickles, fish and dairy products, food safety, quality control and food preservation • Microbes as source of enzyme and energy:Microbial production of industria enzymes, amylase, cellulase, protease, lipase, microbes as a source of bioplastic and biofuels (ethanol, methane and biogas	6.0 h

[27] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[28] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

V	• Pharmaceutical microbiology:Biopharmaceutical production of vaccines, antibiotics, interferons; microbial contamination in pharmaceuticals, sterilization techniques and aseptic processing, quality control and regulatory considerations • Safety and Regulations:Biosafety in industrial microbiology, ethical considerations in microbiology, regulatory guidelines and its compliance	6.0 h

Texts/ References	1. Prescott's Microbiology by Joanne M. Willey, Linda Sherwood, and Christopher J. Woolverton. 2. Industrial Microbiology: An Introduction by Michael J. Waites, Neil L. Morgan, and John S. Rockey. 3. Industrial Microbiology by L. E. Casida Jr. 4. Principles of Fermentation Technology by Whittaker and Stanbury.
Learning Outcomes	Upon completing this course, students will understand the scope and significance of industrial microbiology, including the role of microorganisms in various industrial processes. They will be able to describe microbial growth and physiology, and apply microbial biotechnology to agriculture, environment, pharmaceuticals, and energy. Students will gain insights into food and beverage microbiology, fermentation processes, and microbial contributions to enzyme production and biofuels. They will also be knowledgeable about pharmaceutical microbiology, focusing on biopharmaceuticals, contamination control, sterilization, and regulatory guidelines. This knowledge will prepare students for practical applications and advancements in industrial microbiology.

Course Title	**BOTMD32: Lab Work Based on Paper BOTMD31**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Study of growth behaviour of bacteria and determination of generation time (g) • Enumeration of bacteria from environmental sample using serial dilution method	6.0 h
II	• Qualitative detection of microorganisms producing amylase, cellulose, protease and lipase	6.0 h
III	• Qualitative detection of microorganisms which ferment carbohydrates	6.0 h
IV	• Testing antibiotic sensitivity of microorganisms (Disc diffusion method)	6.0 h
V	• Methylene blue reduction test to assess quality of milk samples	6.0 h

Course Title	**AE3: Offered by Institute/University (2 Credits)**
Category of Course[29]	Major / Minor / Minor (Vocational) / SEC / **AEC-3** / VAC / MD/Internship/Dissertation

[29] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

<table>
<tr><td>Course Title</td><td>BOTSE31: Ethnobotany</td></tr>
<tr><td>Category of Course[30]</td><td>Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation</td></tr>
<tr><td>Credits[31]& Hour of Teaching</td><td><table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>2</td><td>1</td><td>3</td></tr><tr><td>Hour of Teaching</td><td>30</td><td>30</td><td>60</td></tr></table></td></tr>
<tr><td>Course Objectives</td><td>The course aims to provide a comprehensive understanding of ethnobotany by exploring its history, scope, and methodologies. Students will learn about traditional literature, data collection techniques, and the importance of ethnobotanical flora in conservation strategies. The course will cover medico-ethnobotany, focusing on the preparation and uses of traditional medicines, and will delve into the traditional knowledge of plants like Ocimum and Withania. Additionally, students will examine the role of ethnic practices in biodiversity conservation and the relevance of ethnobotany in contemporary research and sustainable development.</td></tr>
<tr><td colspan="2"><table>
<tr><td>Units</td><td>Course Content</td><td>Hours of Teaching:30 h</td></tr>
<tr><td>I</td><td>• History and scope of ethnobotany
• Ethnobotanical studies: Traditional literature and other sources, tools and techniques (Sampling of informants, data collection, interviews & group discussion, data analysis)</td><td>6.0 h</td></tr>
<tr><td>II</td><td>• Importance of ethnobotanical flora and its conservation strategies
• Medico-ethnobotany: ethnomedicine preparation, their uses</td><td>6.0 h</td></tr>
<tr><td>III</td><td>• Traditional knowledge of some plants of ethnobotanical importance: Ocimum, Withania, Andrographis, Terminalia, Phyllanthus, Eclipta, Artemesia, Moringa
• Role of ethnic practices in biodiversity conservation: ethnic groups, religious practices, sacred place, sacred groves, community protocols</td><td>6.0 h</td></tr>
<tr><td>IV</td><td>• Ethnobotany and its relevance in contemporary research
• Importance of ethnobotanical flora and its conservation strategies</td><td>6.0 h</td></tr>
<tr><td>V</td><td>• Traditional Herbal Medicine
• Role of Ethnobotany in sustainable development</td><td>6.0 h</td></tr>
</table></td></tr>
<tr><td>Texts/ References</td><td>1. Maheshwari JK (2019) Ethnobotany and medicinal plants of Indian subcontinent, Scientific publishers, Jodhpur, India.
2. Martinez JL, Munoz- Acevedo A& Rai M (2018) Ethnobotany: application of medicinal plants, CRC Press.
3. Jordan JA, Minnis PE, Eliseus WJ (2014) Plains apache ethnobotany, University of Oklahoma Press.
4. Young KJ & Hopkins WG (2006) Ethnobotany, Facts on File.</td></tr>
<tr><td>Learning Outcomes</td><td>Upon completion of this course, students will have a deep understanding of ethnobotany, including its history, methodologies, and practical applications. They will be proficient in ethnobotanical study techniques such as data collection and analysis, and recognize the significance of ethnobotanical flora for conservation. Students will gain insights into medico-ethnobotany, including traditional medicinal practices and plant uses. They will also appreciate the role of ethnic practices in preserving biodiversity and understand the relevance of ethnobotany in contemporary research and sustainable development. This knowledge will equip them to contribute effectively to ethnobotanical and conservation efforts.</td></tr>
</table>

[30] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[31] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Title	BOTSE32: Lab Work Based on Paper BOTSE31	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Preparation of herbarium of 10 ethnobotanical plants	10.0 h
II	• Local visit to identify the plants used in traditional medicine	10.0 h
III	• People-plant interaction: Collection of data about traditional uses of local plants from common people	10.0 h

SEMESTER: IV

Course Title	**BOTMJ41: Angiosperm Taxonomy**
Category of Course[32]	**Major** / Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[33]**& Hour of Teaching**	<table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr><tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr></table>
Course Objectives	This course aims to provide students with a thorough understanding of angiosperm taxonomy and classification. It covers principles and systems of plant classification, including Bentham and Hooker's and Hutchinson's systems. Students will learn about the distinguishing characteristics and economic importance of various dicot and monocot families. The course also explores botanical nomenclature, numerical taxonomy, and the role of embryology and chemotaxonomy in plant classification. Additionally, students will gain insights into molecular approaches and the APG system of classification. By the end of the course, students will be equipped to apply these concepts to the study and classification of angiosperms.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Systematics: Principles, outline, merits and demerits of classification; Bentham and Hooker's system, Hutchinson system	8.0 h
II	• Distinguishing characteristics of the following families and their economic importance: *a. Dicots: Ranunculaceae, Papaveraceae, Myrtaceae, Apiaceae, Cucurbitaceae, Rubiaceae, Apocynaceae, Asclepiadaceae, Solanaceae, Acanthaceae, Lamiaceae, Euphorbiaceae, Asteraceae* *b. Monocots: Zingeberaceae,Poaceae, Cyperaceae*	20.0 h
III	• Botanical nomenclature: International code of nomenclature; principles: rules and recommendations; priority; typification; rules of effective and valid publications; retention and choice of names • Numerical Taxonomy: characters and attributes, OTUs, coding, cluster analysis, merits and demerits	8.0 h
IV	• Embryology in relation to taxonomy • Chemotaxonomy: Role of phytochemicals (non-protein amino acids, alkaloids, betalins, cynogenicglucosides) andtechniques in chemotaxonomy	6.0 h
V	• Basic idea on molecular approaches to angiosperm taxonomy; APG system of classification	3.0 h

[32] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[33] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Texts/ References	1. Singh V, Jain DK (2012) Taxonomy of Angiosperms, 2nd edition, Rastogi Publication, Meerut. 2. Pandey BP (2007) Angiosperms- Taxonomy, Embryology and Anatomy, S Chand and Co, New Delhi. 3. Chopra GL (1984) Angiosperms: Systematic and Life Cycle, Pradeep Publications, Jalandhar. 4. Singh G (1999) Plant Systematics: Theory and Practices, Oxford and IBH Publishing Co, New Delhi. 5. Judd WS, Christopher S, Campbell, Kellogg AE, Stevens PF (1999) Plant Systematics: A Phylogenetic Approach, Sinauer Associates Inc Publishers, Sunderland, Massachusetts. 6. Simpson MG (2006) Plant Systematics, Elsevier Academic Press,Amsterdam. 7. Verma BK (2011) Introduction to Taxonomy of Angiosperms, PHI Learning, New Delhi.
Learning Outcomes	Upon completing this course, students will be able to understand and apply principles of plant systematics, including various classification systems like Bentham and Hooker's and Hutchinson's. They will be proficient in identifying and describing the characteristics and economic importance of major dicot and monocot families. Students will grasp the principles of botanical nomenclature, numerical taxonomy, and the role of embryology and chemotaxonomy in plant classification. Additionally, they will have an overview of molecular approaches and the APG classification system. This knowledge will enable students to effectively study, classify, and apply taxonomy to angiosperms.

Course Title	**BOTMJ42: Lab Work Based on Paper BOTMJ41**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Taxonomic description of plants of the following families: *Ranunculaceae, Papaveraceae, Myrtaceae, Apiaceae, Cucurbitaceae, Rubiaceae, Apocynaceae, Asclepiadaceae, Solanaceae, Acanthaceae, Lamiaceae, Euphorbiaceae,Asteraceae, Zingeberaceae,Poaceae, Cyperaceae*	20.0 h
II	• Plant collection: submission of at least 25 herbarium sheets	10.0 h

Course Title	**BOTMJ43: Gymnosperm and Paleobotany**
Category of Course[34]	**Major** / Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[35]**& Hour of Teaching**	(see table below)
Course Objectives	This course aims to provide students with a comprehensive understanding of gymnosperms and paleobotany. It covers the characteristic features and classification of gymnosperms, including detailed accounts of Progymnospermopsida, Pteridospermales, and Glossopteridales. Students will study the morphology, anatomy, and reproduction of key gymnosperms such as Cycas, Pinus, and Ephedra. The course also introduces paleobotany, including the geological timescale, evolutionary trends, and the origin of plant groups. Additionally, students will learn about fossils, their types, the process of fossilization, and their significance in understanding plant evolution.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	**Course Content**	**Hours of Teaching: 45 h**
I	Characteristic features and classification of Gymnosperms General accounts of Progymnospermopsida and Pteridospermales (*Lyginopteris*plant; *Sphenopteris* leaf, *Lygonopteris*stem, *Crossotheca* male organ *and Lagenostoma* female organ)	6.0 h

[34] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[35] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

II	Glossopteridales: General account of Glossopteris plant (*Vertebraria* root, *Araucarioxilon* trunk, *Glossopteris* leaf, *Glossotheca*male organ, *Dictyopteridium* female organ)	7.0 h
III	Morphology, anatomy and reproduction of *Cycas*, *Pinus* and *Ephedra*	16.0 h
IV	• An elementary knowledge of Paleobotany: Geological timescale • Evolutionary trends; origin and evolution of different plant groups	8.0 h
V	Fossils, Types of fossils, Process of fossilization, Importance of fossils	8.0 h

Texts/ References	1. Sporne, K.R. (1967) Morphology of Gymnosperms, BI Publication, New Delhi. 2. Bhatnagar, S.P and Moitra, A. (1996) Gymnosperms, New Age International Pvt Ltd Publishers, New Delhi. 3. Arnold, C. A. (1947) An Introduction to Palaeobotany. Mc Graw Hill Publ. Co. 4. Andrews, H. N. (1967) Studies in Palaeobotany. Wiley, New York. 5. Stewart, W. N. and Rothwell, G.W. (1993) Palaeobotany and Evolution of Plants. CambridgeUniversity Press. 6. Taylor, T. N., Taylor, E. L. and Krings, M. (2009) Palaeobotany, the biology and evolution offossil plants. Elsevier Inc. 7. Beck, C. B. (1988) Origin and evolution of gymnosperms. Columbia University Press. 8. Chamberlain, C. J. (1934) Gymnsoperms – structure and evolution. Chicago Univ. Press.
Learning Outcomes	Upon completing this course, students will be able to identify and classify gymnosperms, including Progymnospermopsida, Pteridospermales, and Glossopteridales. They will understand the morphology, anatomy, and reproduction of key gymnosperms like Cycas, Pinus, and Ephedra. Students will gain foundational knowledge in paleobotany, including the geological timescale, evolutionary trends, and the origin of plant groups. They will also be able to describe fossil types, the fossilization process, and the significance of fossils in studying plant evolution. This comprehensive understanding will equip students to apply paleobotanical concepts to the study of plant history and evolution.

Course Title	**BOTMJ44: Lab Work Based on Paper BOTMJ43**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• *Cycas* - General morphology of vegetative and reproductive structures, anatomy (coralloid root,leaflet, rachis, permanent slides), and reproduction (megasporophyll, male cone, section ofmicrosporophyll, permanent slides of ovule and microsporophyll)	10.0 h
II	• *Pinus* - General morphology of vegetative and reproductive structures, anatomy (T.S of shoots,needle, permanent slides), reproduction (male cone, female cone, sections of megasporophyll and microsporophyll, permanent slides)	10.0 h
III	• *Ephedra* - General morphology of vegetative and reproductive structures, anatomy (stem, reproductive structures)	10.0 h

Course Title	**BOTMJ45: Plant Ecology**
Category of Course[36]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation

Credits[37] & Hour of Teaching

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Course Objectives	This course aims to provide students with a foundational understanding of plant ecology. It covers the abiotic and biotic components of the environment, including interactions among plants, animals, and microorganisms. Students will learn about plant adaptations to water, light, and temperature, and explore population dynamics and community ecology, focusing on analytical and synthetic characters. The course also includes ecosystem ecology, covering ecosystem structure, function, and biogeochemical cycles. Additionally, students will study ecological succession, including its processes, mechanisms, and patterns such as hydrosere and xerosere. This comprehensive knowledge will enable students to analyze and interpret ecological interactions and processes in plant communities.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Introduction to ecology • Abiotic environment: Atmosphere, temperature, water, light and soil (structure and soil profile)	8.0 h
II	• Biotic environment: Interactions among plants, animals and man; interactions among plants growing in a community; interactions among plants and microorganisms	8.0 h
III	• Plant adaptations in response to water availability, light and temperature • Population: Concepts, attributes and dynamics	9.0 h
IV	• Community Ecology: Analytical and synthetic characters (Frequency, density, cover, IVI, life forms, biological spectrum, phenology, sociability)	8.0 h
V	• Ecosystem Ecology: Ecosystem structure (abiotic and biotic components, food chain, food web, ecological pyramids); ecosystem function (models of energy flow), biogeochemical cycles (carbon, nitrogen, and phosphorus) • Ecological succession: General process, mechanisms and patterns (Hydrosere and Xerosere)	12.0 h

Texts/ References	1. Odum EP, Barrett GW (2005) Fundamentals of Ecology, Thomson Ed. Brooks/Cole, Cengage Learning India Pvt. Ltd, New Delhi. 2. Odum EP (1983) Basic Ecology Ed. Saunders College Pub, New York. 3. Smith TM, Smith RL (2008) Elements of Ecology, 7th edition, Benjamin- Cummings New York. 4. Miller GT (2004) Essentials of Ecology, 3rd edition, Brooks, Cole, New York. 5. Smith RL (2001) Ecology and Field Biology, Benjamin Cummings, New York. 6. Vallero D (2014) Fundamentals of Air Pollution, 3rd edition, Academic Press, Elsevier, Burlington. 7. Klassen C, Watkins JB (2010) Essentials of Toxicolgy, 2nd edition, McGraw- Hill Companies, New York. 8. Singh JS, Singh SP, Gupta SR (2014) Ecology Environmental Science and Conservation, S Chand & Co, New Delhi. 9. Mueller-Dombois D, Ellenberg H (1974), Aims and Methods of Vegetation Ecology. Wiley & Sons., NY, USA
Learning	Upon completing this course, students will be equipped to analyze and interpret the complex

[36] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[37] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Outcomes	interactions within plant ecology. They will understand the influence of abiotic factors like atmosphere, temperature, water, light, and soil structure on plant life. Students will gain insights into biotic interactions among plants, animals, and microorganisms, and how plants adapt to environmental variables such as water availability, light, and temperature. They will be able to describe population dynamics, community ecology metrics, and ecosystem functions, including energy flow and biogeochemical cycles. Additionally, students will grasp the processes and patterns of ecological succession, enabling them to assess and manage plant communities effectively.

Course Title	**BOTMJ46: Lab Work Based on Paper BOTMJ45**

Units	**Practicals**	**Hours of Teaching:30 h**
I	• To study soil properties by spot tests: texture, pH, CO_3^{--}, NO_3^-, base deficiency and reductivity	6.0 h
II	• Demonstration of the working and use of the instruments for measuring the environmental factors: *Temperature: Ordinary Thermometer, Maximum-Minimum Thermometer.* *Moisture: Rain Gauge, Psychrometer* *Wind velocity: Anemometer* *Light intensity: Lux meter*	6.0 h
III	• Determination of minimal size of sample lot by species-area curve • Charting of vegetation to find out coverage	6.0 h
IV	• Study of community structure by quadrat method with respect to following: *Density* *Frequency*	6.0 h
V	• Study of following ecological adaptations: *Hydrophytes: Nymphaea leaf, Hydrilla stem* *Xerophytes: Nerium leaf, Muehlenbekia, Casuarina* *Epiphytes: Vanda* *Parasites: Cuscuta with host*	6.0 h

Course Title	**BOTMJ47: Microbial Resource**
Category of Course[38]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[39]**& Hour of Teaching**	(see table below)
Course Objectives	This course aims to provide students with an in-depth understanding of microbial resources and their applications for human welfare. It covers the role of microbes in food production, including fermented foods, probiotics, and single-cell proteins, as well as their use in the production of enzymes, amino acids, and biofuels. Students will explore the medical applications of microbes, such as antibiotic production and resistance. The course also addresses microbial contributions to crop improvement, waste management through composting, and bioremediation. By the end of the course, students will be equipped to apply microbial knowledge to various industries and environmental solutions.

	Theory	Practical	Cumulative
Credits	1	1	2
Hour of Teaching	15	30	45

[38] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[39] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Units	Course Content	Hours of Teaching:15 h
I	• Introduction to microbial world applications for human welfare • Microbes as edible foods: Microorganism involved in fermented foods production-fermented milk, Probiotics, Cheese production, Production of alcoholic beverages; Single Cell Protein (SCP): microorganisms used, raw material used as substrate, condition for growth and production; nutritive value and uses of SCP; Baker's yeast: Production and mold cultures; Edible mushroom and nutritional value	2.0 h
II	• Microbes in industries: Microorganism involved in the production of enzymes, amino acids, organic acids, biopolymers and biofuel; biotransformation	3.0 h
III	• Microbes as medical products: Microorganism as a source of antibiotics, mode of action of antibiotics and microbial resistance against antibiotics	3.0 h
IV	• Crop improvement using microbes: Nitrogen fixing microorganisms (Cyanobacteria and Bactria), biological nitrogen fixation, bio pesticides, phosphate solubilizing bacteria, mycorrhiza, plant growth promoting rhizobacteria (PGPR)	3.0 h
V	• Microbes in composting of solid wastes: Waste water treatment, anaerobic digestion, methane production • Bioremediation and microbes: Process and organisms involved, Constraints and priorities of bioremediation, microbial degradation of hydrocarbons, microbial leaching of ores and metals, biotransformation of xenobiotics, biodegradable plastics and super bug	4.0 h

Texts/ References	1. Dubey RC, Maheshwari DK (2022). A Textbook of Microbiology. S Chand and Company Ltd, New Delhi. 2. Madigan MT, Martinko JM, Bender KS, Buckley DH, Stahl DA (2014) Brock Biology of Microorganism, 14th edition, Benjamin Cummings, New York. 3. Stanier RY, Ingraham JL, Wheelis ML, Painter PR (1987) General Microbiology, 5th edition, MacMillan, New Jersey. 4. Talaro KP, Chess B (2011) Foundation in Microbiology, 8th edition, MacGraw-Hill, New York 5. Willey JM, Sherwood L, Woolverton CJ (2013) Prescott's Microbiology, 9th edition, McGraw-Hill, New York. 6. PD Sharma (2017). Microbiology. Rastogi Publications, Meerut 7. Dubey RC, Maheshwari DK (2012) A Textbook of Microbiology, Revised edition, S Chand and Company Ltd, New Delhi.
Learning Outcomes	Upon completing this course, students will be able to identify and apply microbial resources for human welfare across various sectors. They will understand the role of microbes in food production, including fermented products and single-cell proteins, and their applications in industry for enzyme and biofuel production. Students will grasp the use of microbes in medicine, focusing on antibiotics and resistance. They will also learn how microbes contribute to crop improvement and environmental management, including composting, bioremediation, and waste treatment. This knowledge will enable students to leverage microbial technologies for sustainable and innovative solutions in multiple fields.

Course Title	BOTMJ48: Lab Work Based on Paper BOTMJ47	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Study of free living nitrogen fixing cyanobacteria: *Nostoc*, *Anabaena*, *Aulosira*	6.0 h
II	• Study of symbiotic nitrogen fixers in *Cycas* and *Azolla*	6.0 h
III	• Study and enumeration of soil bacteria/microflora using MPN • Methylene blue reduction test of milk	6.0 h
IV	• Demonstration of certain metabolic activates: *Fermentation of carbohydrates* *Catalase activity*	6.0 h
V	• Determination of minimum inhibitory concentrations of antibiotics and toxins	6.0 h

Course Title	**BOTMN41: Angiosperm Taxonomy**
Category of Course[40]	Major / **Minor**/ Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[41]& Hour of Teaching	(see table below)
Course Objectives	This course aims to provide students with a thorough understanding of angiosperm taxonomy and classification. It covers principles and systems of plant classification, including Bentham and Hooker's and Hutchinson's systems. Students will learn about the distinguishing characteristics and economic importance of various dicot and monocot families. The course also explores botanical nomenclature, numerical taxonomy, and the role of embryology and chemotaxonomy in plant classification. Additionally, students will gain insights into molecular approaches and the APG system of classification. By the end of the course, students will be equipped to apply these concepts to the study and classification of angiosperms.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Systematics: Principles, outline, merits and demerits of classification; Bentham and Hooker's system, Hutchinson system	8.0 h
II	• Distinguishing characteristics of the following families and their economic importance: *a. Dicots: Ranunculaceae, Papaveraceae, Myrtaceae, Apiaceae, Cucurbitaceae, Rubiaceae, Apocynaceae, Asclepiadaceae, Solanaceae, Acanthaceae, Lamiaceae, Euphorbiaceae, Asteraceae* *b. Monocots: Zingeberaceae,Poaceae, Cyperaceae*	20.0 h
III	• Botanical nomenclature: International code of nomenclature; principles: rules and recommendations; priority; typification; rules of effective and valid publications; retention and choice of names • Numerical Taxonomy: characters and attributes, OTUs, coding, cluster analysis, merits and demerits	8.0 h
IV	• Embryology in relation to taxonomy • Chemotaxonomy: Role of phytochemicals (non-protein amino acids, alkaloids, betalins, cynogenicglucosides) andtechniques in chemotaxonomy	6.0 h
V	• Basic idea on molecular approaches to angiosperm taxonomy; APG system of classification	3.0 h

[40] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[41] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Texts/ References	1. Singh V, Jain DK (2012) Taxonomy of Angiosperms, 2nd edition, Rastogi Publication, Meerut. 2. Pandey BP (2007) Angiosperms- Taxonomy, Embryology and Anatomy, S Chand and Co, New Delhi. 3. Chopra GL (1984) Angiosperms: Systematic and Life Cycle, Pradeep Publications, Jalandhar. 4. Singh G (1999) Plant Systematics: Theory and Practices, Oxford and IBH Publishing Co, New Delhi. 5. Judd WS, Christopher S, Campbell, Kellogg AE, Stevens PF (1999) Plant Systematics: A Phylogenetic Approach, Sinauer Associates Inc Publishers, Sunderland, Massachusetts. 6. Simpson MG (2006) Plant Systematics, Elsevier Academic Press,Amsterdam. 7. Verma BK (2011) Introduction to Taxonomy of Angiosperms, PHI Learning, New Delhi.
Learning Outcomes	Upon completing this course, students will be able to understand and apply principles of plant systematics, including various classification systems like Bentham and Hooker's and Hutchinson's. They will be proficient in identifying and describing the characteristics and economic importance of major dicot and monocot families. Students will grasp the principles of botanical nomenclature, numerical taxonomy, and the role of embryology and chemotaxonomy in plant classification. Additionally, they will have an overview of molecular approaches and the APG classification system. This knowledge will enable students to effectively study, classify, and apply taxonomy to angiosperms.
Course Title	**BOTMN42: Lab Work Based on Paper BOTMN41**

Units	**Practicals**	**Hours of Teaching:30 h**
I	• Taxonomic description of plants of the following families: *Ranunculaceae, Papaveraceae, Myrtaceae, Apiaceae, Cucurbitaceae, Rubiaceae, Apocynaceae, Asclepiadaceae, Solanaceae, Acanthaceae, Lamiaceae, Euphorbiaceae,Asteraceae, Zingeberaceae,Poaceae, Cyperaceae*	20.0 h
II	• Plant collection: submission of at least 25 herbarium sheets	10.0 h

Course Title	**BOTMN43: Gymnosperm and Paleobotany**
Category of Course[42]	Major / **Minor**/ Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[43]**& Hour of Teaching**	

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Course Objectives	This course aims to provide students with a comprehensive understanding of gymnosperms and paleobotany. It covers the characteristic features and classification of gymnosperms, including detailed accounts of Progymnospermopsida, Pteridospermales, and Glossopteridales. Students will study the morphology, anatomy, and reproduction of key gymnosperms such as Cycas, Pinus, and Ephedra. The course also introduces paleobotany, including the geological timescale, evolutionary trends, and the origin of plant groups. Additionally, students will learn about fossils, their types, the process of fossilization, and their significance in understanding plant evolution.

Units	**Course Content**	**Hours of Teaching: 45 h**
I	• Characteristic features and classification of Gymnosperms • General accounts of Progymnospermopsida and Pteridospermales (*Lyginopteris*plant;*Sphenopteris* leaf, *Lygonopteris*stem, *Crossotheca* male organ *and Lagenostoma* female organ)	6.0 h

[42] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[43] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

II	• Glossopteridales: General account of Glossopteris plant (*Vertebraria* root, *Araucarioxilon* trunk, *Glossopteris* leaf, *Glossotheca*male organ, *Dictyopteridium* female organ)	7.0 h
III	• Morphology, anatomy and reproduction of *Cycas*, *Pinus* and *Ephedra*	16.0 h
IV	• An elementary knowledge of Paleobotany: Geological timescale • Evolutionary trends; origin and evolution of different plant groups	8.0 h
V	• Fossils, Types of fossils, Process of fossilization, Importance of fossils	8.0 h

Texts/ References	1. Sporne, K.R. (1967) Morphology of Gymnosperms, BI Publication, New Delhi. 2. Bhatnagar, S.P and Moitra, A. (1996) Gymnosperms, New Age International Pvt Ltd Publishers, New Delhi. 3. Arnold, C. A. (1947) An Introduction to Palaeobotany. Mc Graw Hill Publ. Co. 4. Andrews, H. N. (1967) Studies in Palaeobotany. Wiley, New York. 5. Stewart, W. N. and Rothwell, G.W. (1993) Palaeobotany and Evolution of Plants. CambridgeUniversity Press. 6. Taylor, T. N., Taylor, E. L. and Krings, M. (2009) Palaeobotany, the biology and evolution offossil plants. Elsevier Inc. 7. Beck, C. B. (1988) Origin and evolution of gymnosperms. Columbia University Press. 8. Chamberlain, C. J. (1934) Gymnsoperms – structure and evolution. Chicago Univ. Press.
Learning Outcomes	Upon completing this course, students will be able to identify and classify gymnosperms, including Progymnospermopsida, Pteridospermales, and Glossopteridales. They will understand the morphology, anatomy, and reproduction of key gymnosperms like Cycas, Pinus, and Ephedra. Students will gain foundational knowledge in paleobotany, including the geological timescale, evolutionary trends, and the origin of plant groups. They will also be able to describe fossil types, the fossilization process, and the significance of fossils in studying plant evolution. This comprehensive understanding will equip students to apply paleobotanical concepts to the study of plant history and evolution.

Course Title	**BOTMN44: Lab Work Based on Paper BOTMN43**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• *Cycas* - General morphology of vegetative and reproductive structures, anatomy (coralloid root,leaflet, rachis, permanent slides), and reproduction (megasporophyll, male cone, section ofmicrosporophyll, permanent slides of ovule and microsporophyll)	10.0 h
II	• *Pinus* - General morphology of vegetative and reproductive structures, anatomy (T.S of shoots,needle, permanent slides), reproduction (male cone, female cone, sections of megasporophyll and microsporophyll, permanent slides)	10.0 h
III	• *Ephedra* - General morphology of vegetative and reproductive structures, anatomy (stem, reproductive structures)	10.0 h

<table>
<tr><td>Course Title</td><td colspan="3">BOTMN45: Plant Ecology</td></tr>
<tr><td>Category of Course[44]</td><td colspan="3">Major / Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation</td></tr>
<tr><td>Credits[45]& Hour of Teaching</td><td colspan="3">
<table>
<tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr>
<tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr>
<tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr>
</table>
</td></tr>
<tr><td>Course Objectives</td><td colspan="3">This course aims to provide students with a foundational understanding of plant ecology. It covers the abiotic and biotic components of the environment, including interactions among plants, animals, and microorganisms. Students will learn about plant adaptations to water, light, and temperature, and explore population dynamics and community ecology, focusing on analytical and synthetic characters. The course also includes ecosystem ecology, covering ecosystem structure, function, and biogeochemical cycles. Additionally, students will study ecological succession, including its processes, mechanisms, and patterns such as hydrosere and xerosere. This comprehensive knowledge will enable students to analyze and interpret ecological interactions and processes in plant communities.</td></tr>
<tr><td>Units</td><td colspan="2">Course Content</td><td>Hours of Teaching:45 h</td></tr>
<tr><td>I</td><td colspan="2">Introduction to ecology
Abiotic environment: Atmosphere, temperature, water, light and soil (structure and soil profile)</td><td>8.0 h</td></tr>
<tr><td>II</td><td colspan="2">Biotic environment: Interactions among plants, animals and man; interactions among plants growing in a community; interactions among plants and microorganisms</td><td>8.0 h</td></tr>
<tr><td>III</td><td colspan="2">Plant adaptations in response to water availability, light and temperature
Population: Concepts, attributes and dynamics</td><td>9.0 h</td></tr>
<tr><td>IV</td><td colspan="2">Community Ecology: Analytical and synthetic characters (Frequency, density, cover, IVI, life forms, biological spectrum, phenology, sociability)</td><td>8.0 h</td></tr>
<tr><td>V</td><td colspan="2">Ecosystem Ecology: Ecosystem structure (abiotic and biotic components, food chain, food web, ecological pyramids); ecosystem function (models of energy flow), biogeochemical cycles (carbon, nitrogen, and phosphorus)
Ecological succession: General process, mechanisms and patterns (Hydrosere and Xerosere)</td><td>12.0 h</td></tr>
<tr><td>Texts/ References</td><td colspan="3">Odum EP, Barrett GW (2005) Fundamentals of Ecology, Thomson Ed. Brooks/Cole, Cengage Learning India Pvt. Ltd, New Delhi.
Odum EP (1983) Basic Ecology Ed. Saunders College Pub, New York.
Smith TM, Smith RL (2008) Elements of Ecology, 7th edition, Benjamin- Cummings New York.
Miller GT (2004) Essentials of Ecology, 3rd edition, Brooks, Cole, New York.
Smith RL (2001) Ecology and Field Biology, Benjamin Cummings, New York.
Vallero D (2014) Fundamentals of Air Pollution, 3rd edition, Academic Press, Elsevier, Burlington.
Klassen C, Watkins JB (2010) Essentials of Toxicolgy, 2nd edition, McGraw- Hill Companies, New York.
Singh JS, Singh SP, Gupta SR (2014) Ecology Environmental Science and Conservation, S Chand & Co, New Delhi.
Mueller-Dombois D, Ellenberg H (1974), Aims and Methods of Vegetation Ecology. Wiley & Sons., NY, USA</td></tr>
<tr><td>Learning Outcomes</td><td colspan="3">Upon completing this course, students will be equipped to analyze and interpret the complex interactions within plant ecology. They will understand the influence of abiotic factors like</td></tr>
</table>

[44] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[45] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	atmosphere, temperature, water, light, and soil structure on plant life. Students will gain insights into biotic interactions among plants, animals, and microorganisms, and how plants adapt to environmental variables such as water availability, light, and temperature. They will be able to describe population dynamics, community ecology metrics, and ecosystem functions, including energy flow and biogeochemical cycles. Additionally, students will grasp the processes and patterns of ecological succession, enabling them to assess and manage plant communities effectively.

Course Title	**BOTMN46: Lab Work Based on Paper BOTMN45**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• To study soil properties by spot tests: texture, pH, CO_3^{--}, NO_3^{-}, base deficiency and reductivity	6.0 h
II	• Demonstration of the working and use of the instruments for measuring the environmental factors: *Temperature: Ordinary Thermometer, Maximum-Minimum Thermometer* *Moisture: Rain Gauge, Psychrometer* *Wind velocity: Anemometer* *Light intensity: Lux meter*	6.0 h
III	• Determination of minimal size of sample lot by species-area curve • Charting of vegetation to find out coverage	6.0 h
IV	• Study of community structure by quadrat method with respect to following: *Density* *Frequency*	6.0 h
V	• Study of following ecological adaptations: *Hydrophytes: Nymphaea leaf, Hydrilla stem* *Xerophytes: Nerium leaf, Muehlenbekia, Casuarina* *Epiphytes: Vanda* *Parasites: Cuscuta with host*	6.0 h

Course Title	**AE4: Offered by Institute/University (2 Credits)**
Category of Course[46]	Major / Minor / Minor (Vocational) / SEC / **AEC-4**/ VAC / MD/Internship/Dissertation

[46] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

SEMESTER-V

Course Title	**BOTMJ51: Plant Physiology**
Category of Course[47]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[48] **& Hour of Teaching**	

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Course Objectives	This course objective is to provide students with a detailed understanding of key aspects of plant physiology, including water relations, transport mechanisms, and nutrient translocation. Students will explore the processes of phloem transport, transpiration, and photosynthesis, including various pathways and factors affecting these processes. The course also covers respiration, plant photoreceptors, and plant growth regulation, emphasizing hormones and growth regulators. By integrating these concepts, students will gain a comprehensive understanding of plant physiological processes, enabling them to analyze and apply physiological principles to enhance plant growth and development in various agricultural and research contexts.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Plant Water Relations: Permeability, osmosis, water potential • Transport of Water, Solutes and Translocation: Uptake, transport and translocation of water, ions, solutes from soil, passive and active transports	7.0 h
II	• Phloem transport; composition of photo-assimilates translocated by phloem, mechanism of phloem transport, phloem loading and unloading • Transpiration: foliar transpiration, guttation, plant anti-transpirants, factors affecting transpiration, significance of transpiration	8.0 h
III	• Photosynthesis:Light harvesting complexes, absorption and action spectrum, light reaction, dark reaction, photorespiration, Hatch-Slack Pathway, CAM Pathway, comparison of C3, C4 and CAM plants, factors affecting photosynthesis	8.0 h
IV	• Respiration: Glycolysis, oxidative decarboxylation of pyruvate to acetyl CoA, citric acid cycle (Krebs Cycle), regulation of respiratory pathways, respiratory quotient • Plant photoreceptors: Phytochromes, cryptochromes, photoperiodism	12.0 h
V	• Plant growth regulation: Plant hormones and plant growth regulators, auxins, gibberellins, cytokinins, abscisic acid, ethylene, polyamines, salicylic acid, nitric oxide, jasmonates, brassinosteroids • Fruit ripening: Plant senescence, programmed cell death	10.0 h

Texts/ References	1. Buchanan BB, Gruisemm W, Jones RL (2015) Biochemistry and Molecular Biology of Plants, 2nd edition, Wiley Blackwell, New Jersey. 2. Hopkins WG, Huner NPA (2009) Introduction to Plant Physiology, 4th edition, Wiley International edition, John Wiley & Sons, New York. 3. Taiz L, Zeiger E, Moller IM, Murph A (2015) Plant Physiology and Development, 6th edition, Sinurer Associates Inc Publishers, Sunderland, Massachusetts. 4. Frank B. Salisbury and Cleon W. Ross (1985) Plant Physiology Wadsworth Publishing Company, Belmont, California. 5. Robert M. Devlin (3rd edition) (1975) Plant Physiology Van Nostrand Reinhold Company,

[47] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[48] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	New York. 6. Walter Larcher (4th edition) (2003) Physiological Plant Ecology: Ecophysiology and Stress Physiology of Functional Groups Springer Verlag, Berlin. 7. Hans Mohr and Peter Schopfer (2010) Plant Physiology Springer Verlag, Berlin. 8. Edwin Oxlade (2007) Plant Physiology: The Structure of Plants Explained Glmp Ltd, Abergele, United Kingdom. 9. Kochhar SL and Gujral SK (2020) Plant Physiology: Theory and Applications, Second Edition, Springer
Learning Outcomes	Upon completing this course, students will be able to analyze and explain plant water relations, transport mechanisms, and nutrient translocation. They will understand the processes of phloem transport, transpiration, and photosynthesis, including different pathways like C3, C4, and CAM. Students will also grasp the principles of respiration, the role of plant photoreceptors, and the impact of plant hormones on growth regulation. This knowledge will enable them to assess plant physiological processes critically and apply this understanding to optimize plant growth, development, and productivity in agricultural and research settings.

Course Title	**BOTMJ52: Lab Work Based on Paper BOTMJ51**

Units	**Practicals**	**Hours of Teaching:30 h**
I	• Measurement of osmotic concentration of cell sap by plasmolytic method (*Tradescantia* leaf epidermal peel/ potato tuber cylinders) • Effect of temperature, organic solvent and salt on permeability of cells membrane of beet root (*Beta vulgaris*)	6.0 h
II	• Determination of the rate of transpiration by varying environmental factors • Determination of the rate of photosynthesis in terms of oxygen evolution of an aquatic plant (*Hydrilla*): with varying light intensity, wavelengths of light and CO_2 concentrations	6.0 h
III	• Extraction, separation and study of the properties of photosynthetic pigments by paper chromatography	6.0 h
IV	• Isolation of chloroplasts and demonstration of photoreduction of dye [2,6-dichloro-phenol indophenols (DCPIP)	6.0 h
V	• Determination of RQ of different plant tissues (pipette manometer method) • Fruit ripening experiment: ethephon treatment of climacteric and non- climacteric fruits and their observation	6.0 h

Course Title	BOTMJ53: Plant Pathology
Category of Course[49]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[50]& Hour of Teaching	(see table below)
Course Objectives	This course aims to provide a comprehensive understanding of plant pathology, including the historical development and fundamental concepts of plant diseases. Students will learn about various plant pathogens—viruses, bacteria, fungi, and nematodes—along with their characteristics, disease symptoms, and infection cycles. The course covers plant defense mechanisms, disease management strategies including traditional, biological, and chemical methods, and integrated disease management approaches. Students will also explore recent trends and molecular tools for disease diagnosis, equipping them to analyze, manage, and mitigate plant diseases effectively in agricultural and research contexts.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	Course Content	Hours of Teaching:45 h
I	• Introduction to Plant Pathology and Historicaldevelopments • Plant Pathogens: Viruses, Bacteria, Fungi and Nematodes; Characteristics, disease symptoms, mode of infection and disease cycle	10.0 h
II	• Plant defense mechanisms: Structural, biochemical and molecular mechanism	10.0 h
III	• Disease symptoms, cycle and management of Virus (TMV), bacteria (Citrus canker), fungi (Late blight of potato, Powdery mildew of pea, wilt of Arhar and smut of wheat)	7.0 h
IV	• Disease Management Strategies: Traditional, biological and chemical methods of disease control, integrated disease management (IDM), engineering approaches and resistant varieties	10.0 h
V	• Recent Trends and molecular tools in disease diagnosis	8.0 h

Texts/ References	
	1. Madigan MT, Martinko JM, Dunlap PV, Clark DP (2011) Brock Biology of Microorganisms, 13th edition, Pearson Education Inc, New Jersey. 2. Talaro KP, Chess B (2011) Foundation in Microbiology, 8th edition, McGraw-Hill, New York. 3. Agrios GN (2005) Plant Pathology, 5th edition, Academic Press, London. 4. Lucas John A (1998) Plant Pathology and Plant Pathogens, Wiley-Blackwell, CRC Press, New York. 5. Willey JM, Sherwood L, Woolverton CJ (2013) Prescott's Microbiology, 9the edition, McGraw-Hill, New York. 6. Dubey RC, Maheshwari Dk (2012) A Textbook of Microbiology, Revised edition, S Chand and Company Ltd, New Delhi. 7. Singh RS (2009) Plant Diseases, 9th edition, Oxford and IBH Pub Co Pvt Ltd, New Delhi. 8. PD Sharma (2020) Microbiology and Plant Pathology 4th Ed, Rastogi Publications. 9. Kumar S, Behera L, Yadav AS & Sonakar VK (2022). Biotech Laboratory Manual On Plant Pathology Biochemical and Molecular Techniques, Biotech Publisher. 10. Anne Marte Tronsmo, David B. Collinge, Annika Djurle, Lisa Munk, Jonathan Yuen, Arne Tronsmo (2020). Plant pathology and plant diseases, CABI Publication.
Learning Outcomes	Upon completing this course, students will be able to identify and classify plant pathogens including viruses, bacteria, fungi, and nematodes, and understand their modes of infection and disease cycles. They will gain insights into plant defense mechanisms and be able to diagnose and manage plant diseases using traditional, biological, and chemical methods. Students will also be

[49] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[50] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	equipped to apply integrated disease management strategies and utilize recent molecular tools for disease diagnosis. Overall, they will be prepared to address plant pathology challenges effectively in agricultural and research settings.

Course Title	**BOTMJ54: Lab Work Based on Paper BOTMJ53**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Study of the following plant pathogens: *Sclerospora graminicola* *Peronospora parasiticia* *Erysiphe polygoni* *Uromyces pisi* *Melampsora lini* *Ustilago* *Tolyposporium penicillariae* *Fusarium oxysporum*	24.0 h
II	• Symptoms of bacterial blight	2.0 h
III	• Symptoms of mosaic	2.0 h
IV	• Little leaf of brinjal	2.0 h

<table>
<tr><td>Course Title</td><td colspan="2">BOTMJ55: Recombinant DNA Technology</td></tr>
<tr><td>Category of Course[51]</td><td colspan="2"><u>Major</u> / Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation</td></tr>
<tr><td>Credits[52]& Hour of Teaching</td><td colspan="2"><table><tr><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr><tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr><tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr></table></td></tr>
<tr><td>Course Objectives</td><td colspan="2">This course aims to provide comprehensive knowledge in recombinant DNA technology, covering the isolation, analysis, and modification of nucleic acids. Students will learn to utilize DNA modifying enzymes and various molecular tools for gene manipulation, including the CRISPR-Cas system. They will gain practical skills in transgenic plant production, vector construction, and the generation of genomic and cDNA libraries. The course also emphasizes the selection and screening of recombinants and the engineering of vectors for gene expression in different systems. Ultimately, students will be equipped to apply these techniques in research and biotechnology applications.</td></tr>
<tr><td>Units</td><td>Course Content</td><td>Hours of Teaching:45 h</td></tr>
<tr><td>I</td><td>Isolation of DNA (genomic and plasmid) and RNA, qualitative and quantitative analyses of nucleic acids, labeling of nucleic acids, reverse-transcription
DNA modifying enzymes: Endonucleases, exonucleases, ligase, types and nomenclature of restriction enzymes, isoschizomers and neoschizomers, polymerases, Klenow fragment, phosphatase, transcriptase, transferase, topoisomerase, telomerase, in vitro translation</td><td>10.0 h</td></tr>
<tr><td>II</td><td>Transgenic plant production, principle of Agrobacterium mediated gene transfer, Ti and Ri plasmids, CRISPR-Cas system, genetically modified crops and their importance</td><td>6.0 h</td></tr>
<tr><td>III</td><td>Plasmid vectors, cloning and expression vectors, construction of genomic and cDNA libraries, joining of DNA fragments to vectors, restriction digestion, dephosphorylation, cohesive and blunt end ligation, adaptors, linkers, transformation, transduction, conjugation, electroporation</td><td>10.0 h</td></tr>
<tr><td>IV</td><td>Selection, screening and analysis of recombinants, antibiotics and their resistance cassettes, colony PCR, restriction fragments length polymorphism, RAPD</td><td>10.0 h</td></tr>
<tr><td>V</td><td>Basic components of vector and its engineering, shuttle vector, gene expression in bacteria, yeast, insects and mammalian cells, codon optimization, promoters, reporter genes, gene silencing, targeted and random mutagenesis
Biosafety regulations</td><td>9.0 h</td></tr>
</table>

[51] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[52] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Texts/ References	• Brown, T. A. (2013, 2020). Gene Cloning and DNA Analysis: An Introduction. John Wiley & Sons. • Primrose, S. B., & Twyman, R. (2001, 2006). Principles of Gene Manipulation and Genomics. John Wiley & Sons. • Michael Goldberg, Janice Fischer, Leroy Hood, Leland Hartwell, Charles (Chip) Aquadro, Lee Silver and Ann E. Reynolds (2023) Genetics: From Genes to Genomes. McGraw Hill. • Cox MM, Nelson DL (2021) Lehninger Principle of Biochemistry, 8th edition, MacMillan, Worth Publishers, London. • Devlin TM (2011) Textbook of Biochemistry, 7th edition, John Wiley & Sons, Inc, New York. • Dubey RC (2013) A Text Book of Biotechnology 4th edition, S Chand & Co Ltd, New Delhi. • Wilson K, Walker J (2018) Principle & Techniques of Biochemistry and Molecular Biology, 8th edition, Cambridge University Press, Cambridge. **Suggested Websites** https://www.google.co.in/books/edition/Applications_of_Recombinant_DNA_Technolo/zf3EDwAAQBAJ?hl=en&gbpv=1&dq=recombinant%20dna%20technology&pg=PR1&printsec=frontcover https://academic.oup.com/aob/article/105/7/1141/148741 https://dx.doi.org/10.1016/B978-0-12-800180-6.00001-3 https://doi:10.1016/j.molp.2016.05.004 https://doi.org/10.1093/jn/130.5.1081 https://academic.oup.com/jxb/article/68/10/2463/2967464
Learning Outcomes	By the end of this course, students will be proficient in isolating and analyzing nucleic acids, using DNA modifying enzymes, and applying molecular tools such as CRISPR-Cas for gene editing. They will understand the principles and techniques of transgenic plant production, vector construction, and library construction. Students will be able to perform cloning, transformation, and screening of recombinants, and will gain practical experience with vector engineering for gene expression in various systems. Additionally, they will be equipped to apply these skills in biotechnology and genetic research.

Course Title	**BOTMJ56: Lab Work Based on Paper BOTMJ55**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Extraction of genomic DNA • Qualitative and quantitative analyses of extracted genomic DNA	6.0 h
II	• Extraction of plasmid DNA	6.0 h
III	• Qualitative and quantitative analyses of extracted plasmid DNA	6.0 h
IV	• Restriction digestion of plasmid DNA	6.0 h
V	• Transformation and selection of plasmid DNA in *E. coli*	6.0 h

Course Title	**BOTMJ57: Plant and Microbial Techniques**
Category of Course[53]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[54]**& Hour of Teaching**	(see table below)
Course Objectives	This course aims to equip students with essential research methodology skills, covering the entire research process from problem formulation to data collection, analysis, and reporting. It focuses on ethical research practices, effective literature review, diverse data collection methods, and the use of statistical and qualitative analysis tools. The course also enhances students' abilities to present and publish their research findings.

	Theory	Practical	Cumulative
Credits	1	1	2
Hour of Teaching	15	30	45

Units	Course Content	Hours of Teaching:15 h
I	• Principles of light-, bright- and dark-field microscopy, fluorescence and phase-contrast microscopy • Basics of scanning and transmission electron microscopy	3.5 h
II	• Spectrophotometry: UV-Visible, fluorescence and nanodrop spectrophotometer • Principles of centrifugation, sedimentation and equilibrium, factors affecting sedimentation velocity, gradient centrifugation, ultracentrifugation	3.5 h
III	• Chromatography techniques: TLC, HPLC, GC, mass spectrometry • Electrophoresis: agarose gel electrophoresis, SDS-PAGE and 2-D gel electrophoresis	3.0 h
IV	• PCR, site directed mutagenesis, microarray and RNA sequencing • DNA sequencing: dideoxy nucleotide and automated sequencing methods	2.5 h
V	• Basic concept of plant cell and tissue culture, microbial culture and cryopreservation • Bioinformatics tools and their applications: BLAST, multiple sequence alignment	2.5 h

Texts/ References

1. Dubey RC (2013) A Text Book of Biotechnology, 4^{th} edition, S Chand & Co Ltd, New Delhi.
2. Green MR, Sambrook J (2000) Molecular Cloning: A Laboratory Manual, 4^{th} edition, Cold Spring Harbor Laboratory Press, Cold Spring Harbor.
3. Bhojwani SS, Razdan MK (1996) Plant Tissue Culture: Theory and Practice, Elsevier, Amsterdam.
4. Baxevanis AD, Ouellette BFF (2005) Bioinformatics: A practical guide to the analysis of genes and proteins, 3^{rd} edition, John Wiley and Sons, New Jersey.
5. Simpson MG (2010) Plant Systematics, 2^{nd} edition, Elsevier Academic Press, Amsterdam.
6. Singh G (2012) Plant Systematics-Theory and Practice, 3^{rd} edition, Oxford and IBH Publishing Co, New Delhi.
7. Karp G (2010) Cell Biology, 6^{th} edition, John Wiley and Sons, New Jersey.
8. Dubey RC, Maheshwari DK (2016) A Text Book of Microbiology, S Chand & Co Ltd, New Delhi.
9. Brown TA (2007) Gemones-3, 3^{rd} edition, Garland Sci Publishing, New York.
10. Willey JM, Sherwood LM, Woolverton CJ (2014) Prescott's Microbiology, 9^{th} edition, MacGraw Hill International, New York.

Suggested Websites

https://bio-protocol.org/en
https://conductscience.com/microbiology-techniques/

Learning Outcomes

Upon completing this course, students will be able to effectively utilize advanced microscopy techniques, spectrophotometry, and centrifugation methods for analyzing plant and microbial samples. They will

[53] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[54] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	demonstrate proficiency in chromatography, electrophoresis, PCR, and DNA sequencing techniques. Students will gain hands-on experience in plant cell and tissue culture, microbial culture, and cryopreservation. Additionally, they will be adept at using bioinformatics tools for sequence analysis, including BLAST and multiple sequence alignment. This knowledge will enable them to conduct sophisticated research and solve complex problems in plant and microbial sciences.

Course Title	**BOTMJ58: Lab Work Based on Paper BOTMJ57**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Analysis of DNA and RNA samples by spectrophotometer	6.0 h
II	• Demonstration of PCR and Gel Documentation system	6.0 h
III	• Analysis of samples by fluorescence spectrophotometer	6.0 h
IV	• Study of different biological samples by microscopic techniques	6.0 h
V	• Nucleic acids and protein sequence analysis using bioinformatics tools	6.0 h

Course Title	**BOTMV51: Herbarium Preparation and Preservation**	
Category of Course[55]	Major / Minor / **Minor (Vocational-2)**/ SEC / AEC /VAC / MD/Internship/Dissertation	
Credits[56]**& Hour of Teaching**	(see table below)	
Course Objectives	The objective of this course is to provide students with comprehensive knowledge and practical skills in herbarium preparation and preservation. Students will learn the history, objectives, and functions of herbaria, including important global and Indian examples. They will gain expertise in techniques for collecting, pressing, drying, mounting, and labeling plant specimens, as well as preserving and maintaining herbarium materials. The course will also cover the collection and preservation of algae, lichens, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms. Additionally, students will explore botanical nomenclature and the principles of virtual herbaria and digitalization.	
Units	**Course Content**	**Hours of Teaching:45 h**
I	• Introduction to herbarium: History of herbarium, objectives of herbarium • Role of herbarium in teaching and research, types of herbaria, functions of herbarium • Important herbaria of the world and India	10.0 h
II	• Herbarium techniques:Collection of the specimen (plants) • Pressingand drying of the specimen, Herbarium pests, Poisoning of specimens or protection of plants from insects or pests, Mounting, stitching, and labelling of specimen, Storage of herbarium sheets	9.0 h
III	• Maintenance of herbaria:Fumigation, heating, chemical treatment, handling of specimens, challenges in maintenance • Collection, identification, and preservation of different types of algae and lichens • Collection, identification, and preservation of fungi	9.0 h

Credits & Hour of Teaching:

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

[55] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[56] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	• Collection, identification, and preservation of bryophytes • Collection, identification, and preservation of pteridophtes	
IV	• Collection, identification, and preservation of gymnosperms • Collection, identification, and preservation of angiosperms • Botanical Nomenclature: Binomial nomenclature, Taxonomical hierarchy, Concept of ICN, Principles and rules of ICN, Ranks, and names, Typification, and principle of priority	9.0 h
V	• Virtual herbarium: Goal of virtual herbarium, importance, Techniques used, Herbarium based databases, Specimens imaging, and Digitalization	8 h

Texts/ References	1. Forman, Leonard, and Diane M. Bridson (2013). The Herbarium Handbook. Kew: Royal Botanic Gardens. ISBN-10: 1900347431. 2. Metsger, D A, and S C. Byers (1999). Managing the Modern Herbarium: An Interdisciplinary Approach. Society for Preservation. ISBN-10: 0963547623 3. Taxonomic Literature II (TL-2) https://www.sil.si.edu/DigitalCollections/tl-2/ 4. Turland, N. J., et al. (eds.) 2018: International Code of Nomenclature for algae, fungi, and plants (Shenzhen Code) adopted by the Nineteenth International Botanical Congress Shenzhen, China, July 2017. Regnum Vegetabile 159. Print and Web. DOI https://doi.org/10.12705/Code.2018 5. Ainsworth, G., Kirk, P., et al. Ainsworth &Bisby's Dictionary of the Fungi (10th ed.), 2008. Wallingford, CT: CABI. ISBN-13: 978 0 85199 826 8 6. https://www.floridamuseum.ufl.edu/herbarium/methods/vouchers/ 7. https://www.herbsociety.org/file_download/inline/2c81731f-ecd5-4f5d-a142-666830a89ed2 8. https://bsi.gov.in/uploads/userfiles/file/Training/Manual_Herbarium%20Technique.pdf
Learning Outcomes	Upon completing this course, students will gain a comprehensive understanding of herbarium practices, including the historical development, objectives, and types of herbaria and their significance in research and education. They will be skilled in the techniques for collecting, pressing, drying, mounting, and labeling plant specimens, as well as implementing effective preservation methods and addressing maintenance challenges. Students will learn to collect and preserve a variety of plant groups, such as algae, lichens, fungi, bryophytes, pteridophytes, gymnosperms, and angiosperms. Additionally, they will acquire knowledge of botanical nomenclature principles and the International Code of Nomenclature, and be proficient in using virtual herbarium techniques and digital tools for specimen imaging and database management.

Course Title	**BOTMV52: Lab Work Based on Paper BOTMV51**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Local field visit to study and identify the local flora and vegetation • Collection, and preservation of algal and lichens samples • Demonstration of herbarium techniques • Collection, and preservation of fungal samples	10.0 h
II	• Collection, and preservation of bryophytes samples • Collection, and preservation of pteridophyte samples • Collection, and preservation of bryophytes samples • Collection, and preservation of pteridophyte samples	10.0 h
III	• Collection, and preservation of gymnosperm samples • Collection, and preservation of angiosperm samples	10.0 h

Course Title	BOTIN51 (2 Credits)
Category of Course[57]	Major / Minor / Minor (Vocational) / SEC / AEC-4/ VAC / MD/**Internship**/Dissertation

SEMESTER-VI

Course Title	BOTMJ61: Comparative Cryptogams			
Category of Course[58]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation			
Credits[59]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a thorough understanding of comparative cryptogams, focusing on algae, fungi, bryophytes, and pteridophytes. Students will explore the classification systems and general characteristics of major algal classes, including pigmentation, storage products, thallus organization, and reproductive strategies. They will study fungal classifications, structures, reproductive cycles, and evolutionary trends. Additionally, the course covers the classification and life histories of bryophytes and pteridophytes, including specific genera and evolutionary aspects of sporophytes and stelar evolution. By the end, students will gain insights into the diversity, evolutionary adaptations, and ecological significance of these non-flowering plants.			

Units	Course Content	Hours of Teaching: 45 h
I	• Classification (Fritsch's system) and general characteristics of major classes • Pigmentation and storage products • Thallus organization and evolutionary tendencies	8.0 h
II	• Reproduction and life history types with reference to Chlorophyceae, Phaeophyceae, Rhodophyceae and Cyanophyceae	8.0 h
III	• General features of fungi and their classification (Ainsworth's system) • Structure, reproduction and life cycles of representative classes of fungi	9.0 h
IV	• Types of fungal spores and mode of their liberation • Evolutionary trends in fungi	7.0 h
V	• Classification of Bryophyta and Pteridophyta • Life histories of bryophytes with reference to *Cyathodium*, *Notothylus*, *Sphagnum* • General account of evolution of sporophyta • Life history of *Psilotum*and*Marsilea*General account on stelar evolution in Pteridophyta	13.0 h

Texts/	1. Fritsch FE (1935) The Structure and Reproduction of the Algae, Vol I, Cambridge University Press, Cambridge.

[57] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[58] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[59] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

References	2. Fritsch FE (1945) The Structure and Reproduction of the Algae, Vol II, Cambridge University Press, Cambridge. 3. Morris I (1967) Introduction to Algae, Hutchinson University Library, London. 4. Smith GM (1972) Cryptogamic Botany Vol I and II, Tata McGraw Hill Publishing Co Ltd, New Delhi. 5. Dixon PS (1973) Biology of the Rhodophyta, Oliver & Boyd, UK 6. Singh V, Pande PC, Jain DK (2010) A Text Book of Botany (Algae, Fungi, Brophyta, Pteridophyta), Pub Rastogi Publication, Meerut. 7. Bold HC, Wynne MJ (1985) Introduction to the Algae, 2nd edition, Prentice-Hall Inc, New Jersey. 8. South GR, Whittick A (1998) Introduction to Phycology, Blackwell Scientific Publication, London. 9. Gangulee HC, Kar AK (2011) College Botany Vol II (Algae, Fungi, Brophyta, Pteridophyta), New Central Book Agency, Kolkata. 10. Lee RE (2018) Phycology, 5th edition, Cambridge University Press, Cambridge. 11. Sambamurty AVSS (2019) A text Book of Algae, Dreamtech Press 9389447178 12. Sharma OP (2017) Algae, McGraw Hill Education 978-0070681941 13. Dubey HC (2012) An Introduction to Fungi: 4th Edition. Scientific Publishers. 14. Watkinson SC, Boddy L and Money N (2015) The Fungi, Third Edition. Academic Press, Elsevier. 15. Petersen JH (2013) The Kingdom of Fungi, Princeton University Press, USA. 16. Kavanagh K. (2018) Fungi Biology and Applications,Third Edition. John Wiley 17. Aneja KR (2015) An Introduction to Mycology. New Age International Private Limited. New Delhi. 18. Sharma OP (2017) Bryophytea, McGraw Hill Education 2017 978-1259062872 19. Alam A (2019) Text Book of Bryophyta, Dreamtech Press 978-9389307221 20. Rashid A (2015). An Introduction to Bryophyta. 1st edition, Vikas Publishing House Pvt. Ltd., New Delhi. 21. Parihar NS (1996). Biology and Morphology of Pteridophytes. Central book Depot. Allahabad. 22. Sambamurthy AVSS (2020) A Textbook of Bryophytes, Pteridophytes, Gymnosperms and Paleobotany. Dreamtech Press India Pvt Ltd. Delhi 23. Rashid A (2011) An Introduction to Pteridopyta, 2nd edition, (Reprint), Vikas Publishing House Pvt Ltd, Noida. 24. Rashid A. (2013) An Introduction to Pteridophyta: Diversity, Development, Differentiation, Second edition. VikasPublishing House PVT LMT, Noida, India. 25. Vashishta PC,Sinha AK, Kumar K (2021) Botany for degree students-Pteridophyta. Vikas Publishing House Pvt. Ltd., Ghaziabad. **Suggested Websites** 1. https://www.algaebase.org/ 2. https://www.psaalgae.org/ 3. http://www.cyanodb.cz/ 4. http://bryophytes.plant.siu.edu/
Learning Outcomes	Upon completing this course, students will gain a comprehensive understanding of comparative cryptogams, including algae, fungi, bryophytes, and pteridophytes. They will be able to classify and describe major algal classes, including their pigmentation, storage products, thallus organization, and reproductive mechanisms. Students will learn about fungal classifications, structures, reproductive cycles, and evolutionary trends. They will also acquire knowledge of the classification and life histories of bryophytes and pteridophytes, including specific genera and evolutionary developments in sporophytes and stelar structures. This course will enhance their understanding of the diversity, adaptations, and ecological roles of these non-flowering plants.

Course Title	BOTMJ62: Lab Work Based on Paper BOTMJ61	
Units	**Practicals**	**Hours of Teaching:30 h**
I	Study of the following genera of Algae: *Pandorina, Eudorina, Scenedesmus, Hydrodictyon, Ulva, Pithophora, Coleochaete, Oedocladium, Cosmarium, Codium, Bryopsis, Padina, Batrachospermum, Gloeotrichia, Aulosira*	10.0 h
II	Study of the following genera of Fungi: *Saccharomyces,Synchytrium, Protomyces, Peronospora,Phyllactinia, Puccinia, Ustilago, Helminthosporium, Alternaria*	10.0 h
III	Study of the following genera of Bryophyta: *Cyathodium, Notothylus, Sphagnum*	5.0 h
IV	Pteridophyta: *Psilotum, Marsilea*	5.0 h

Course Title	**BOTMJ63: Comparative Phanerogams**
Category of Course[60]	**Major** / Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[61] **& Hour of Teaching**	(see below)
Course Objectives	This course aims to provide a comprehensive understanding of phanerogams by exploring the classification, morphology, anatomy, and reproductive strategies of gymnosperms and angiosperms. Students will delve into the phylogenetic trends and world distribution of gymnosperms, including Cycadales, Ginkgoales, Coniferales, and Gnetales. They will also examine the classification and distinguishing characteristics of key angiosperm families, along with their economic importance. The course will integrate numerical taxonomy and molecular approaches, including Hutchinson's classification system and the Angiosperm Phylogeny Group (APG) system, to provide insights into the evolutionary relationships and phylogenetic developments within these plant groups.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	**Course Content**	**Hours of Teaching: 45 h**
I	• Classification of Gymnosperm (K R Sporne) and general account of morphology, anatomy and reproduction of the following: *Cycadales (Zamia), Ginkgoales (Ginkgo), Coniferales (Biota) and Gnetales (Gnetum)*	14.0 h
II	• General account of *Williamsonia* and *Pentaxyl* • Phylogenetic trends in Gymnosperms • World distribution of living Gymnosperms	8.0 h
III	• Classification of Angiosperms (Hutchinson's) and general account of numerical taxonomy	7.0 h
IV	• Distinguishing characters of the following families and their economic importance: *Mangoliaceae, Rutaceae, Combretaceae, Asteraceae, Convolvulaceae, Scrophulariaceae, Verbenaceae, Polygonaceae, Euphorbiaceae, Zingiberaceae, Liliaceae and Cyperaceae*	12.0 h

[60] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[61] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

V	• Molecular approach to taxonomy and Angiosperm phylogeny group (APG)	4.0 h

Texts/ References	1. Sporne KR (1967) Morphology of Gymnosperms, BI Publication, New Delhi. 2. Singh V, Jain DK (2012) Taxonomy of Angiosperms, 2nd edition, Rastogi Publication, Meerut. 3. Bhojwani SS, Bhatnagar SP (2009) Embryology of Angiosperms, Vikash Publishing House, New Delhi. 4. Singh Gurucharan (1999) Plant Systematics- Theory and Practices, Oxfordand IBH Publishing Co, New Delhi. 5. Judd WS, Christopher S, Campbell, Kellogg, AE, Stevens, PF (1999) Plant Systematics: A Phylogenetic Approach, Sinauer Associates Inc Publishers, Sunderland. 6. Simpson MG (2006) Plant Systematics, Elsevier Academic Press, Amsterdam. 7. Bhojwani SS, Bhatnagar SP and Dantu PK (2015) The Embryology of Angiosperms, 6th Edition, Vikas Publishing House Pvt Ltd, Noida. 8. Johri, BM (2011) Embryology of Angiosperms, Springer Berlin Heidelberg. 9. Bhatnagar, S.P and Moitra, A. (1996) Gymnosperms, New Age International Pvt. Ltd. Publishers, New Delhi. 10. Beck, C. B. (1988) Origin and evolution of gymnosperms. Columbia University Press. Chamberlain, C. J. (1934) Gymnsoperms – structure and evolution. Chicago Univ. Press
Learning Outcomes	Upon completing this course, students will be proficient in classifying and understanding the morphology, anatomy, and reproductive strategies of both gymnosperms and angiosperms. They will be able to analyze and compare the evolutionary trends and global distribution of gymnosperms, including Cycadales, Ginkgoales, Coniferales, and Gnetales. Students will also gain expertise in the classification and economic significance of key angiosperm families, and apply numerical taxonomy and molecular techniques, including Hutchinson's classification and the Angiosperm Phylogeny Group (APG) system, to elucidate the phylogenetic relationships within these plant groups.

Course Title	**BOTMJ64: Lab Work Based on Paper BOTMJ63**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Study of morphology, anatomy and reproductive structure of *Zamia*, *Biota*,*Ginkgo* and*Gnetum*	10.0 h
II	• Study of fossils, *viz.*, *Williamsonia* (*Bucklandia*, *Ptilophyllum*) and *Pentaxylon*	10.0 h
III	• Taxonomic description of plant families: *Rutaceae, Combretaceae, Asteraceae, Convolvulaceae, Scrophulariaceae,Verbenaceae, Polygonaceae, Euphorbiaceae, Liliaceae, Cyperaceae*	5.0 h
IV	• Plant collection: Identification, preservation and submission of at least 25 herbarium sheets	5.0 h

Course Title	**BOTMJ65: Biodiversity and Biostatistics**		
Category of Course[62]	**Major** / Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation		
Credits[63]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75

[62] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[63] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Objectives	This course aims to provide students with a comprehensive understanding of biodiversity and biostatistics. Students will explore various levels of biodiversity—genetic, species, and ecosystem—and examine its distribution, ecosystem functions, and services. They will learn about the threats to biodiversity and strategies for its conservation. In the biostatistics section, students will gain skills in sampling methods, experimental design, and statistical analyses, including central tendency, correlation, regression, and analysis of variance. The course will also introduce multivariate methods and statistical software for data analysis, equipping students with essential tools for research and conservation efforts.

Units	Course Content	Hours of Teaching:45 h
I	• Introduction to biodiversity • Levels of biodiversity: Genetic, species and ecosystem	10.0 h
II	• Biodiversity and ecosystem functions: Concepts and Hypotheses • Biodiversity and ecosystem services: Provisioning, regulatory, supporting and cultural	5.0 h
III	• Threats to biodiversity: Causes of biodiversity loss, species extinction, vulnerability of species to extinction, IUCN threat categories, Red data book • Strategies for biodiversity conservation: Principles of biodiversity conservation, *in-situ* and *ex-situ* conservation strategies	8.0 h
IV	• Sampling methods and experimental design • Concepts of central tendency, normal distribution and variability • Contingency tables and chi-square test • Comparison of means: t-test, multiple range tests	10.0 h
V	• Correlation and regression analyses • Analysis of variance: One-way • Introduction to multivariate methods: Parametric and non-parametric ordination, statistical packages for data analyses	12.0 h

Texts/ References	1. Heywood VH and Watson RT. 1995. Global Biodiversity Assessment. UNEP, Cambridge University Press. 2. Singh JS, Singh SP and Gupta SR. 2014. Ecology, Environmental Science and Resource Conservation. S. Chand & Company Pvt. Ltd. New Delhi 3. Magurran, A.E. 2004. Measuring Biological Diversity. Blackwell Publishing, Oxford 4. Snedecor GW and Cochran WG. 1989. Statistical Methods. 8th Edition, Iowa State University Press, Ames. 5. Infan AK and Khanum A. 2020. Fundamentals of Biostatistics. Ukaa Publications, Hyderabad.
Learning Outcomes	Upon completing this course, students will have a thorough understanding of biodiversity at genetic, species, and ecosystem levels, including its distribution, functions, and services. They will be equipped to analyze biodiversity threats and implement conservation strategies. Additionally, students will acquire practical skills in biostatistics, including sampling methods, experimental design, and statistical analyses such as central tendency, correlation, regression, and variance. They will also be familiar with multivariate methods and statistical software, enabling them to effectively analyze and interpret complex ecological and biodiversity data for research and conservation applications.

Course Title	BOTMJ66: Lab Work Based on Paper BOTMJ65	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Analysis of species-area relations • Analysis of herbaceous species richness in protected and unprotected grasslands	6.0 h
II	• Analysis of herbaceous species evenness in protected and unprotected grasslands • To assess alpha diversity of protected and unprotected grasslands	6.0 h
III	• To assess beta diversity in protected and unprotected grasslands • To analyze gamma diversity across grassland communities • To estimate mean, standard deviation, standard error and variance	6.0 h
IV	• To estimate chi-square test for association analysis • To estimate t-test for differentiating the means • Determination of one way analysis of variance	6.0 h
V	• To establish correlation between variables • To establish regression equation between variables	6.0 h

Course Title	BOTMJ67: Cytogenetics and Evolutionary Processes
Category of Course[64]	**Major** / Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[65] **& Hour of Teaching**	(see table below)
Course Objectives	The objective of this course is to provide a comprehensive understanding of cytogenetics and evolutionary processes by exploring chromosome structure, karyotyping, and chromosomal rearrangements. Students will learn about chromosome manipulation techniques, the impact of numerical chromosomal variations, and the principles of mutagenesis. The course will cover chromatin remodeling and epigenetics, emphasizing their roles in gene regulation and evolutionary biology. Additionally, students will delve into quantitative genetics, focusing on inheritance patterns and QTL analysis, and study population and evolutionary genetics, including gene pool dynamics, Hardy-Weinberg equilibrium, and speciation processes.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	Course Content	Hours of Teaching:45 h
I	• Chromosome structure and karyotype:Physical, chemical, and molecular organization of chromosomes; Classical and modern concepts**,** Karyotype symmetry and asymmetry, Karyotype classification and estimation, Karyotype evolutionary factors and trends in natural plant population	5.0 h
II	• Banding techniques and chromosome manipulations:Classical banding techniques, The International System for Human Cytogenomic Nomenclature (ISCN), chromosome linearization and differentiation of selected gymnosperms and angiosperms, Genomic level or whole genome manipulations, Chromosome level (addition and substitution) manipulations, applications	7.0 h
III	• Threats to biodiversity: Causes of biodiversity loss, speChromosome rearrangements: Meiotic configuration and genetic consequences of	11.0 h

64 SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

65 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	deletion, duplication, inversion, and translocation. Permanent translocation heterozygote (PTH) • Source and consequences of numerical variation in chromosomes: Monosomic, nullisomic, trisomics; autopolyploids, allopolyploids,Status of allopolyploids in plant evolution • cies extinction, vulnerability of species to extinction, IUCN threat categories, Red data book • Strategies for biodiversity conservation: Principles of biodiversity conservation, *in-situ* and *ex-situ* conservation strategies	
IV	• Mutagens and mutagenesis:Basic concepts of mutation, Mutation types, Molecular basis of mutations, Chemical mutagens (Base analogs, deaminating, hydroxylating, alkylating, and intercalating agents), and Physical mutagens (Ionising and Non-ionising radiations) • Chromatin remodelling and epigenetics: Concept and scope of epigenetics, Components of chromatin remodelling complexes, Assembly and disassembly of nucleosome, DNA methylation and histone modifications, Techniques for studying epigenetics mechanisms, and the biological significance of epigenetics: genomic imprinting, paramutation, and silencing	12.0 h
V	• Quantitative genetics:Concept of quantitative inheritance, genes, and environment, the multiple factor hypothesis, partitioning the phenotypic variance, Broad-Sense Heritability, Narrow-Sense Heritability, quantitative trait loci, and QTL analysis • Population and evolutionary genetics:Population, gene pool, allele frequency, Hardy-Weinberg principle (Allele frequencies and genotype frequencies), selection, mutation, migration, genetic drift, linkage disequilibrium, and speciation	10.0 h

Texts/ References	1. Stebbins GL (1971) Chromosome Evolution in Higher Plants, Edward Arnold LTD, London 2. Swanson CP, Mertz TF, Young WJ (1982) Cytogenetics: The Chromosomes in Division, Inheritance, and Evolution, 2nd edition, Englewood Cliff, Prentice-Hall, New Jersey. 3. Clark MS, Wall WJ (1996) Chromosomes: The Complex Code, Chapman & Hall, London. 4. Sharma AK, Sharma A (1985) Advances in Chromosome and Cell Genetics, Oxford & IBH Publishing Co, Kolkata. 5. Krebs JE, Lewin B, Goldstein ES (2011) Genes X, Sudburry, Masschusetts. 6. Brooker RJ (2011) Genetics: Analysis and Principles, 4th edition, McGraw-Hill Education, New Delhi. 7. De Robertis EMF (2011) Cell and Molecular Biology, 8th edition, Lippincott Williams & Wilkins, New York. 8. Singh BD (2014) Genetics, Kalyani Publishers, Kolkata. 9. Alberts B (2014) Molecular Biology of the Cell, 6th edition, Garland Science Pub, New York. 10. David CA, et al., (2007) Epigenetics, 2nd edition, Cold Spring Harbor Laboratory Press, New York. 11. Spillane C, McKeown PC (2014) Plant Epigenetics and Epigenomics: Methods and Protocol, Springer Publisher, London
Learning Outcomes	Upon completing this course, students will be proficient in analyzing chromosome structure and karyotype, understanding chromosomal rearrangements and manipulations. They will gain insight into numerical chromosomal variations and their evolutionary implications, and become familiar with mutagenesis, chromatin remodeling, and epigenetics. The course will enable students to grasp quantitative genetics concepts, such as inheritance patterns and QTL analysis, and apply these principles to real-world genetic studies. Additionally, students will understand population genetics dynamics, including gene pool, allele frequencies, and evolutionary processes like selection, mutation, and speciation.

Course Title	BOTMJ68: Lab Work Based on Paper BOTMJ67	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Preparation of metaphase squashes and karyotype analysis in *Allium*, *Vicia*, and *Aloe* • Karyotype analysis using microphotographs of metaphase squashes and preparation of ideograms of suitable plant materials	6.0 h
II	• To study the effects of Colchicine: a) C-mitosis, b) polyploidy	6.0 h
III	• To study the permanent translocation heterozygote in suitable plant material	6.0 h
IV	• Study of; a) mitotic index and b) mitotic phase index	6.0 h
V	• Demonstration of QTL mapping • Demonstration of Hardy-Weinberg equilibrium using available data	6.0 h

Course Title	**BOTMV61: Plant Micropropagation**
Category of Course[66]	Major / Minor / **Minor (Vocational-3)**/ SEC / AEC /VAC/ MD/Internship/Dissertation
Credits[67] **& Hour of Teaching**	(see table below)
Course Objectives	This course aims to provide students with a comprehensive understanding of plant micropropagation techniques. Students will learn the historical development, principles, and laboratory organization for in vitro plant culture. They will master media preparation, aseptic techniques, and explant handling. The course covers cellular totipotency, micropropagation stages, and various methods such as meristem culture, organogenesis, and somatic embryogenesis. Students will also address common problems, including contamination and somaclonal variations, and explore advanced topics like haploid plant production and industrial applications. By the end, students will be equipped to apply these techniques in both research and commercial settings.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Introduction to *In vitro*plant micropropagation: Historical perspectives, principles of plant tissue culture, Laboratory organization • Media preparation: Constituents, Types of media, Steps of media preparation, Precaution	9.0 h
II	• Maintenance of aseptic conditions, Explant selection and preparation, Surface sterilization • Cellular totipotency and differentiation: Concept of cellular totipotency and differentiation, role of plant growth regulators	9.0 h
III	• Stages of micropropagation: Preparatory, Culture initiation, Multiplication, Plantlet formation, Hardening and acclimatization • Types of micropropagation: Meristem culture, Organogenesis and Somatic embryogenesis	9.0 h

[66] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[67] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

IV	• Problems associated with micropropagation: Contamination, recalcitrance, vitrification, off types, high costs, somaclonal variations • Haploid plant production: parasexual hybridization significance of Androgensis and Gynogenesis, In-vitro fertilization	9.0 h
V	• Industrial and other applications of micropropagation: Production of disease-free plants, Suspension cultures, Hairy root cultures.Use of bioreactorsfor commercial production of bioactive compounds	9.0 h

Texts/ References	1. Bhojwani SS & Dantu PK (2013) Plant Tissue Culture: An Introductory Text, Springer. 2. Park S (2021) Plant Tissue Culture: Techniques and Experiments, Fourth Edition, Academic Press, Elsevier. 3. Razdan MK (2003) Introduction to Plant Tissue Culture, Science Publishers, USA.
Learning Outcomes	Upon completing this course, students will have developed a thorough understanding of plant micropropagation techniques and their applications. They will be proficient in media preparation, aseptic techniques, and explant handling. Students will grasp the concepts of cellular totipotency and differentiation, and be able to navigate the stages of micropropagation, including meristem culture, organogenesis, and somatic embryogenesis. They will also identify and address common issues such as contamination and somaclonal variations. Additionally, students will be knowledgeable about haploid plant production and the industrial applications of micropropagation, including the use of bioreactors for bioactive compound production.

Course Title	**BOTMV62: Lab Work Based on Paper BOTMV61**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Stock preparation of Murashige and Skoog Medium	5.0 h
II	• Media preparation	5.0 h
III	• Explant preparation and surface sterilization	10.0 h
IV	• To initiate culture by inoculating explants in different combinations of auxins and cytokinins	10.0 h

Honours without Research Degree

SEMESTER-VII

Course Title	**BOTMJ71: Plant Biochemistry and Molecular Biology**
Category of Course	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation
Credits& Hour of Teaching	

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Course Objectives	The course objective is to provide students with a comprehensive understanding of plant biochemistry and molecular biology. Students will learn the biosynthesis of key carbon compounds, including sucrose, starch, and cellulose, and delve into sulfur and phosphorus metabolism, including ATP synthesis. They will gain insight into nitrogen metabolism, including the biochemistry of nitrogen fixation and nitrate reduction. The course will cover the structure and replication of DNA, RNA transcription and processing, and protein synthesis and modification. Additionally, students will explore enzyme mechanisms and molecular tools and techniques such as electrophoresis and cloning vectors.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Biosynthesis of carbon compounds: sucrose, starch, cellulose • Sulphur and phosphorus metabolism: Activation and assimilation of sulphur, energy-rich phosphorus compounds; ATP synthesis	10.0 h

II	• Nitrogen metabolism: components, types and substrate of enzyme nitrogenase, biochemistry of nitrogen fixation; Nitrate metabolism: Uptake and reduction into ammonia, ammonia assimilation	8.0 h
III	• DNA: Structure and properties of different forms of DNA, DNA replication in prokaryotes and eukaryotes, DNA damage and repair	8.0 h
IV	• RNA:Structure and properties of different forms of RNA, transcription in prokaryotes and eukaryotes, mRNA processing-5' cap formation-3'end processing and polyadenylation, splicing, nuclear export of mRNA	6.0 h
V	• Protein: Basic aspects of protein conformation, translation in prokaryotes and eukaryotes (activation of amino acids, initiation, elongation, termination and release of peptides), post-translational modification of proteins • Enzymes: Mechanism of enzyme action, coenzymes, allosteric enzyme, isozymes • Molecular tools and techniques: Basic principles of electrophoresis and blotting, cloning vectors, molecular cloning of nucleic acid fragments	13.0 h

Texts/References	1. Berg JM et al. (2018) Biochemistry, 8th edition, WH Freeman & Company, New York. 2. Cox MM, Nelson DL (2021) Lehninger Principle of Biochemistry, 8th edition MacMillan, Worth Publishers,London. 3. Devlin TM (2011) Textbook of Biochemistry, 7th edition, John Wiley & Sons, Inc, New York. 4. Voet D, Voet JG and Pratt CW (2018) Biochemistry, 5th Edition, John Wiley & Sons, Inc, New York. 5. Zubay GL (1998) Biochemistry, 4th edition, Wm C Brown Publishers, Dubuque, Iowa. 6. Dubey RC (2013) A Text Book of Biotechnology 4th edition, S Chand & Co Ltd, New Delhi. 7. Wilson K, Walker J (2018) Principle & Techniques of Biochemistry and Molecular Biology, 8th edition, Cambridge University Press, Cambridge. 8. Bowsher C & Tobin A (2021) Plant Biochemistry, 2nd Edition, Taylor & Francis Inc, CRC press. 9. Arthur M. Lesk (2016) Introduction to Protein Science Architecture Function and Genomics, Oxford University press, London **Suggested websites** https://academic.oup.com/aob/article/105/7/1141/148741 https://dx.doi.org/10.1016/B978-0-12-800180-6.00001-3 https://doi:10.1016/j.molp.2016.05.004 https://doi.org/10.1093/jn/130.5.1081 https://academic.oup.com/jxb/article/68/10/2463/2967464 https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=2 https://epgp.inflibnet.ac.in/epgpdata/uploads/epgp_content/food_technology/food_chemistry/16. protein_ structure_and_denaturation/et/40_et_m16.pdf https://www.youtube.com/watch?v=MODnIkQvyz0 https://www.youtube.com/watch?v=AeVDoDp3llI
Learning Outcomes	Upon completion of this course, students will be able to understand and describe the biosynthesis of essential carbon compounds like sucrose, starch, and cellulose, and elucidate sulfur and phosphorus metabolism, including ATP synthesis. They will gain insights into nitrogen metabolism, focusing on nitrogen fixation and nitrate reduction. Students will be proficient in explaining DNA structure, replication, and repair, RNA transcription, processing, and translation, and protein synthesis and post-translational modifications. Additionally, they will be familiar with enzyme mechanisms and molecular techniques such as electrophoresis and molecular cloning, enabling practical applications in plant biochemistry and molecular biology.

Course Title	BOTMJ72: Lab Work Based on Paper BOTMJ71	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Chromatographic separation and comparative analysis of photosynthetic pigments in plants • Chromatographic separation of amino acids	6.0 h
II	• Preparation of standard graph and estimation of DNA by diphenylamine test • Preparation of standard graph and estimation of RNA by Orcinol test	6.0 h
III	• Preparation of standard graph and estimation of protein by Bradford method • Study on the effects of light/dark on the nitrate reductase enzyme	6.0 h
IV	• To study the time dependent activity of urease enzyme. • Study on the effects of different concentrations of urea on urease activity	6.0 h
V	• Determination of different concentrations of urease enzyme on its activity • Preparation of standard graph and estimation of carbohydrates by anthrone reagent method • Demonstration of electrophoretic and blotting instruments	6.0 h

Course Title	**BOTMJ73: Plant Biotechnology and Genetic Engineering**			
Category of Course[68]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[69]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of plant biotechnology and genetic engineering, covering historical developments, model plants, and applications. Students will learn about recombinant DNA technology, including vectors, restriction enzymes, and genomic libraries. They will explore plant cell, tissue, and organ culture techniques, including totipotency, media preparation, and regeneration protocols. The course will cover plant transformation techniques, including both indirect and direct methods. Students will examine transgenic plants and their applications, gene expression regulation, and advanced gene editing technologies like ZFNs, TALENs, and CRISPR/Cas systems, addressing their applications, limitations, and ethical considerations.			

Units	Course Content	Hours of Teaching: 45 h
I	• Plant Biotechnology: Introduction, History, Model plants, Scope and applications • Recombinant DNA technology: Vectors, restriction enzymes, molecular probes, genomic and cDNA libraries	8.0 h
II	• Plant Cell, Tissue and Organ Culture *(a) History, concept of totipotency, nutritional requirements of culture, mediapreparation, aseptic condition, applications and limitations* *(b) Regeneration Protocols: Single Cell Culture, Organogenesis and Somatic Embryogenesis, Factors and significance* *(c) Protoplast Culture and Fusion: Protoplast isolation and culture, Protoplast fusion, methods and factors, Applications*	8.0 h
III	• Techniques for plant transformation	8.0 h

[68] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[69] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	(a) Indirect gene transfer methods (Agrobacterium tumefaciens, and Agrobacterium rhizogenes mediated transfer) *(b) Direct Gene transfer methods (electroporation, particle bombardment, microinjection, PEG, silicon carbide)*	
IV	• Transgenic plants, some genetically modified plants (Flavr Savr tomato, Bt brinjal, Bt corn, Bt cotton, Golden rice, virus resistant plants, pest resistant plants) and molecular forming, applications and limitations (ethical and environmental issues)	9.0 h
V	• Gene expression and its regulation in plants: Transcription factors and cis-regulatory elements (CREs), epigenetic mechanisms of transcriptional regulation, translational control, programmed DNA rearrangements. • Gene editing and its applications: Zinc-finger nucleases (ZFNs), transcription • activator-like effector nucleases (TALENs), CRISPR/Cas systems	12.0 h

Texts/ References	1. Bhojwani SS & Dantu PK (2013) Plant Tissue Culture: An Introductory Text, Springer. 2. Park S (2021) Plant Tissue Culture: Techniques and Experiments, Fourth Edition, Academic Press, Elsevier. 3. Razdan MK (2003) Introduction to Plant Tissue Culture, Science Publishers, USA 4. Slater A, Scott NW and Fowler MR (2008) Plant biotechnology: the genetic manipulation of plants, second edition, Oxford University Press. 5. Sahni S, Prasad BD and Kumar P (2017) Plant Biotechnology: Transgenics, Stress 6. Management and Biosafety Issues, CRC Press.
Learning Outcomes	Upon completing this course, students will be proficient in plant biotechnology and genetic engineering techniques. They will understand the historical development and applications of plant biotechnology, including recombinant DNA technology and plant cell culture methods. Students will gain expertise in various plant transformation techniques, both indirect and direct, and be familiar with the production and use of transgenic plants. They will also learn about gene expression regulation and advanced gene editing technologies, such as ZFNs, TALENs, and CRISPR/Cas systems. Additionally, students will address the applications, limitations, and ethical considerations of these biotechnological advances.

Course Title	**BOTMJ74: Lab Work Based on Paper BOTMJ73**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Medium preparation of Murashige and Skoog • Surface sterilization and inoculation of nodal explants on MS medium	6.0 h
II	• To raise suspension culture by using friable calli	6.0 h
III	• To initiate hairy root culture	6.0 h
IV	• Floral dip method of plant transformation (*Arabidopsis thaliana*) and screening for gene transfer (plasmid isolation & transformation, screening through PCR and agarose gel electrophoresis)	6.0 h
V	• Synthetic seed preparation	6.0 h

Course Title	**BOTMJ75: Field Excursion**
Category of Course[70]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation
Credits[71]& Hour of Teaching	

	Theory	Practical	Cumulative
Credits	2		2
Hour of Teaching	30		30

Course Objectives	Field excursions will provide immersive, hands-on learning experiences that bridge theoretical knowledge and real-world applications. These trips are designed to enhance students' understanding of ecological, geological, botanical, and environmental concepts through direct observation and interaction with natural habitats. By engaging in fieldwork, students develop essential skills in data collection, species identification, and environmental assessment, fostering critical thinking and problem-solving abilities. Field excursions also promote teamwork, collaboration, and effective communication, as students often work in groups to conduct research and share findings. Moreover, these experiences help students appreciate the complexity and interconnectedness of natural systems, encouraging a deeper commitment to environmental conservation and sustainability. Ultimately, field excursions enrich the educational journey by making abstract concepts tangible and inspiring a lasting passion for scientific inquiry and exploration.
Learning Outcomes	Field excursions for BSc students offer hands-on learning experiences, enhancing their understanding of ecological systems, plant identification, and data collection methods. These trips provide practical insights into theoretical concepts, foster observational and analytical skills, and encourage environmental awareness. By interacting directly with nature, students gain a deeper appreciation for biodiversity and conservation. Additionally, fieldwork promotes teamwork, critical thinking, and problem-solving abilities, preparing students for advanced research and professional careers in biological sciences.

Course Title	**BOTMJ76: Photoecophysiology**
Category of Course[72]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation
Credits[73]& Hour of Teaching	

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Course Objectives	This course aims to provide a comprehensive understanding of photoecophysiology, focusing on light-regulated growth and development in cyanobacteria and plants. Students will explore the roles of photoreceptors, light quality, and light signaling in plant and microbial adaptation. Key topics include molecular mechanisms of photodamage and protection, light harvesting processes, and carbon concentrating mechanisms. Through this course, students will gain insights into how light influences physiological processes, signal transduction, and adaptation strategies in diverse photosynthetic organisms.

Units	Course Content	Hours of Teaching: 45 h
I	• Light-regulated growth and development in cyanobacteria: Phytochrome-related photosensor and light perception, Photomorphogenesis, Complementary chromatic acclimation, Light quality and quantity dependent changes in light-harvesting complex, Signal transduction, Photoregulation of morphology	9.5 h

[70] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[71] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

[72] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[73] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

II	• Molecular mechanisms of photodamage and photoprotection: Photo-induced damage to microbes; genetical, biochemical and molecular aspects of mycosporine-like amino acids (MAAs) and scytonemin production, ecological and economical implications	9.5 h
III	• Light-regulated plant growth and development: Photoreceptors and light perception, Physiological responses, Photomorphogenesis, Circadian clock, Signal transduction, Seed germination, Hypocotyl elongation	6.5 h
IV	• Light signalling dynamics in plants: Phytochromes, Cryptochromes, PIFs, Phototropins, UV resistance locus8 (UVR8), HY5, COP1, SUMOylation, Chloroplast movement • Light harvesting in plants and cyanobacteria: Photosynthetic pigments and phycobilisomes, State transition, photochemical quenching (PQ), non-photochemical quenching (NPQ), photoinhibition, orange carotenoid protein (OCP), xanthophyll cycle, principle of PAM fluorometry	13.0 h
V	• Carbon concentrating mechanisms (CCM) in plants and cyanobacteria: Photorespiration, Components of CCM, Pyrenoids, Carboxysome, RuBISCO, Carbonic anhydrase	6.5 h

Texts/References	1. Ruban, A., Foyer, C., & Murchie, E. (Eds.). (2022). Photosynthesis in Action: Harvesting Light, Generating Electrons, Fixing Carbon. Academic Press. 2. Taiz L, Zeiger E (2015) Plant Physiology and Development, 7th edition, Sinauer Associates Inc. Publishers, Sunderland. 3. Hopkins WG, Hunter NPA (2011) Introduction to Plant Physiology, Wiley International edition, John Wiley & Sons, New Jersey. 4. Buchanan B, GruissemW, Jones RL (2002) Biochemistry and Molecular Biology of Plants, American Society of Plant Biologists, Rockville. 5. Devlin R.M. (1983) Plant Physiology, Cengage Learning Inc., London. 6. Salisbury F.B., Ross C.W. (2004) Plant Physiology, Wadsworth, California
Learning Outcomes	Upon completing this course, students will have a thorough understanding of the principles and mechanisms underlying light-regulated growth and development in cyanobacteria and plants. They will be able to explain the roles of various photoreceptors and light signaling pathways in photomorphogenesis and adaptation. Students will gain insights into the molecular mechanisms of photodamage and photoprotection, including the production of protective compounds. They will also understand light harvesting processes, including the functions of photosynthetic pigments and carbon concentrating mechanisms. Overall, students will be equipped to analyze and interpret the effects of light on plant and microbial physiology.

Course Title	**BOTMJ77: Lab Work Based on Paper BOTMJ76**

Units	**Practicals**	**Hours of Teaching:30 h**
I	• Effects of light and dark on seed germination • Hypocotyl length measurements in light and dark grown seedlings	6.0 h
II	• Photosynthetic pigments analysis in light and dark grown seedlings • Growth measurement of cyanobacteria under white light, blue light, red light and UVR	6.0 h
III	• Extraction and quantification of photosynthetic pigments from cyanobacteria grown under different light conditions	6.0 h
IV	• Extraction and spectrophotometric analysis of mycosporine-like amino acids in cyanobacteria	6.0 h
V	• Extraction and spectrophotometric analysis of scytonemin in cyanobacteria	6.0 h

Course Title	BOTMJ78: Research Methodology
Category of Course[74]	Major/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[75] & Hour of Teaching	

	Theory	Practical	Cumulative
Credits	2	0	2
Hour of Teaching	30	0	30

Course Objectives	This course aims to equip students with essential research methodology skills, covering the entire research process from problem formulation to data collection, analysis, and reporting. It focuses on ethical research practices, effective literature review, diverse data collection methods, and the use of statistical and qualitative analysis tools. The course also enhances students' abilities to present and publish their research findings.

Units	Course Content	Hours of Teaching:30 h
I	Introduction to Research Methodology Understanding research, its significance, and different types of research, Steps involved in the research process, Ethical considerations and plagiarism, Identifying and defining research problems, Types of research designs	6.0 h
II	Literature Review Role and significance of literature review in research, Identifying and accessing various sources of literature. Methods for effective literature review, Importance and methods of citation and referencing	4.0 h
III	Data Collection Methods Techniques for collecting primary data (surveys, interviews, observations, experiments), Utilizing secondary data sources, understanding different sampling techniques (random, stratified, cluster), Designing and using questionnaires, interview guides, and other tools, Reliability and validity in data collection	7.0 h
IV	Data Analysis and Interpretation Statistical methods for analysing quantitative data, Techniques for analysing qualitative data, Introduction to statistical software (SPSS, R) and qualitative analysis tools, making sense of data and drawing conclusions, Effective ways to present data using tables, graphs, and charts	7.0 h
V	Research Reporting and Presentation Structure and components of a research report (introduction, methodology, results, discussion, conclusion), Guidelines for writing a thesis or dissertation, writing research papers and articles for journals, Techniques for presenting research findings orally and visually, Understanding the publication process and choosing appropriate journals	6.0 h

Texts/References	1. Research Methodology: Methods and Techniques" by C.R. Kothari and Gaurav Garg 2. Research Methods for Business Students" by Mark Saunders, Philip Lewis, and Adrian Thornhill 3. The Craft of Research" by Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams 4. Research Design: Qualitative, Quantitative, and Mixed Methods Approaches" by John W. Creswell and J. David Creswell 5. Qualitative Inquiry and Research Design: Choosing Among Five Approaches" by John W. Creswell and Cheryl N. Poth 6. Research Methods: The Essential Knowledge Base" by William M. Trochim and James P. Donnelly 7. Research Methods in Education" by Louis Cohen, Lawrence Manion, and Keith Morrison 8. Practical Research: Planning and Design" by Paul D. Leedy and Jeanne Ellis Ormrod

[74] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[75] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	9. Introduction to Research Methods: A Hands-On Approach" by Bora Pajo **Suggested Websites** https://bio-protocol.org/en https://conductscience.com/microbiology-techniques
Learning Outcomes	By the end of this course, students will proficiently design, conduct, and analyze research, adhering to ethical standards. They will be adept at literature review, data collection, and using analytical tools, enabling them to write structured research reports and present their findings effectively.

Course Title	**BOTMN71: Plant Biochemistry and Molecular Biology**		
Category of Course	Major / **Minor**/ Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation		
Credits & Hour of Teaching			

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Course Objectives	The course objective is to provide students with a comprehensive understanding of plant biochemistry and molecular biology. Students will learn the biosynthesis of key carbon compounds, including sucrose, starch, and cellulose, and delve into sulfur and phosphorus metabolism, including ATP synthesis. They will gain insight into nitrogen metabolism, including the biochemistry of nitrogen fixation and nitrate reduction. The course will cover the structure and replication of DNA, RNA transcription and processing, and protein synthesis and modification. Additionally, students will explore enzyme mechanisms and molecular tools and techniques such as electrophoresis and cloning vectors.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Biosynthesis of carbon compounds: sucrose, starch, cellulose • Sulphur and phosphorus metabolism: Activation and assimilation of sulphur, energy-rich phosphorus compounds; ATP synthesis	10.0 h
II	• Nitrogen metabolism: components, types and substrate of enzyme nitrogenase, biochemistry of nitrogen fixation; Nitrate metabolism: Uptake and reduction into ammonia, ammonia assimilation	8.0 h
III	• DNA: Structure and properties of different forms of DNA, DNA replication in prokaryotes and eukaryotes, DNA damage and repair	8.0 h
IV	• RNA:Structure and properties of different forms of RNA, transcription in prokaryotes and eukaryotes, mRNA processing-5' cap formation-3'end processing and polyadenylation, splicing, nuclear export of mRNA	6.0 h
V	• Protein: Basic aspects of protein conformation, translation in prokaryotes and eukaryotes (activation of amino acids, initiation, elongation, termination and release of peptides), post-translational modification of proteins • Enzymes: Mechanism of enzyme action, coenzymes, allosteric enzyme, isozymes • Molecular tools and techniques: Basic principles of electrophoresis and blotting, cloning vectors, molecular cloning of nucleic acid fragments	13.0 h

Texts/References	1. Berg JM et al. (2018) Biochemistry, 8th edition, WH Freeman & Company, New York. 2. Cox MM, Nelson DL (2021) Lehninger Principle of Biochemistry, 8th edition MacMillan, Worth Publishers,London. 3. Devlin TM (2011) Textbook of Biochemistry, 7th edition, John Wiley & Sons, Inc, New York. 4. Voet D, Voet JG and Pratt CW (2018) Biochemistry, 5th Edition, John Wiley & Sons, Inc, New York. 5. Zubay GL (1998) Biochemistry, 4th edition, Wm C Brown Publishers, Dubuque, Iowa. 6. Dubey RC (2013) A Text Book of Biotechnology 4th edition, S Chand & Co Ltd, New Delhi. 7. Wilson K, Walker J (2018) Principle & Techniques of Biochemistry and Molecular Biology, 8th edition, Cambridge University Press, Cambridge. 8. Bowsher C & Tobin A (2021) Plant Biochemistry, 2nd Edition, Taylor & Francis Inc, CRC press. 9. Arthur M. Lesk (2016) Introduction to Protein Science Architecture Function and Genomics, Oxford University press, London **Suggested websites** https://academic.oup.com/aob/article/105/7/1141/148741 https://dx.doi.org/10.1016/B978-0-12-800180-6.00001-3 https://doi:10.1016/j.molp.2016.05.004 https://doi.org/10.1093/jn/130.5.1081 https://academic.oup.com/jxb/article/68/10/2463/2967464 https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=2 https://epgp.inflibnet.ac.in/epgpdata/uploads/epgp_content/food_technology/food_chemistry/16. protein_ structure_and_denaturation/et/40_et_m16.pdf https://www.youtube.com/watch?v=MODnIkQvyz0 https://www.youtube.com/watch?v=AeVDoDp3llI
Learning Outcomes	Upon completion of this course, students will be able to understand and describe the biosynthesis of essential carbon compounds like sucrose, starch, and cellulose, and elucidate sulfur and phosphorus metabolism, including ATP synthesis. They will gain insights into nitrogen metabolism, focusing on nitrogen fixation and nitrate reduction. Students will be proficient in explaining DNA structure, replication, and repair, RNA transcription, processing, and translation, and protein synthesis and post-translational modifications. Additionally, they will be familiar with enzyme mechanisms and molecular techniques such as electrophoresis and molecular cloning, enabling practical applications in plant biochemistry and molecular biology.

Course Title	**BOTMN72: Lab Work Based on Paper BOTMN71**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Chromatographic separation and comparative analysis of photosynthetic pigments in plants • Chromatographic separation of amino acids	6.0 h
II	• Preparation of standard graph and estimation of DNA by diphenylamine test • Preparation of standard graph and estimation of RNA by Orcinol test	6.0 h
III	• Preparation of standard graph and estimation of protein by Bradford method • Study on the effects of light/dark on the nitrate reductase enzyme	6.0 h
IV	• To study the time dependent activity of urease enzyme • Study on the effects of different concentrations of urea on urease activity	6.0 h
V	• Determination of different concentrations of urease enzyme on its activity • Preparation of standard graph and estimation of carbohydrates by anthrone reagent method • Demonstration of electrophoretic and blotting instruments	6.0 h

Course Title	**BOTMN73: Plant Biotechnology and Genetic Engineering**			
Category of Course[76]	Major / **Minor**/ Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[77]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of plant biotechnology and genetic engineering, covering historical developments, model plants, and applications. Students will learn about recombinant DNA technology, including vectors, restriction enzymes, and genomic libraries. They will explore plant cell, tissue, and organ culture techniques, including totipotency, media preparation, and regeneration protocols. The course will cover plant transformation techniques, including both indirect and direct methods. Students will examine transgenic plants and their applications, gene expression regulation, and advanced gene editing technologies like ZFNs, TALENs, and CRISPR/Cas systems, addressing their applications, limitations, and ethical considerations.			

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Plant Biotechnology: Introduction, History, Model plants, Scope and applications • Recombinant DNA technology: Vectors, restriction enzymes, molecular probes, genomic and cDNA libraries	8.0 h
II	• Plant Cell, Tissue and Organ Culture (a) History, concept of totipotency, nutritional requirements of culture, mediapreparation, aseptic condition, applications and limitations (b) Regeneration Protocols: Single Cell Culture, Organogenesis and Somatic embryogenesis, Factors and significance (c) Protoplast Culture and Fusion: Protoplast isolation and culture, Protoplast fusion, methods and factors, applications	8.0 h
III	• Techniques for plant transformation *(a) Indirect gene transfer methods (Agrobacterium tumefaciens, and Agrobacterium rhizogenes mediated transfer)* *(b) Direct Gene transfer methods (electroporation, particle bombardment, microinjection, PEG, silicon carbide)*	8.0 h
IV	• Transgenic plants, some genetically modified plants (Flavr Savr tomato, Bt brinjal, Bt corn, Bt cotton, Golden rice, virus resistant plants, pest resistant plants) and molecular forming, applications and limitations (ethical and environmental issues)	9.0 h
V	• Gene expression and its regulation in plants: Transcription factors and cis-regulatory elements (CREs), epigenetic mechanisms of transcriptional regulation, translational control, programmed DNA rearrangements. • Gene editing and its applications: Zinc-finger nucleases (ZFNs), transcription • activator-like effector nucleases (TALENs), CRISPR/Cas systems	12.0 h

Texts/References	
	1. Bhojwani SS & Dantu PK (2013) Plant Tissue Culture: An Introductory Text, Springer. 2. Park S (2021) Plant Tissue Culture: Techniques and Experiments, Fourth Edition, Academic Press, Elsevier. 3. Razdan MK (2003) Introduction to Plant Tissue Culture, Science Publishers, USA 4. Slater A, Scott NW and Fowler MR (2008) Plant biotechnology: the genetic manipulation of plants, second edition, Oxford University Press. 5. Sahni S, Prasad BD and Kumar P (2017) Plant Biotechnology: Transgenics, Stress 6. Management and Biosafety Issues, CRC Press.

[76] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[77] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Learning Outcomes	Upon completing this course, students will be proficient in plant biotechnology and genetic engineering techniques. They will understand the historical development and applications of plant biotechnology, including recombinant DNA technology and plant cell culture methods. Students will gain expertise in various plant transformation techniques, both indirect and direct, and be familiar with the production and use of transgenic plants. They will also learn about gene expression regulation and advanced gene editing technologies, such as ZFNs, TALENs, and CRISPR/Cas systems. Additionally, students will address the applications, limitations, and ethical considerations of these biotechnological advances.

Course Title	**BOTMN74: Lab Work Based on Paper BOTMN73**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Medium preparation of Murashige and Skoog • Surface sterilization and inoculation of nodal explants on MS medium 6 h	6.0 h
II	• To raise suspension culture by using friable calli	6.0 h
III	• To initiate hairy root culture	6.0 h
IV	• Floral dip method of plant transformation (*Arabidopsis thaliana*) and screening for gene transfer (plasmid isolation & transformation, screening through PCR and agarose gel electrophoresis)	6.0 h
V	• Synthetic seed preparation	6.0 h

Course Title	**BOTMN76: Photoecophysiology**			
Category of Course[78]	Major/ **Minor** / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[79]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of photoecophysiology, focusing on light-regulated growth and development in cyanobacteria and plants. Students will explore the roles of photoreceptors, light quality, and light signaling in plant and microbial adaptation. Key topics include molecular mechanisms of photodamage and protection, light harvesting processes, and carbon concentrating mechanisms. Through this course, students will gain insights into how light influences physiological processes, signal transduction, and adaptation strategies in diverse photosynthetic organisms.			

Units	**Course Content**	**Hours of Teaching: 45 h**
I	• Light-regulated growth and development in cyanobacteria: Phytochrome-related photosensor and light perception, Photomorphogenesis, Complementary chromatic acclimation, Light quality and quantity dependent changes in light-harvesting complex, Signal transduction, Photoregulation of morphology	9.5 h
II	• Molecular mechanisms of photodamage and photoprotection: Photo-induced damage to microbes; genetical, biochemical and molecular aspects of mycosporine-like amino acids (MAAs) and scytonemin	9.5 h

[78] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[79] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	production, ecological and economical implications	
III	• Light-regulated plant growth and development: Photoreceptors and light perception, Physiological responses, Photomorphogenesis, Circadian clock, Signal transduction, Seed germination, Hypocotyl elongation	6.5 h
IV	• Light signalling dynamics in plants: Phytochromes, Cryptochromes, PIFs, Phototropins, UV resistance locus8 (UVR8), HY5, COP1, SUMOylation, Chloroplast movement • Light harvesting in plants and cyanobacteria: Photosynthetic pigments and phycobilisomes, State transition, photochemical quenching (PQ), non-photochemical quenching (NPQ), photoinhibition, orange carotenoid protein (OCP), xanthophyll cycle, principle of PAM fluorometry	13.0 h
V	• Carbon concentrating mechanisms (CCM) in plants and cyanobacteria: Photorespiration, Components of CCM, Pyrenoids, Carboxysome, RuBISCO, Carbonic anhydrase	6.5 h

Texts/References	1. Ruban, A., Foyer, C., & Murchie, E. (Eds.). (2022). Photosynthesis in Action: Harvesting Light, Generating Electrons, Fixing Carbon. Academic Press. 2. Taiz L, Zeiger E (2015) Plant Physiology and Development, 7th edition, Sinauer Associates Inc. Publishers, Sunderland. 3. Hopkins WG, Hunter NPA (2011) Introduction to Plant Physiology, Wiley International edition, John Wiley & Sons, New Jersey. 4. Buchanan B, GruissemW, Jones RL (2002) Biochemistry and Molecular Biology of Plants, American Society of Plant Biologists, Rockville. 5. Devlin R.M. (1983) Plant Physiology, Cengage Learning Inc., London. 6. Salisbury F.B., Ross C.W. (2004) Plant Physiology, Wadsworth, California
Learning Outcomes	Upon completing this course, students will have a thorough understanding of the principles and mechanisms underlying light-regulated growth and development in cyanobacteria and plants. They will be able to explain the roles of various photoreceptors and light signaling pathways in photomorphogenesis and adaptation. Students will gain insights into the molecular mechanisms of photodamage and photoprotection, including the production of protective compounds. They will also understand light harvesting processes, including the functions of photosynthetic pigments and carbon concentrating mechanisms. Overall, students will be equipped to analyze and interpret the effects of light on plant and microbial physiology.

Course Title	**BOTMN77: Lab Work Based on Paper BOTMN76**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Effects of light and dark on seed germination • Hypocotyl length measurements in light and dark grown seedlings	6.0 h
II	• Photosynthetic pigments analysis in light and dark grown seedlings • Growth measurement of cyanobacteria under white light, blue light, red light and UVR	6.0 h
III	• Extraction and quantification of photosynthetic pigments from cyanobacteria grown under different light conditions	6.0 h
IV	• Extraction and spectrophotometric analysis of mycosporine-like amino acids in cyanobacteria	6.0 h
V	• Extraction and spectrophotometric analysis of scytonemin in cyanobacteria	6.0 h

SEMESTER-VIII

<table>
<tr><td>Course Title</td><td colspan="4">BOTMJ81: Microbiology</td></tr>
<tr><td>Category of Course[80]</td><td colspan="4">Major/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation</td></tr>
<tr><td rowspan="3">Credits[81]& Hour of Teaching</td><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr>
<tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr>
<tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr>
<tr><td>Course Objectives</td><td colspan="4">This course aims to provide a comprehensive understanding of microbiology, covering the fundamentals of Archaea and Bacteria, including key genera such as Halobacteria and Agrobacterium. Students will explore microbial growth, nutrition, and energy metabolism, and delve into the structure and lifecycle of various viruses. The course will examine modes of genetic exchange, operon regulation, and quorum sensing. Additionally, students will gain insights into the applications of microbes in antibiotics production, bioremediation, and fermentation technology. The course will also address nitrogen fixation processes and Rhizobium-legume interactions, emphasizing their impact on soil fertility and agricultural productivity.</td></tr>
</table>

Units	Course Content	Hours of Teaching:45 h
I	• Introduction to Archaea and Bacteria: A general account of *Halobacteria, Thermoplasma, Agrobacterium* • Growth and nutrition: Growth kinetics, Batch and continuous cultures, Nutritional classification of microorganisms, Assimilation of Sulfur and phosphate	10.0 h
II	• Energy Metabolism: Autotrophs and heterotrophs, hydrogen bacteria, purple non-sulphur bacteria and cyanobacteria • A detailed account of different types of viruses; structure of bacteriophages, LPP1, TMV and retroviruses. Lytic cycle in T4 phages and its regulation; lysogeny and its regulation in lambda phage; a brief account of viroids and prions	11.0 h
III	• Modes of genetic exchange in microbes: Transformation, Transduction, Conjugation, Evolutionary Significance • Repression and activation of Lac operon	7.0 h
IV	• Quorum sensing: Role of Quorum sensing in bioluminescence and virulence • Role of Microbes in preparation of antibiotics, secondary metabolites, bioremediation and soil fertility	8.0 h
V	• Fermentation technology for production of lactic acid and acetic acid • Nitrogen fixation by free-living and symbiotic microorganisms, Rhizobium-legume interaction, genetics of nodulation	9.0 h

Texts/References	1. Michael T. Madigan, Kelly S. Bender, Daniel H. Buckley, W. Matthew Sattley and David A. Stahl (2019). Brock Biology of Microorganisms, 15th edition (Global Edition, 15/E) 2. Stanier RY, Ingraham JL, Wheelis ML, Painter RR (1992). General Microbiology, 5th edition, MacMillan, Hampshire & London. 3. Talaro KP, Chess B (2011). Foundation in Microbiology, 8th edition, MacGraw-Hill, New York. 4. Dubey, R.C. and Maheshwari, D. K. (1999, reprint 2016). A text book of Microbiology(1stEdn), S. Chand & company, Pvt. Ltd., New Delhi, India

[80] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[81] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	5. Frobisher, Hinsdill, Crabtree and Goodheart (1974). Fundamentals of Microbiology. Saunders' International Students Edition. Toppan Company Ltd. Singapore 6. Joanne M. Willey, Linda M. Sherwood, C.J. Woolverton (2017). Prescott's Microbiology. 10th Edition. McGraw Hills. International Edition. Singapore. 7. Michael J. Pelczer JR, ECS Chan and NoelR. Krieg (1986, reprint 2011). Microbiology, Vth Edition, Tata McGraw Hills, India
Learning Outcomes	Upon completing this course, students will be equipped with a thorough understanding of microbial diversity, including Archaea, Bacteria, and various viruses. They will grasp key concepts in microbial growth, nutrition, and energy metabolism, and will be familiar with microbial genetic exchange and operon regulation. Students will understand the role of quorum sensing in microbial behavior and its applications in bioluminescence and virulence. They will also acquire practical knowledge in microbial applications, such as antibiotics production, bioremediation, and fermentation technologies, and will comprehend the processes and significance of nitrogen fixation and Rhizobium-legume interactions in soil fertility.

Course Title	**BOTMJ82: Lab Work Based on Paper BOTMJ81**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Staining of microorganisms and endospore • Preparation of culture medium, sterilization and cultivation of bacteria	5.0 h
II	• Study of growth of unicellular microorganism by turbidometric method • Methylene blue reduction test of milk	10.0 h
III	• Demonstration of certain metabolic activities: *Fermentation of carbohydrates, Catalase activity*	5.0 h
IV	• Determination of minimum inhibitory concentrations and LC_{50} of antibiotics.	5.0 h
V	• Determination of beta-galactosidase activity for degradation of ONPG in *E. coli* grown in different combination of sugar	5.0 h

Course Title	**BOTMJ83: Stress Biology**			
Category of Course[82]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[83] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of stress biology in extremophiles, cyanobacteria, and plants. Students will explore the molecular phylogenetics of these organisms, mechanisms of reactive oxygen species (ROS) generation and control, and the impact of abiotic stresses such as drought, salinity, and temperature. Emphasis will be placed on the photosynthetic system's adaptations, functional genomics, proteomics, and metabolomics related to stress responses. Additionally, the course covers gene mining for stress tolerance and the application of green chemistry for biofuel and bioactive compound production using photosynthetic organisms.			

[82] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[83] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Units	Course Content	Hours of Teaching: 45 h
I	• Distribution of extremophiles, cyanobacteria and plants across environmental gradients, molecular phylogenetic analysis • Reactive oxygen species (ROS), generation, impacts on nucleic acid, proteins, lipids and its molecular control and anti-oxidant processe	11.0 h
II	• Abiotic stress tolerance and signaling: Drought, salinity, ultraviolet radiation, light, temperatures, desiccation and heavy metals, HSPs	9.0 h
III	• Photosynthetic system: light harvesting complex, chlorophyll, phycobiliproteins, carotenoids and light harvesting proteins (LHCs), genetic regulation and adjustments during abiotic stresses.	9.0 h
IV	• Functional genomics, proteomics and metabolomics of cyanobacteria and plants with reference to stress responsive gene/proteins	5.5 h
V	• Gene mining from cyanobacterial and plant model system for the development of stress tolerant crops • Green chemistry using photosynthetic organisms for production of biofuels and bioactive compounds	10.5 h

Texts/References	1. Bryant DA (1995) The Molecular Biology of Cyanobacteria, Kluwer Academic Publisher, Berlin. 2. WhittonBA, Potts M (2000) Ecology of Cyanobacteria - Their diversity in Time and Space, Kluwer Academic Publishers, Berlin. 3. Walters, Dale, Adrian Newton, and Gary Lyon. Induced resistance for plant defence. Blackwell, 2007. 4. Burdon, Roy H. *Genes and the Environment*. CRC Press, 1999. 5. Buchanan, Bob B., Wilhelm Gruissem, and Russell L. Jones, eds. Biochemistry and molecular biology of plants. John wiley & sons, 2015. 6. Lazar, Thomas. "Taiz, L. and Zeiger, E. Plant physiology. 3rd edn." (2003): 750-751. 7. Salisbury, F. B., and Ross, C. W. (1992). Plant Physiology, Wadsworth Pub. *Com., Inc., Belmont, California-USA.*
Learning Outcomes	Upon completing this course, students will gain a comprehensive understanding of stress biology in extremophiles, cyanobacteria, and plants. They will be proficient in analyzing reactive oxygen species (ROS) and their impact on cellular macromolecules, mastering mechanisms of abiotic stress tolerance, and understanding the adaptations of photosynthetic systems under stress. Students will also be skilled in utilizing functional genomics, proteomics, and metabolomics to investigate stress responses, apply gene mining for developing stress-tolerant crops, and leverage green chemistry for biofuel and bioactive compound production.

Course Title	**BOTMJ84: Lab Work Based on Paper BOTMJ83**

Units	Practicals	Hours of Teaching:30 h
I	• Demonstration of CYANOBASE, Phytozome, Tair, and retrieving of gene sequences for multiple sequence alignment for candidate gene for making phylogenetic tree using (MEGA 5.0, CLUSTAL X etc) • Determination of LC_{50} doses of selected abiotic stresses in cyanoacterium *Anabaena* sp PCC 7120 and *Arabidopsis thaliana*	10.0 h
II	• Determination of lipid per-oxidation and hydrogen peroxide content in stressed and non-stressed photosynthetic organisms • SDS-PAGE profiling of controlled and selected abiotic stress exposed model photosynthetic organisms in order to monitor stressed induced proteins	5.0 h
III	• DNA isolation, primer designing and PCR amplification of 16S and 18S rDNA sequences from photosynthetic organisms	5.0 h
IV	• Lipid assay from stressed and non-stressed photosynthetic organisms	5.0 h

V	• Study of fluorescence spectra of control and stressed model cyanobacterium and plant using PAM fluorometry	5.0 h

Course Title	**BOTMJ85: Ecology, Environmental Pollution and Toxicology**
Category of Course[84]	Major/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation
Credits[85] **& Hour of Teaching**	Theory / Practical / Cumulative — Credits: 3 / 1 / 4; Hour of Teaching: 45 / 30 / 75
Course Objectives	The course aims to provide a comprehensive understanding of ecological principles, environmental pollution, and toxicology. Students will explore population dynamics, community structures, ecosystem functions, and ecological succession. They will analyze the sources and impacts of water, air, and soil pollution on ecosystems, and gain insights into ecotoxicology, including toxicity types, dose-response relationships, and biomagnification. This course equips students with the knowledge to assess environmental challenges and develop strategies for mitigating pollution and its effects on ecological systems.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	Course Content	Hours of Teaching: 45 h
I	• Population: Patterns of population dispersion, population growth, *r* and *k*-selection strategies, population regulation, mechanisms of differentiation • Community: Continuum and discreet community concepts, community characteristics and their analyses, species diversity and indices, concept of ecological niche	10.5 h
II	• Ecosystem: Concept, components and organization; primary productivity and its measurement; energy flow • Ecological succession: Mechanisms; individualistic and holistic models of succession; trends of succession, concept of climax	9.5 h
III	• Water Pollution: Sources and kinds, impact of pollution on aquatic ecosystems, eutrophication	7.5 h
IV	• Air Pollution: Sources and kinds, impact of air pollutants (SO_2, NO_2 and particulate) on plants and ecosystems • Soil Pollution: Sources and kinds, impact of heavy metals and pesticides on plants and ecosystems	9.0 h
V	• Ecotoxicology: Concept of toxicity, types of toxicants, dose-response relationship, chronic and acute toxicity, indices of toxicity, factors affecting toxicity; biomagnification	8.5 h

Texts/References

1. Klassen C, Watkins JB (2010) Essentials of Toxicolgy, 2nd edition, McGraw- Hill Companies, New York.
2. Singh JS, Singh SP, Gupta SR (2014) Ecology Environmental Science and Conservation, S Chand & Co, New Delhi.
3. Begon, M., Townsend, C. R. 2021) Ecology: From Individuals to Ecosystems. United Kingdom: Wiley.
4. Bradshaw RHW and Sykes MT. 2014. Ecosystem Dynamics: From the Past to the Future. United Kingdom: Wiley.
5. Molles Jr MC and Laursen A. 2020. Ecology: Concepts & Applications Author(s): 5th Canadian Edition McGraw-Hill
6. Mueller-Dombois D and Ellenberg 1974. Aims and Methods of Vegetation Ecology. John Wiley & Sons. New York
7. Odum Eugene Pleasants 1967. Fundamentals of Ecology. United States: Saunders.

[84] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[85] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	8. Raffaelli DG, Frid CLJ, 2010. Ecosystem Ecology: A New Synthesis. Cambridge University Press
Learning Outcomes	Upon completing this course, students will be able to understand and analyze various ecological concepts, including population dynamics, community structures, and ecosystem functions. They will gain insights into ecological succession and primary productivity. Students will also be proficient in identifying sources and impacts of water, air, and soil pollution, and assessing their effects on ecosystems. Additionally, they will acquire knowledge in ecotoxicology, including the types of toxicants, dose-response relationships, and the concept of biomagnification. This comprehensive understanding will enable students to address environmental challenges and contribute to pollution mitigation and ecological conservation efforts.

Course Title	**BOTMJ86: Lab Work Based on Paper BOTMJ85**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Determination of the minimal size of sample plot for vegetation analysis by species-area curve method	6.0 h
II	• Study of frequency of herbaceous species in grassland and to compare the frequency distribution with Raunkiaer's Standard Frequency Diagram. • Estimation of the density and abundance of species in protected and grazed grasslands	6.0 h
III	• Estimation of Importance Value Index (IVI) for grassland species on the basis of frequency, density and biomass • Determination of water holding capacity of polluted soil samples	6.0 h
IV	• Determination of foliar dust deposition in samples collected from different sites exposed to air pollution • Assessment of leaf area injury in foliar samples collected from sites exposed to air pollution	6.0 h
V	• Measurement of the dissolved oxygen content of different water samples • Estimation of the total alkalinity of different water samples	6.0 h

Course Title	**BOTMJ87: Plant Diseases and Protection**		
Category of Course[86]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation		
Credits[87] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	The objective of this course is to provide students with a comprehensive understanding of plant diseases, including historical epidemics, physiology of parasitism, and the role of effectors and elicitors in disease development. Students will learn about factors influencing disease progression and study specific diseases such as green ear disease and rust. The course will cover the molecular mechanisms of plant immunity, pathogen virulence, and modern engineering techniques for disease resistance, including RNA interference and gene editing. Additionally, students will gain insights into epidemiology, disease management strategies, and the importance of biosecurity and intellectual property in plant protection.		

[86] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[87] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Units	Course Content	Hours of Teaching: 45 h
I	• Historical account of major plant disease epidemics • Physiology of Parasitism and role of effectors in host-pathogen interaction and disease development • Role of elicitor (enzyme, toxin, growth regulator, and polysaccharides) in plant disease development	10.5 h
II	• Factors regulating disease development and disease cycle of following diseases: Green ear disease of bajara, downey mildew of cucurbits, rust of linseed, smut of bajara , wilt of Arhar, root knot nematodes, viral/viriods disease, little leaf of brinjal	15.0 h
III	• Molecular basis of plant Immunity and pathogen virulence: Effector-triggered immunity (ETI), Pattern-triggered immunity (PTI), and pathogen-associated molecular patterns (PAMPs)	10.5 h
IV	• Engineering for disease resistance: RNA interferences, *Agrobacterium* mediated plant transformation, gene editing	4.5 h
V	• Epidemiology, disease spread and management: Biosecurity, Intellectual property rights (IPR) and Intellectual Property Management (IPM)	4.5 h

Texts/References	1. Agrios GN (2005) Plant Pathology (Vth edition) Academic Press, Burlington. 2. Dickinson CM (2003) Molecular Plant Pathology, Bios Scientific Publisher, Oxford. 3. Robert N, Trigiano, Windham MT, Windham AS (2003) Plant Pathology: Concepts and Laboratory Exercises, CRC Press, Boca Raton, USA. 4. Singh RS (2008) Plant Diseases, Oxford and IBH Publishing Co Pvt Ltd, New Delhi. 5. Singh RS (2008) Principles of Plant Pathology, Oxford and IBH Publishing Co Pvt Ltd, New Delhi. 6. Dhingra OD, James B, Sinclair (1995) Basic Plant Pathology Methods, CRC Publication, Boca Raton, USA. 7. PD Sharma (2020) Microbiology and Plant Pathology 4th Ed, Rastogi Publications. 8. R Maheshwary (2016) Fungi: Experimental Methods In Biology, Second Edition. CRC Press. 9. Anne Marte Tronsmo, David B. Collinge, Annika Djurle, Lisa Munk, Jonathan Yuen, Arne Tronsmo (2020). Plant pathology and plant diseases, CABI Publication.
Learning Outcomes	Upon completing this course, students will understand the historical and physiological aspects of plant diseases, including the role of effectors and elicitors in disease development. They will gain knowledge of disease-specific factors and cycles for various plant diseases and grasp the molecular mechanisms underlying plant immunity and pathogen virulence. Students will also learn about advanced techniques for engineering disease resistance, such as RNA interference and gene editing. Furthermore, they will be equipped with insights into epidemiology, disease management strategies, biosecurity, and intellectual property rights related to plant protection.

Course Title	**BOTMJ88: Lab Work Based on Paper BOTMJ87**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Study of symptoms and cause of following plant diseases, namely wart of potato, downy mildew of cucurbits, wart of sesame, stem gall on coriander, linseed rust, tikka disease of groundnut, red rot of sugarcane, bacterial blight of rice, yellow vein mosaic of bhindi, and other local diseases on crops	10.0 h
II	• Preparation of media and slants for isolation of pathogens	5.0 h
III	• *In-vitro* Isolation of pathogenic fungi • Identification of pathogenic organism of local disease using Koch's postulates	5.0 h
IV	• To Study synergism/antagonism between pathogen in culture medium	5.0 h
V	• Symptoms of bacterial blight, mosaic and little leaf of brinjal	5.0 h

Course Title	BOTMN81: Microbiology
Category of Course[88]	Major/ **Minor** / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation
Credits[89] & Hour of Teaching	Theory / Practical / Cumulative — Credits: 3 / 1 / 4; Hour of Teaching: 45 / 30 / 75
Course Objectives	This course aims to provide a comprehensive understanding of microbiology, covering the fundamentals of Archaea and Bacteria, including key genera such as Halobacteria and Agrobacterium. Students will explore microbial growth, nutrition, and energy metabolism, and delve into the structure and lifecycle of various viruses. The course will examine modes of genetic exchange, operon regulation, and quorum sensing. Additionally, students will gain insights into the applications of microbes in antibiotics production, bioremediation, and fermentation technology. The course will also address nitrogen fixation processes and Rhizobium-legume interactions, emphasizing their impact on soil fertility and agricultural productivity.

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Units	Course Content	Hours of Teaching:45 h
I	• Introduction to Archaea and Bacteria: A general account of *Halobacteria, Thermoplasma, Agrobacterium* • Growth and nutrition: Growth kinetics, Batch and continuous cultures, Nutritional classification of microorganisms, Assimilation of Sulfur and phosphate	10.0 h
II	• Energy Metabolism: Autotrophs and heterotrophs, hydrogen bacteria, purple non-sulphur bacteria and cyanobacteria • A detailed account of different types of viruses; structure of bacteriophages, LPP1, TMV and retroviruses. Lytic cycle in T4 phages and its regulation; lysogeny and its regulation in lambda phage; a brief account of viroids and prions	11.0 h
III	• Modes of genetic exchange in microbes: Transformation, Transduction, Conjugation, Evolutionary Significance • Repression and activation of Lac operon	7.0 h
IV	• Quorum sensing: Role of Quorum sensing in bioluminescence and virulence • Role of Microbes in preparation of antibiotics, secondary metabolites, bioremediation and soil fertility	8.0 h
V	• Fermentation technology for production of lactic acid and acetic acid • Nitrogen fixation by free-living and symbiotic microorganisms, Rhizobium-legume interaction, genetics of nodulation	9.0 h

Texts/References

1. Michael T. Madigan, Kelly S. Bender, Daniel H. Buckley, W. Matthew Sattleyand David A. Stahl (2019). Brock Biology of Microorganisms, 15th edition (Global Edition, 15/E).
2. Stanier RY, Ingraham JL, Wheelis ML, Painter RR (1992). General Microbiology, 5th edition, MacMillan, Hampshire & London.
3. Talaro KP, Chess B (2011). Foundation in Microbiology, 8th edition, MacGraw-Hill, New York.
4. Dubey, R.C. and Maheshwari, D. K. (1999, reprint 2016). A text book of Microbiology(1stEdn), S. Chand & company, Pvt. Ltd., New Delhi, India
5. Frobisher, Hinsdill, Crabtree and Goodheart (1974). Fundamentals of Microbiology. Saunders' International Students Edition. Toppan Company Ltd. Singapore
6. Joanne M. Willey, Linda M. Sherwood, C.J. Woolverton (2017). Prescott's Microbiology. 10thEdition. McGraw Hills. International Edition. Singapore.

[88] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[89] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	7. Michael J. Pelczer JR, ECS Chan and NoelR. Krieg (1986, reprint 2011). Microbiology, Vth Edition, Tata McGraw Hills, India
Learning Outcomes	Upon completing this course, students will be equipped with a thorough understanding of microbial diversity, including Archaea, Bacteria, and various viruses. They will grasp key concepts in microbial growth, nutrition, and energy metabolism, and will be familiar with microbial genetic exchange and operon regulation. Students will understand the role of quorum sensing in microbial behavior and its applications in bioluminescence and virulence. They will also acquire practical knowledge in microbial applications, such as antibiotics production, bioremediation, and fermentation technologies, and will comprehend the processes and significance of nitrogen fixation and Rhizobium-legume interactions in soil fertility.

Course Title	**BOTMN82: Lab Work Based on Paper BOTMN81**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Staining of microorganisms and endospore • Preparation of culture medium, sterilization and cultivation of bacteria	5.0 h
II	• Study of growth of unicellular microorganism by turbidometric method • Methylene blue reduction test of milk	1.0.0 h
III	• Demonstration of certain metabolic activities: *Fermentation of carbohydrates, Catalase activity*	5.0 h
IV	• Determination of minimum inhibitory concentrations and LC_{50} of antibiotics	5.0 h
V	• Determination of beta-galactosidase activity for degradation of ONPG in *E. coli* grown in different combination of sugar	5.0 h

Course Title	**BOTMN83: Stress Biology**
Category of Course[90]	Major/ **Minor**/ Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation
Credits[91] & Hour of Teaching	

	Theory	Practical	Cumulative
Credits	3	1	4
Hour of Teaching	45	30	75

Course Objectives	This course aims to provide a comprehensive understanding of stress biology in extremophiles, cyanobacteria, and plants. Students will explore the molecular phylogenetics of these organisms, mechanisms of reactive oxygen species (ROS) generation and control, and the impact of abiotic stresses such as drought, salinity, and temperature. Emphasis will be placed on the photosynthetic system's adaptations, functional genomics, proteomics, and metabolomics related to stress responses. Additionally, the course covers gene mining for stress tolerance and the application of green chemistry for biofuel and bioactive compound production using photosynthetic organisms.

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Distribution of extremophiles, cyanobacteria and plants across environmental gradients, molecular phylogenetic analysis • Reactive oxygen species (ROS), generation, impacts on nucleic acid, proteins, lipids and its molecular control and anti-oxidant processes	11.0 h
II	• Abiotic stress tolerance and signaling: Drought, salinity, ultraviolet radiation, light, temperatures, desiccation and heavy metals	9.0 h
III	• Photosynthetic system: light harvesting complex, chlorophyll, phycobiliproteins, carotenoids and light harvesting proteins (LHCs), genetic regulation and adjustments during abiotic stresses	9.0 h
IV	• Functional genomics, proteomics and metabolomics of cyanobacteria and plants with reference to stress responsive gene/proteins	5.5 h
V	• Gene mining from cyanobacterial and plant model system for the development of stress tolerant crops • Green chemistry using photosynthetic organisms for production of biofuels and bioactive compounds	10.5 h

Texts/References	1. Bryant DA (1995) The Molecular Biology of Cyanobacteria, Kluwer Academic Publisher, Berlin. 2. WhittonBA**,** Potts M (2000) Ecology of Cyanobacteria **-** Their diversity in Time and Space, Kluwer Academic Publishers, Berlin. 3. Walters, Dale, Adrian Newton, and Gary Lyon. Induced resistance for plant defence. Blackwell, 2007. 4. Burdon, Roy H. *Genes and the Environment*. CRC Press, 1999. 5. Buchanan, Bob B., Wilhelm Gruissem, and Russell L. Jones, eds. Biochemistry and molecular biology of plants. John wiley & sons, 2015. 6. Lazar, Thomas. "Taiz, L. and Zeiger, E. Plant physiology. 3rd edn." (2003): 750-751. 7. Salisbury, F. B., and Ross, C. W. (1992). Plant Physiology, Wadsworth Pub. *Com., Inc., Belmont, California-USA*.
Learning Outcomes	Upon completing this course, students will gain a comprehensive understanding of stress biology in extremophiles, cyanobacteria, and plants. They will be proficient in analyzing reactive oxygen species (ROS) and their impact on cellular macromolecules, mastering mechanisms of abiotic stress tolerance, and understanding the adaptations of photosynthetic systems under stress. Students will also be skilled in utilizing functional genomics, proteomics, and metabolomics to investigate stress responses, apply gene mining for developing stress-tolerant crops, and leverage green chemistry for biofuel and bioactive compound production.

[90] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[91] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Title	BOTMN84: Lab Work Based on Paper BOTMN83	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Demonstration of CYANOBASE, Phytozome, Tair, and retrieving of gene sequences for multiple sequence alignment for candidate gene for making phylogenetic tree using (MEGA 5.0, CLUSTAL X etc) • Determination of LC_{50} doses of selected abiotic stresses in cyanoacterium *Anabaena* sp PCC 7120 and *Arabidopsis thaliana*	10.0 h
II	• Determination of lipid per-oxidation and hydrogen peroxide content in stressed and non-stressed photosynthetic organisms • SDS-PAGE profiling of controlled and selected abiotic stress exposed model photosynthetic organisms in order to monitor stressed induced proteins	5.0 h
III	• DNA isolation, primer designing and PCR amplification of 16S and 18S rDNA sequences from photosynthetic organisms	5.0 h
IV	• Lipid assay from stressed and non-stressed photosynthetic organisms	5.0 h
V	• Study of fluorescence spectra of control and stressed model cyanobacterium and plant using PAM fluorometry	5.0 h

Course Title	BOTMN85: Ecology,Pollution and Toxicology			
Category of Course[92]	Major/ **Minor**/ Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[93] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	The course aims to provide a comprehensive understanding of ecological principles, environmental pollution, and toxicology. Students will explore population dynamics, community structures, ecosystem functions, and ecological succession. They will analyze the sources and impacts of water, air, and soil pollution on ecosystems, and gain insights into ecotoxicology, including toxicity types, dose-response relationships, and biomagnification. This course equips students with the knowledge to assess environmental challenges and develop strategies for mitigating pollution and its effects on ecological systems.			

Units	Course Content	Hours of Teaching:45 h
I	• Population: Patterns of population dispersion, population growth, *r* and *k*-selection strategies, population regulation, mechanisms of differentiation • Community: Continuum and discreet community concepts, community characteristics and their analyses, species diversity and indices, concept of ecological niche	10.5 h
II	• Ecosystem: Concept, components and organization; primary productivity and its measurement; energy flow • Ecological succession: Mechanisms; individualistic and holistic models	9.5 h

[92] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[93] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	of succession; trends of succession, concept of climax	
III	• Water Pollution: Sources and kinds, impact of pollution on aquatic ecosystems, eutrophication	7.5 h
IV	• Air Pollution: Sources and kinds, impact of air pollutants (SO_2, NO_2 and particulate) on plants and ecosystems • Soil Pollution: Sources and kinds, impact of heavy metals and pesticides on plants and ecosystems	9.0 h
V	• Ecotoxicology: Concept of toxicity, types of toxicants, dose-response relationship, chronic and acute toxicity, indices of toxicity, factors affecting toxicity; biomagnification	8.5 h

Texts/References	1. Klassen C, Watkins JB (2010) Essentials of Toxicolgy, 2nd edition, McGraw- Hill Companies, New York. 2. Singh JS, Singh SP, Gupta SR (2014) Ecology Environmental Science and Conservation, S Chand & Co, New Delhi. 3. Begon, M., Townsend, C. R. 2021) Ecology: From Individuals to Ecosystems. United Kingdom: Wiley. 4. Bradshaw RHW and Sykes MT. 2014. Ecosystem Dynamics: From the Past to the Future. United Kingdom: Wiley. 5. Molles Jr MC and Laursen A. 2020. Ecology: Concepts & Applications Author(s): 5th Canadian Edition McGraw-Hill 6. Mueller-Dombois D and Ellenberg 1974. Aims and Methods of Vegetation Ecology. John Wiley & Sons. New York 7. Odum Eugene Pleasants 1967. Fundamentals of Ecology. United States: Saunders. 8. Raffaelli DG, Frid CLJ, 2010. Ecosystem Ecology: A New Synthesis. Cambridge University Press
Learning Outcomes	Upon completing this course, students will be able to understand and analyze various ecological concepts, including population dynamics, community structures, and ecosystem functions. They will gain insights into ecological succession and primary productivity. Students will also be proficient in identifying sources and impacts of water, air, and soil pollution, and assessing their effects on ecosystems. Additionally, they will acquire knowledge in ecotoxicology, including the types of toxicants, dose-response relationships, and the concept of biomagnification. This comprehensive understanding will enable students to address environmental challenges and contribute to pollution mitigation and ecological conservation efforts.

Course Title	**BOTMN86: Lab Work Based on Paper BOTMN85**

Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Determination of the minimal size of sample plot for vegetation analysis by species-area curve method	6.0 h
II	• Study of frequency of herbaceous species in grassland and to compare the frequency distribution with Raunkiaer's Standard Frequency Diagram • Estimation of the density and abundance of species in protected and grazed grasslands	6.0 h
III	• Estimation of Importance Value Index (IVI) for grassland species on the basis of frequency, density and biomass • Determination of water holding capacity of polluted soil samples	6.0 h
IV	• Determination of foliar dust deposition in samples collected from different sites exposed to air pollution • Assessment of leaf area injury in foliar samples collected from sites exposed to air pollution	6.0 h

V	• Measurement of the dissolved oxygen content of different water samples • Estimation of the total alkalinity of different water samples	6.0 h

Course Title	**BOTMN87: Plant Diseases and Protection**
Category of Course[94]	Major/ **Minor** / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation
Credits[95] **& Hour of Teaching**	| Theory | Practical | Cumulative Credits | 3 | 1 | 4 Hour of Teaching | 45 | 30 | 75
Course Objectives	The objective of this course is to provide students with a comprehensive understanding of plant diseases, including historical epidemics, physiology of parasitism, and the role of effectors and elicitors in disease development. Students will learn about factors influencing disease progression and study specific diseases such as green ear disease and rust. The course will cover the molecular mechanisms of plant immunity, pathogen virulence, and modern engineering techniques for disease resistance, including RNA interference and gene editing. Additionally, students will gain insights into epidemiology, disease management strategies, and the importance of biosecurity and intellectual property in plant protection.

Units	Course Content	Hours of Teaching:45 h
I	• Historical account of major plant disease epidemics • Physiology of Parasitism and role of effectors in host-pathogen interaction and disease development • Role of elicitor (enzyme, toxin, growth regulator, and polysaccharides) in plant disease development	10.5 h
II	• Factors regulating disease development and disease cycle of following diseases: Green ear disease of bajara, downey mildew of cucurbits, rust of linseed, smut of bajara , wilt of Arhar, root knot nematodes, viral/viriods disease, little leaf of brinjal	15.0 h
III	• Molecular basis of plant Immunity and pathogen virulence: Effector-triggered immunity (ETI), Pattern-triggered immunity (PTI), and pathogen-associated molecular patterns (PAMPs)	10.5 h
IV	• Engineering for disease resistance: RNA interferences, *Agrobacterium* mediated plant transformation, gene editing,	4.5 h
V	• Epidemiology, disease spread and management: Biosecurity, Intellectual property rights (IPR) andIntellectual Property Management (IPM)	4.5 h

Texts/References	1. Agrios GN (2005) Plant Pathology (Vth edition) Academic Press, Burlington. 2. Dickinson CM (2003) Molecular Plant Pathology, Bios Scientific Publisher, Oxford. 3. Robert N, Trigiano, Windham MT, Windham AS (2003) Plant Pathology: Concepts and Laboratory Exercises, CRC Press, Boca Raton, USA. 4. Singh RS (2008) Plant Diseases, Oxford and IBH Publishing Co Pvt Ltd, New Delhi. 5. Singh RS (2008) Principles of Plant Pathology, Oxford and IBH Publishing Co Pvt Ltd, New Delhi. 6. Dhingra OD, James B, Sinclair (1995) Basic Plant Pathology Methods, CRC Publication, Boca Raton, USA. 7. PD Sharma (2020) Microbiology and Plant Pathology 4th Ed, Rastogi Publications. 8. R Maheshwary (2016) Fungi: Experimental Methods In Biology, Second Edition. CRC Press.

[94] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[95] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	9. Anne Marte Tronsmo, David B. Collinge, Annika Djurle, Lisa Munk, Jonathan Yuen, Arne Tronsmo (2020). Plant pathology and plant diseases, CABI Publication.
Learning Outcomes	Upon completing this course, students will understand the historical and physiological aspects of plant diseases, including the role of effectors and elicitors in disease development. They will gain knowledge of disease-specific factors and cycles for various plant diseases and grasp the molecular mechanisms underlying plant immunity and pathogen virulence. Students will also learn about advanced techniques for engineering disease resistance, such as RNA interference and gene editing. Furthermore, they will be equipped with insights into epidemiology, disease management strategies, biosecurity, and intellectual property rights related to plant protection.

Course Title	**BOTMN88: Lab Work Based on Paper BOTMN87**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Study of symptoms and cause of following plant diseases, namely wart of potato, downy mildew of cucurbits, wart of sesame, stem gall on coriander, linseed rust, tikka disease of groundnut, red rot of sugarcane, bacterial blight of rice, yellow vein mosaic of bhindi, and other local diseases on crops	10.0 h
II	• Preparation of media and slants for isolation of pathogens	5.0 h
III	• *In-vitro* Isolation of pathogenic fungi. • Identification of pathogenic organism of local disease using Koch's postulates	5.0 h
IV	• To Study synergism /antagonism between pathogen in culture medium	5.0 h
V	• Symptoms of bacterial blight, mosaic and little leaf of brinjal	5.0 h

Honours with Research Degree

SEMESTER-VII

Course Title	**BOTMJ71R: Plant Biochemistry and Molecular Biology**			
Category of Course[96]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[97] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	The course objective is to provide students with a comprehensive understanding of plant biochemistry and molecular biology. Students will learn the biosynthesis of key carbon compounds, including sucrose, starch, and cellulose, and delve into sulfur and phosphorus metabolism, including ATP synthesis. They will gain insight into nitrogen metabolism, including the biochemistry of nitrogen fixation and nitrate reduction. The course will cover the structure and replication of DNA, RNA transcription and processing, and protein synthesis and modification. Additionally, students will explore enzyme mechanisms and molecular tools and techniques such as electrophoresis and cloning vectors.			

[96] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[97] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Units	Course Content	Hours of Teaching:45 h
I	• Biosynthesis of carbon compounds: sucrose, starch, cellulose • Sulphur and phosphorus metabolism: Activation and assimilation of sulphur, energy-rich phosphorus compounds; ATP synthesis	8.0 h
II	• Nitrogen metabolism: components, types and substrate of enzyme nitrogenase, biochemistry of nitrogen fixation; Nitrate metabolism: Uptake and reduction into ammonia, ammonia assimilation	8.0 h
III	• DNA: Structure and properties of different forms of DNA, DNA replication in prokaryotes and eukaryotes, DNA damage and repair	8.0 h
IV	• RNA:Structure and properties of different forms of RNA, transcription in prokaryotes and eukaryotes, mRNA processing-5' cap formation-3'end processing and polyadenylation, splicing, nuclear export of mRNA	8.0 h
V	• Protein: Basic aspects of protein conformation, translation in prokaryotes and eukaryotes (activation of amino acids, initiation, elongation, termination and release of peptides), post-translational modification of proteins • Enzymes: Mechanism of enzyme action, coenzymes, allosteric enzyme, isozymes • Molecular tools and techniques: Basic principles of electrophoresis and blotting, cloning vectors, molecular cloning of nucleic acid fragments	13.0 h

Texts/References	1. Berg JM et al. (2018) Biochemistry, 8th edition, WH Freeman & Company, New York. 2. Cox MM, Nelson DL (2021) Lehninger Principle of Biochemistry, 8th edition MacMillan, Worth Publishers,London. 3. Devlin TM (2011) Textbook of Biochemistry, 7th edition, John Wiley & Sons, Inc, New York. 4. Voet D, Voet JG and Pratt CW (2018) Biochemistry, 5th Edition, John Wiley & Sons, Inc, New York. 5. Zubay GL (1998) Biochemistry, 4th edition, Wm C Brown Publishers, Dubuque, Iowa. 6. Dubey RC (2013) A Text Book of Biotechnology 4th edition, S Chand & Co Ltd, New Delhi. 7. Wilson K, Walker J (2018) Principle & Techniques of Biochemistry and Molecular Biology, 8th edition, Cambridge University Press, Cambridge. 8. Bowsher C & Tobin A (2021) Plant Biochemistry, 2nd Edition, Taylor & Francis Inc, CRC press. 9. Arthur M. Lesk (2016) Introduction to Protein Science Architecture Function and Genomics,Oxford University press, London **Suggested websites** https://academic.oup.com/aob/article/105/7/1141/148741 https://dx.doi.org/10.1016/B978-0-12-800180-6.00001-3 https://doi:10.1016/j.molp.2016.05.004 https://doi.org/10.1093/jn/130.5.1081 https://academic.oup.com/jxb/article/68/10/2463/2967464 https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=2 https://epgp.inflibnet.ac.in/epgpdata/uploads/epgp_content/food_technology/food_chemistry/16.protein_ structure_and_denaturation/et/40_et_m16.pdf https://www.youtube.com/watch?v=MODnIkQvyz0 https://www.youtube.com/watch?v=AeVDoDp3llI
Learning Outcomes	Upon completion of this course, students will be able to understand and describe the biosynthesis of essential carbon compounds like sucrose, starch, and cellulose, and elucidate sulfur and phosphorus metabolism, including ATP synthesis. They will gain insights into nitrogen metabolism, focusing on nitrogen fixation and nitrate reduction. Students will be proficient in explaining DNA structure, replication, and repair, RNA transcription, processing, and translation, and protein synthesis and post-translational modifications. Additionally, they will be familiar with enzyme mechanisms and molecular techniques such as electrophoresis and molecular cloning, enabling practical applications in plant biochemistry and molecular biology.

Course Title	BOTMJ72R: Lab Work Based on Paper BOTMJ71R	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Chromatographic separation and comparative analysis of photosynthetic pigments in plants • Chromatographic separation of amino acids	6.0 h
II	• Preparation of standard graph and estimation of DNA by diphenylamine test • Preparation of standard graph and estimation of RNA by Orcinol test	6.0 h
III	• Preparation of standard graph and estimation of protein by Bradford method • Study on the effects of light/dark on the nitrate reductase enzyme	6.0 h
IV	• To study the time dependent activity of urease enzyme. • Study on the effects of different concentrations of urea on urease activity	6.0 h
V	• Determination of different concentrations of urease enzyme on its activity • Preparation of standard graph and estimation of carbohydrates by anthrone reagent method • Demonstration of electrophoretic and blotting instruments	6.0 h

Course Title	**BOTMJ73R: Plant Biotechnology and Genetic Engineering**			
Category of Course[98]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[99] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of plant biotechnology and genetic engineering, covering historical developments, model plants, and applications. Students will learn about recombinant DNA technology, including vectors, restriction enzymes, and genomic libraries. They will explore plant cell, tissue, and organ culture techniques, including totipotency, media preparation, and regeneration protocols. The course will cover plant transformation techniques, including both indirect and direct methods. Students will examine transgenic plants and their applications, gene expression regulation, and advanced gene editing technologies like ZFNs, TALENs, and CRISPR/Cas systems, addressing their applications, limitations, and ethical considerations.			

Units	**Course Content**	**Hours of Teaching: 45 h**
I	• Plant Biotechnology: Introduction, History, Model plants, Scope and applications • Recombinant DNA technology: Vectors, restriction enzymes, molecular probes, genomic and cDNA libraries	8.5 h
II	• Plant Cell, Tissue and Organ Culture (a) *History, concept of totipotency, nutritional requirements of culture, mediapreparation, aseptic condition, applications and limitations* (b) *Regeneration Protocols: Single Cell Culture, Organogenesis and Somatic embryogenesis, factors and significance*	8.5 h

[98] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[99] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	(c) Protoplast Culture and Fusion: Protoplast isolation and culture, Protoplastfusion, methods and factors, applications	
III	• Techniques for plant transformation *(a) Indirect gene transfer methods (Agrobacterium tumefaciens, and Agrobacterium rhizogenes mediated transfer)* *(b)Direct Gene transfer methods (electroporation, particle bombardment, microinjection, PEG, silicon carbide)*	8.0 h
IV	• Transgenic plants, some genetically modified plants (Flavr Savr tomato, Bt brinjal, Bt corn, Bt cotton, Golden rice, virus resistant plants, pest resistant plants) and molecular forming, applications and limitations (ethical and environmental issues)	8.0 h
V	• Gene expression and its regulation in plants: Transcription factors and cis-regulatory elements (CREs), epigenetic mechanisms of transcriptional regulation, translational control, programmed DNA rearrangements. • Gene editing and its applications: Zinc-finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs), CRISPR/Cas systems	12.0 h

Texts/ References	1. Bhojwani SS & Dantu PK (2013) Plant Tissue Culture: An Introductory Text, Springer. 2. Park S (2021) Plant Tissue Culture: Techniques and Experiments, Fourth Edition, Academic Press, Elsevier. 3. Razdan MK (2003) Introduction to Plant Tissue Culture, Science Publishers, USA 4. Slater A, Scott NW and Fowler MR (2008) Plant biotechnology: the genetic manipulation of plants, second edition, Oxford University Press. 5. Sahni S, Prasad BD and Kumar P (2017) Plant Biotechnology: Transgenics, Stress 6. Management and Biosafety Issues, CRC Press.
Learning Outcomes	Upon completing this course, students will be proficient in plant biotechnology and genetic engineering techniques. They will understand the historical development and applications of plant biotechnology, including recombinant DNA technology and plant cell culture methods. Students will gain expertise in various plant transformation techniques, both indirect and direct, and be familiar with the production and use of transgenic plants. They will also learn about gene expression regulation and advanced gene editing technologies, such as ZFNs, TALENs, and CRISPR/Cas systems. Additionally, students will address the applications, limitations, and ethical considerations of these biotechnological advances.

Course Title	**BOTMJ74R: Lab Work Based on paper BOTMJ73R**

Units	**Practicals**	**Hours of Teaching:30 h**
I	• Medium preparation of Murashige and Skoog • Surface sterilization and inoculation of nodal explants on MS medium	6.0 h
II	• To raise suspension culture by using friable calli	6.0 h
III	• To initiate hairy root culture	6.0 h
IV	• Floral dip method of plant transformation (*Arabidopsis thaliana*) and screening for gene transfer (plasmid isolation & transformation, screening through PCR and agarose gel electrophoresis)	6.0 h
V	• Synthetic seed preparation	6.0 h

<table>
<tr><td>Course Title</td><td colspan="4">BOTMJ75R: Ecosystem Dynamics</td></tr>
<tr><td>Category of Course[100]</td><td colspan="4">Major/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation</td></tr>
<tr><td rowspan="3">Credits[101]& Hour of Teaching</td><td></td><td>Theory</td><td>Practical</td><td>Cumulative</td></tr>
<tr><td>Credits</td><td>3</td><td>1</td><td>4</td></tr>
<tr><td>Hour of Teaching</td><td>45</td><td>30</td><td>75</td></tr>
<tr><td>Course Objectives</td><td colspan="4">This course aims to provide students with a comprehensive understanding of ecosystem dynamics, including the organization and development of communities, mechanisms of ecological succession, and the evolution of ecosystems. Students will explore energy flow models, primary productivity, and nutrient cycling within ecosystems. The course covers ecosystem stability, complexity, and sustainability, and examines the interactions between human activities and ecosystem dynamics. By integrating theoretical concepts and practical applications, students will develop insights into maintaining and managing ecosystem health and functionality.</td></tr>
</table>

Units	Course Content	Hours of Teaching:45 h
I	• Introduction: Introduction to ecosystem dynamics • Community organization: Holistic, Reductionist, continuum concepts and related theories, Synthetic approach	9.0 h
II	• Ecosystem development: Mechanism and concepts of ecological succession, pattern of ecosystem attributes along successional stages, Concept and models of succession	9.5 h
III	• Ecosystem processes: Energy flow models, Net primary productivity, functional metabolic ratio, ecological efficiencies, decomposition, herbivory and carnivory	9.5 h
IV	• Nutrient dynamics: Nutrient cycling and models, nutrient import, export and retention, nutrient use efficiency, nutrient resorption and retranslocation, nutrient conserving adaptation in oligotrophic soils	9.5 h
V	• Ecosystem stability, complexity and sustainability • Man and ecosystem dynamics	7.5 h

<table>
<tr><td>Texts/ References</td><td>1. Begon, M., Townsend, C. R. 2021) Ecology: From Individuals to Ecosystems. United Kingdom: Wiley.
2. Bradshaw RHW and Sykes MT. 2014. Ecosystem Dynamics: From the Past to the Future. United Kingdom: Wiley.
3. Molles Jr MC and Laursen A. 2020. Ecology: Concepts and Applications Author(s): 5th Canadian Edition McGraw-Hill
4. Mueller-Dombois D and Ellenberg 1974. Aims and Methods of Vegetation Ecology. John Wiley & Sons. New York
5. Odum Eugene Pleasants 1967. Fundamentals of Ecology. United States: Saunders.
6. Raffaelli DG, Frid CLJ, 2010. Ecosystem Ecology: A New Synthesis. Cambridge University Press
7. Singh JS, Singh SP and Gupta SR. 2014. Ecology, Environmental Science and Conservation. S. Chand & Company Pvt. Ltd., New Delhi</td></tr>
<tr><td>Learning Outcomes</td><td>Upon completing this course, students will be able to understand and analyze ecosystem dynamics, including community organization, ecological succession, and ecosystem development. They will gain insights into energy flow models, nutrient cycling, and ecosystem processes such as decomposition and herbivory. Students will also learn to evaluate ecosystem stability, complexity, and sustainability, and understand the impact of human activities on ecosystems. Through this knowledge, they will be equipped to apply concepts in managing and conserving ecosystems effectively.</td></tr>
</table>

[100] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[101] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Title	BOTMJ76R: Lab work based on paper BOTMJ75R	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Estimation of herbaceous species biomass • Analysis of species composition (based on IVI)	10.0 h
II	• Quantification of dominance	5.0 h
III	• Estimation of soil carbon	5.0 h
IV	• Estimation of total nitrogen	5.0 h
V	• Estimation of Chlorophyll content	5.0 h

Course Title	BOTMJ77R: Field Excursion			
Category of Course[102]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[103]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	2		2
	Hour of Teaching	30		30
Course Objectives	Field excursions will provide immersive, hands-on learning experiences that bridge theoretical knowledge and real-world applications. These trips are designed to enhance students' understanding of ecological, geological, botanical, and environmental concepts through direct observation and interaction with natural habitats. By engaging in fieldwork, students develop essential skills in data collection, species identification, and environmental assessment, fostering critical thinking and problem-solving abilities. Field excursions also promote teamwork, collaboration, and effective communication, as students often work in groups to conduct research and share findings. Moreover, these experiences help students appreciate the complexity and interconnectedness of natural systems, encouraging a deeper commitment to environmental conservation and sustainability. Ultimately, field excursions enrich the educational journey by making abstract concepts tangible and inspiring a lasting passion for scientific inquiry and exploration.			
Learning Outcomes	Field excursions for BSc students offer hands-on learning experiences, enhancing their understanding of ecological systems, plant identification, and data collection methods. These trips provide practical insights into theoretical concepts, foster observational and analytical skills, and encourage environmental awareness. By interacting directly with nature, students gain a deeper appreciation for biodiversity and conservation. Additionally, fieldwork promotes teamwork, critical thinking, and problem-solving abilities, preparing students for advanced research and professional careers in biological sciences.			

[102] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[103] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Course Title	BOTMJ78R: Research Methodology
Category of Course[104]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC / MD/Internship/Dissertation
Credits[105]& Hour of Teaching	(see below)
Course Objectives	This course aims to equip students with essential research methodology skills, covering the entire research process from problem formulation to data collection, analysis, and reporting. It focuses on ethical research practices, effective literature review, diverse data collection methods, and the use of statistical and qualitative analysis tools. The course also enhances students' abilities to present and publish their research findings.

	Theory	Practical	Cumulative
Credits	2	0	2
Hour of Teaching	30	0	30

Units	Course Content	Hours of Teaching:30 h
I	Introduction to Research Methodology Understanding research, its significance, and different types of research, Steps involved in the research process, Ethical considerations and plagiarism, Identifying and defining research problems, Types of research designs	6.0 h
II	Literature Review Role and significance of literature review in research, Identifying and accessing various sources of literature. Methods for effective literature review, Importance and methods of citation and referencing	6.0 h
III	Data Collection Methods Techniques for collecting primary data (surveys, interviews, observations, experiments), Utilizing secondary data sources, understanding different sampling techniques (random, stratified, cluster), Designing and using questionnaires, interview guides, and other tools, Reliability and validity in data collection	6.0 h
IV	Data Analysis and Interpretation Statistical methods for analysing quantitative data, Techniques for analysing qualitative data, Introduction to statistical software (SPSS, R) and qualitative analysis tools, making sense of data and drawing conclusions, Effective ways to present data using tables, graphs, and charts	6.0 h
V	Research Reporting and Presentation Structure and components of a research report (introduction, methodology, results, discussion, conclusion), Guidelines for writing a thesis or dissertation, writing research papers and articles for journals, Techniques for presenting research findings orally and visually, Understanding the publication process and choosing appropriate journals	6.0 h

Texts/ References	1. Research Methodology: Methods and Techniques" by C.R. Kothari and Gaurav Garg 2. Research Methods for Business Students" by Mark Saunders, Philip Lewis, and Adrian Thornhill 3. The Craft of Research" by Wayne C. Booth, Gregory G. Colomb, and Joseph M. Williams 4. Research Design: Qualitative, Quantitative, and Mixed Methods Approaches" by John W.

[104] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[105] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	Creswell and J. David Creswell 5. Qualitative Inquiry and Research Design: Choosing Among Five Approaches" by John W. Creswell and Cheryl N. Poth 6. Research Methods: The Essential Knowledge Base" by William M. Trochim and James P. Donnelly 7. Research Methods in Education" by Louis Cohen, Lawrence Manion, and Keith Morrison 8. Practical Research: Planning and Design" by Paul D. Leedy and Jeanne Ellis Ormrod 9. Introduction to Research Methods: A Hands-On Approach" by Bora Pajo **Suggested Websites** https://bio-protocol.org/en https://conductscience.com/microbiology-techniques
Learning Outcomes	By the end of this course, students will proficiently design, conduct, and analyze research, adhering to ethical standards. They will be adept at literature review, data collection, and using analytical tools, enabling them to write structured research reports and present their findings effectively.

Course Title	**BOTMN71R: Plant Biochemistry and Molecular Biology**			
Category of Course[106]	Major/ **Minor** / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[107]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	The course objective is to provide students with a comprehensive understanding of plant biochemistry and molecular biology. Students will learn the biosynthesis of key carbon compounds, including sucrose, starch, and cellulose, and delve into sulfur and phosphorus metabolism, including ATP synthesis. They will gain insight into nitrogen metabolism, including the biochemistry of nitrogen fixation and nitrate reduction. The course will cover the structure and replication of DNA, RNA transcription and processing, and protein synthesis and modification. Additionally, students will explore enzyme mechanisms and molecular tools and techniques such as electrophoresis and cloning vectors.			

Units	**Course Content**	**Hours of Teaching: 45 h**
I	Biosynthesis of carbon compounds: sucrose, starch, cellulose Sulphur and phosphorus metabolism: Activation and assimilation of sulphur, energy-rich phosphorus compounds; ATP synthesis	8.0 h
II	Nitrogen metabolism: components, types and substrate of enzyme nitrogenase, biochemistry of nitrogen fixation; Nitrate metabolism: Uptake and reduction into ammonia, ammonia assimilation	8.0 h
III	DNA: Structure and properties of different forms of DNA, DNA replication in prokaryotes and eukaryotes, DNA damage and repair	8.0 h
IV	RNA:Structure and properties of different forms of RNA, transcription in prokaryotes and eukaryotes, mRNA processing-5' cap formation- 3'end processing and polyadenylation, splicing, nuclear export of mRNA	8.0 h
V	Protein: Basic aspects of protein conformation, translation in prokaryotes and eukaryotes (activation of amino acids,	13.0 h

[106] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[107] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	initiation, elongation, termination and release of peptides), post-translational modification of proteins Enzymes: Mechanism of enzyme action, coenzymes, allosteric enzyme, isozymes Molecular tools and techniques: Basic principles of electrophoresis and blotting, cloning vectors, molecular cloning of nucleic acid fragments	

Texts/ References	Cox MM, Nelson DL (2021) Lehninger Principle of Biochemistry, 8th edition MacMillan, Worth Publishers,London. Devlin TM (2011) Textbook of Biochemistry, 7th edition, John Wiley & Sons, Inc, New York. Voet D, Voet JG and Pratt CW (2018) Biochemistry, 5th Edition, John Wiley & Sons, Inc, New York. Zubay GL (1998) Biochemistry, 4th edition, Wm C Brown Publishers, Dubuque, Iowa. Dubey RC (2013) A Text Book of Biotechnology 4th edition, S Chand & Co Ltd, New Delhi. Wilson K, Walker J (2018) Principle & Techniques of Biochemistry and Molecular Biology, 8th edition, Cambridge University Press, Cambridge. Bowsher C & Tobin A (2021) Plant Biochemistry, 2nd Edition, Taylor & Francis Inc, CRC press. Arthur M. Lesk (2016) Introduction to Protein Science Architecture Function and Genomics, Oxford University press, London. Berg JM et al. (2018) Biochemistry, 8th edition, WH Freeman & Company, New York. **Suggested websites** https://academic.oup.com/aob/article/105/7/1141/148741 https://dx.doi.org/10.1016/B978-0-12-800180-6.00001-3 https://doi:10.1016/j.molp.2016.05.004 https://doi.org/10.1093/jn/130.5.1081 https://academic.oup.com/jxb/article/68/10/2463/2967464 https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=2 https://epgp.inflibnet.ac.in/epgpdata/uploads/epgp_content/food_technology/food_chemistry/16.protein_ structure_and_denaturation/et/40_et_m16.pdf https://www.youtube.com/watch?v=MODnIkQvyz0 https://www.youtube.com/watch?v=AeVDoDp3llI
Learning Outcomes	Upon completion of this course, students will be able to understand and describe the biosynthesis of essential carbon compounds like sucrose, starch, and cellulose, and elucidate sulfur and phosphorus metabolism, including ATP synthesis. They will gain insights into nitrogen metabolism, focusing on nitrogen fixation and nitrate reduction. Students will be proficient in explaining DNA structure, replication, and repair, RNA transcription, processing, and translation, and protein synthesis and post-translational modifications. Additionally, they will be familiar with enzyme mechanisms and molecular techniques such as electrophoresis and molecular cloning, enabling practical applications in plant biochemistry and molecular biology.

Course Title	**BOTMN72R: Lab Work Based on Paper BOTMN71R**

Units	**Practicals**	**Hours of Teaching:30 h**
I	• Chromatographic separation and comparative analysis of photosynthetic pigments in plants • Chromatographic separation of amino acids	6.0 h
II	• Preparation of standard graph and estimation of DNA by diphenylamine test • Preparation of standard graph and estimation of RNA by Orcinol test	6.0 h
III	• Preparation of standard graph and estimation of protein by Bradford method • Study on the effects of light/dark on the nitrate reductase enzyme	6.0 h
IV	• To study the time dependent activity of urease enzyme • Study on the effects of different concentrations of urea on urease activity	6.0 h

V	• Determination of different concentrations of urease enzyme on its activity • Preparation of standard graph and estimation of carbohydrates by anthrone reagent method • Demonstration of electrophoretic and blotting instruments	6.0 h

Course Title	**BOTMN73R: Plant Biotechnology and Genetic Engineering**			
Category of Course[108]	Major/**Minor** / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[109]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of plant biotechnology and genetic engineering, covering historical developments, model plants, and applications. Students will learn about recombinant DNA technology, including vectors, restriction enzymes, and genomic libraries. They will explore plant cell, tissue, and organ culture techniques, including totipotency, media preparation, and regeneration protocols. The course will cover plant transformation techniques, including both indirect and direct methods. Students will examine transgenic plants and their applications, gene expression regulation, and advanced gene editing technologies like ZFNs, TALENs, and CRISPR/Cas systems, addressing their applications, limitations, and ethical considerations.			

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Plant Biotechnology: Introduction, History, Model plants, Scope and applications • Recombinant DNA technology: Vectors, restriction enzymes, molecular probes, genomic and cDNA libraries	8.5 h
II	• Plant Cell, Tissue and Organ Culture *(a) History, concept of totipotency, nutritional requirementsof culture, mediapreparation, aseptic condition, applications and limitations* *(b) Regeneration Protocols: Single Cell Culture, Organogenesis and SomaticEmbryogenesis, Factors and significance* *(c) Protoplast Culture and Fusion: Protoplast isolation and culture, Protoplastfusion, methods and factors, applications*	8.5 h
III	• Techniques for plant transformation *(a) Indirect gene transfer methods (Agrobacterium tumefaciens, and Agrobacterium rhizogenes mediated transfer)* *(b) Direct Gene transfer methods (electroporation, particle bombardment, microinjection, PEG, silicon carbide)*	8.0 h
IV	• Transgenic plants, some genetically modified plants (Flavr Savr tomato, Bt brinjal, Bt corn, Bt cotton, Golden rice, virus resistant plants, pest resistant plants) and molecular forming, applications and limitations (ethical and environmental issues)	8.0 h
V	• Gene expression and its regulation in plants: Transcription factors and cis-regulatory elements (CREs), epigenetic mechanisms of transcriptional regulation, translational control, programmed DNA rearrangements • Gene editing and its applications: Zinc-finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs), CRISPR/Cas systems	12.0 h

[108] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[109] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Texts/References	1. Bhojwani SS & Dantu PK (2013) Plant Tissue Culture: An Introductory Text, Springer. 2. Park S (2021) Plant Tissue Culture: Techniques and Experiments, Fourth Edition, Academic Press, Elsevier. 3. Razdan MK (2003) Introduction to Plant Tissue Culture, Science Publishers, USA 4. Slater A, Scott NW and Fowler MR (2008) Plant biotechnology: the genetic manipulation of plants, second edition, Oxford University Press. 5. Sahni S, Prasad BD and Kumar P (2017) Plant Biotechnology: Transgenics, Stress 6. Management and Biosafety Issues, CRC Press.
Learning Outcomes	Upon completing this course, students will be proficient in plant biotechnology and genetic engineering techniques. They will understand the historical development and applications of plant biotechnology, including recombinant DNA technology and plant cell culture methods. Students will gain expertise in various plant transformation techniques, both indirect and direct, and be familiar with the production and use of transgenic plants. They will also learn about gene expression regulation and advanced gene editing technologies, such as ZFNs, TALENs, and CRISPR/Cas systems. Additionally, students will address the applications, limitations, and ethical considerations of these biotechnological advances.

Course Title	**BOTMN74R: Lab Work Based on Paper BOTMN73R**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• Medium preparation of Murashige and Skoog • Surface sterilization and inoculation of nodal explants on MS medium	6.0 h
II	• To raise suspension culture by using friable calli.	6.0 h
III	• To initiate hairy root culture	6.0 h
IV	• Floral dip method of plant transformation (*Arabidopsis thaliana*) and screening for gene transfer (plasmid isolation & transformation, screening through PCR and agarose gel electrophoresis)	6.0 h
V	• Synthetic seed preparation	6.0 h

Course Title	**BOTMN75R: Ecosystem Dynamics**			
Category of Course[110]	Major/ **Minor**/ Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[111]**& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide students with a comprehensive understanding of ecosystem dynamics, including the organization and development of communities, mechanisms of ecological succession, and the evolution of ecosystems. Students will explore energy flow models, primary productivity, and nutrient cycling within ecosystems. The course covers ecosystem stability, complexity, and sustainability, and examines the interactions between human activities and ecosystem dynamics. By integrating theoretical concepts and practical applications, students will develop insights into maintaining and managing ecosystem health and functionality.			

110 SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

111 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

Units	Course Content	Hours of Teaching:45 h
I	• Introduction: Introduction to ecosystem dynamics. • Community organization: Holistic, Reductionist, continuum concepts and related theories, Synthetic approach	9.0 h
II	• Ecosystem development: Mechanism and concepts of ecological succession, pattern of ecosystem attributes along successional stages, Evolution of ecosystem	9.5 h
III	• Ecosystem processes: Energy flow models, Net primary productivity, functional metabolic ratio, ecological efficiencies, decomposition, herbivory and carnivory	9.5 h
IV	• Nutrient dynamics: Nutrient cycling and models, nutrient import, export and retention, nutrient use efficiency, nutrient resorption and retranslocation, nutrient conserving adaptation in oligotrophic soils	9.5 h
V	• Ecosystem stability, complexity and sustainability • Man and ecosystem dynamics	7.5 h

Texts/ References	1. Begon, M., Townsend, C. R. 2021) Ecology: From Individuals to Ecosystems. United Kingdom: Wiley. 2. Bradshaw RHW and Sykes MT. 2014. Ecosystem Dynamics: From the Past to the Future. United Kingdom: Wiley. 3. Molles Jr MC and Laursen A. 2020. Ecology: Concepts and Applications Author(s): 5th Canadian Edition McGraw-Hill 4. Mueller-Dombois D and Ellenberg 1974. Aims and Methods of Vegetation Ecology. John Wiley & Sons. New York 5. Odum Eugene Pleasants 1967. Fundamentals of Ecology. United States: Saunders. 6. Raffaelli DG, Frid CLJ, 2010. Ecosystem Ecology: A New Synthesis. Cambridge University Press 7. Singh JS, Singh SP and Gupta SR. 2014. Ecology, Environmental Science and Conservation. S. Chand & Company Pvt. Ltd., New Delhi
Learning Outcomes	Upon completing this course, students will be able to understand and analyze ecosystem dynamics, including community organization, ecological succession, and ecosystem development. They will gain insights into energy flow models, nutrient cycling, and ecosystem processes such as decomposition and herbivory. Students will also learn to evaluate ecosystem stability, complexity, and sustainability, and understand the impact of human activities on ecosystems. Through this knowledge, they will be equipped to apply concepts in managing and conserving ecosystems effectively.

Course Title	**BOTMN76R: Lab Work Based on Paper BOTMN75R**	
Units	**Practicals**	**Hours of Teaching: 30 h**
I	• Estimation of herbaceous species biomass • Analysis of species composition (based on IVI)	10.0 h
II	• Quantification of dominance	5.0 h
III	• Estimation of soil carbon	5.0 h
IV	• Estimation of total nitrogen	5.0 h
V	• Estimation of Chlorophyll content	5.0 h

SEMESTER-VIII

Course Title	BOTDS81R: Dissertation			
Category of Course[112]	Major / Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/**Dissertation**			
Credits[113]& Hour of Teaching		Theory	Practical	Cumulative
	Credits			12

Course Title	BOTMJ82R: Aquatic Ecology			
Category of Course[114]	**Major**/ Minor / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[115]& Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of aquatic ecosystems, including hydrological cycles, primary productivity, and food web dynamics. Students will explore the characteristics and ecological processes of lakes, rivers, estuaries, and oceans, focusing on topics such as thermal stratification, eutrophication, and nutrient limitations. The course will also cover the diversity of aquatic organisms, including phytoplankton, zooplankton, and benthic communities, and address the ecological significance and conservation of coral reefs and other marine biota. Through this course, students will develop skills to analyze and manage aquatic environments effectively.			

Units	Course Content	Hours of Teaching:45 h
I	• Hydrological cycle; Watershed hydrology • Aquatic primary productivity and its measurement	10.0 h
II	• Aquatic food webs; trophic cascade • Lakes: General characteristics, ecological zonation in lakes, thermal stratification, water circulation	11.0 h
III	• Diversity of phytoplankton and zooplankton, concept of nutrient limitation • General account of benthic and periphytic communities	8.0 h
IV	• Eutrophication: Causes and consequences, indices of eutrophy, control of eutrophication • Rivers: Characteristics of lotic systems; River continuum concept	7.0 h
V	• Estuaries: General account of estuaries, types and significance • Oceans: Ecological zonation in marine environment, marine biota, coral reefs	9.0 h

Texts/ References	
	1. Singh JS, Singh SP, Gupta SR (2014) Ecology Environmental Science and Conservation, S Chand & Co, New Delhi. 2. Barnes RSK and Mann KH 1991. Fundamentals of aquatic ecology. Wiley and Blackwell. 3. Doods WK. and Whiles MR, Freshwater Ecology, 3rd Edition Concepts and Environmental

112 SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

113 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

114 SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

115 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

	Applications of Limnology. Elesiever. 4. Polunin NVC 2010. Aquatic ecosystems, Cambridge university press.
Learning Outcomes	Upon completion of this course, students will be able to analyze and interpret the dynamics of aquatic ecosystems, including the hydrological cycle, primary productivity, and trophic interactions. They will understand the ecological characteristics of lakes, rivers, estuaries, and oceans, with insights into thermal stratification, nutrient limitations, and eutrophication. Students will gain knowledge about the diversity of aquatic organisms, including phytoplankton, zooplankton, and benthic communities. They will also be equipped to assess and address environmental issues related to aquatic systems, such as eutrophication and conservation of marine habitats like coral reefs.

Course Title	**BOTMJ83R: Lab Work Based on Paper BOTMJ82R**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• To determine the pH of given water samples • To estimate free CO_2 in given water samples	6.0 h
II	• To estimate the chloride concentration in given water samples • To determine the total hardness of the water samples	6.0 h
III	• To measure the dissolve oxygen (DO) in water samples • To estimate the nitrate concentration in water samples	6.0 h
IV	• To estimate the phosphate concentration in given water samples • To measure the biological oxygen demand (BOD) in given water samples	6.0 h
V	• To estimate the primary productivity in given water samples	6.0 h

Course Title	**BOTMN84R: Aquatic Ecology**			
Category of Course[116]	Major/ **Minor** / Minor (Vocational) / SEC / AEC /VAC/ MD/Internship/Dissertation			
Credits[117] **& Hour of Teaching**		Theory	Practical	Cumulative
	Credits	3	1	4
	Hour of Teaching	45	30	75
Course Objectives	This course aims to provide a comprehensive understanding of aquatic ecosystems, including hydrological cycles, primary productivity, and food web dynamics. Students will explore the characteristics and ecological processes of lakes, rivers, estuaries, and oceans, focusing on topics such as thermal stratification, eutrophication, and nutrient limitations. The course will also cover the diversity of aquatic organisms, including phytoplankton, zooplankton, and benthic communities, and address the ecological significance and conservation of coral reefs and other marine biota. Through this course, students will develop skills to analyze and manage aquatic environments effectively.			

Units	**Course Content**	**Hours of Teaching:45 h**
I	• Hydrological cycle; Watershed hydrology • Aquatic primary productivity and its measurement	10.0 h
II	• Aquatic food webs; trophic cascade • Lakes: General characteristics, ecological zonation inlakes, thermal stratification, water circulation	11.0 h

[116] SEC – Skill Enhancement Course; AEC – Ability Enhancement Course; VAC – Value Added Course; MD – Multidisciplinary Course

[117] 1 Credit (Theory) = 15 Hours; 1 Credit (Practical) = 30 Hours

III	• Diversity of phytoplankton and zooplankton, concept of nutrient limitation • General account of benthic and periphytic communities	8.0 h
IV	• Eutrophication: Causes and consequences, indices of eutrophy, control of eutrophication • Rivers: Characteristics of lotic systems; River continuum concept.	7.0 h
V	• Estuaries: General account of estuaries, types and significance • Oceans: Ecological zonation in marine environment, marine biota, coral reefs	9.0 h

Texts/ References	1. Singh JS, Singh SP, Gupta SR (2014) Ecology Environmental Science and Conservation, S Chand & Co, New Delhi. 2. Barnes RSK and Mann KH 1991. Fundamentals of aquatic ecology. Wiley and Blackwell. *3. Doods WK. and Whiles MR, Freshwater Ecology, 3rd Edition Concepts and Environmental Applications of Limnology. Elesiever.* 4. Polunin NVC 2010. Aquatic ecosystems, Cambridge university press.
Learning Outcomes	Upon completion of this course, students will be able to analyze and interpret the dynamics of aquatic ecosystems, including the hydrological cycle, primary productivity, and trophic interactions. They will understand the ecological characteristics of lakes, rivers, estuaries, and oceans, with insights into thermal stratification, nutrient limitations, and eutrophication. Students will gain knowledge about the diversity of aquatic organisms, including phytoplankton, zooplankton, and benthic communities. They will also be equipped to assess and address environmental issues related to aquatic systems, such as eutrophication and conservation of marine habitats like coral reefs.

Course Title	**BOTMN85R: Lab Work Based on Paper BOTMN84R**	
Units	**Practicals**	**Hours of Teaching:30 h**
I	• To determine the pH of given water samples • To estimate free CO_2 in given water samples	6.0 h
II	• To estimate the chloride concentration in given water samples • To determine the total hardness of the water samples	6.0 h
III	• To measure the dissolve oxygen (DO) in water samples • To estimate the nitrate concentration in water samples	6.0 h
IV	• To estimate the phosphate concentration in given water samples • To measure the biological oxygen demand (BOD) in given water samples	6.0 h
V	• To estimate the primary productivity in given water samples	6.0 h